



Epoka University

Faculty of Architecture and Engineering

Department of Computer Engineering

Program : Software Engineering

Course : Software Modeling and Design

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**Software Modeling and Design Project - Airport Management
System**

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1. Executive Summary

1.1 Project Overview

The aim of the Airport Management System is to efficiently manage and coordinate the main operations of an airport. The system is designed to handle flight scheduling, passenger management, staff coordination, ticketing, and airport services in an integrated and organized way. It aims to improve operational efficiency, reduce errors, enhance security, and provide a better experience for passengers and airport staff.

2. Product/Service Description

2.1 Product Context

The Airport Management System is an enterprise-level system that operates as part of a larger airport ecosystem rather than as an independent application. It serves as a central hub through which multiple airport stakeholders exchange operational information.

The system interacts with various internal and external systems including:

- Airline reservation and booking systems
- Flight Information Display Systems (FIDS)
- Airport Operational Database (AODB)
- Baggage handling and tracking systems
- Security monitoring and surveillance systems
- Access control systems
- Human Resource Management systems
- Finance and billing systems
- Maintenance management systems
- Asset management systems
- Immigration and customs systems
- Notification and communication services

The system collects, processes, stores, and distributes information related to flights, passengers, employees, assets, airport facilities, and operational activities. It acts as a "single source of truth" for airport operations and enables coordination between departments.

2.2 User Characteristics

1. Administrator

Responsible for overall system configuration and management. The administrator manages user accounts, assigns permissions and roles, maintains system settings, and oversees the correct functioning of the system.

2. Operations Staff

Responsible for operational planning and coordination. This role manages gate and stand allocation, oversees flight scheduling, and monitors operational data through the Airport Operational Database.

3. Ground Handling Staff

Manages activities related to aircraft services on the ground. Responsibilities include flight handling, baggage management, apron operations, and coordination with airline and airport staff during aircraft arrival and departure.

4. Airline Staff

Handles airline-related services such as passenger check-in, boarding procedures, ticket validation, and seat allocation. They also manage passenger lists for their airline flights.

5. Security Staff

Ensures airport safety and compliance with security regulations. Responsibilities include access control, surveillance monitoring, passenger and staff security checks, and maintaining overall airport safety.

6. Maintenance Staff

Responsible for maintaining airport infrastructure and equipment. Tasks include asset management, equipment maintenance, repair operations, and facility management.

7. Finance Staff

Responsible for managing the financial operations of the airport. This includes handling billing and invoicing for airport services, tracking payments from airlines and passengers, managing service fees, and generating financial reports.

8. Human Resources (HR) Staff

Responsible for managing employee-related processes. This includes maintaining employee records, managing staff schedules and shifts, tracking attendance, and ensuring staff training and certification compliance.

9. Asset Management Staff

Responsible for managing airport assets such as equipment, vehicles, and IT systems. This role ensures proper asset registration, tracking, allocation, and lifecycle management to support efficient airport operations.

3. Requirements

3.1 Functional Requirements

Req#	Requirement	Comments	Priority
BR_01	Administrator shall be able to create, update, deactivate user accounts, and assign roles and permissions to users because user access must be managed securely and different users require different levels of system access.	Business Process = User Management	1
BR_02	Administrator shall be able to modify system configuration settings because the system must adapt to operational and organizational requirements.	Business Process = System Configuration	2
BR_03	Administrator shall be able to monitor system activity logs because user actions and system events must be tracked for security and auditing.	Business Process = System Monitoring	2
BR_04	Administrator shall be able to manage airport reference data (gates, stands, airport codes) because operational data must remain accurate and up to date.	Business Process = Data Maintenance	2
BR_05	Operations Staff shall be able to assign gates to scheduled flights because aircraft must be directed to the correct airport gate.	Business Process = Gate Management	1
BR_06	Operations Staff shall be able to create and update flight schedules because airport operations depend on accurate scheduling information.	Business Process = Flight Scheduling	1
BR_07	Operations Staff shall be able to assign aircraft stands because aircraft parking and ground operations must be coordinated efficiently.	Business Process = Stand Allocation	1
BR_08	Operations Staff shall be able to monitor flight status (arrivals, departures, delays) because operational planning requires up-to-date flight information.	Business Process = Flight Monitoring	2
BR_09	Operations Staff shall be able to update operational records for flights, gates, and stands because airport operations must be accurately documented.	Business Process = Operations Management	2
BR_10	Operations Staff shall be able to detect scheduling conflicts between flights, gates, and stands because operational overlaps must be avoided.	Business Process = Conflict Detection	1
BR_11	Security Staff shall be able to monitor airport surveillance cameras because continuous visual monitoring is required to detect suspicious activities and ensure passenger safety.	Business Process = Surveillance Monitoring	1
BR_12	Security Staff shall be able to control and verify access to restricted airport areas because only authorized personnel should enter secure zones.	Business Process = Access Control	1
BR_13	Security Staff shall be able to perform passenger and staff security checks because safety regulations require screening before entering secure airport areas.	Business Process = Security Screening	1

BR_14	Security Staff shall be able to record and report security incidents because all security events must be documented for investigation and auditing.	Business Process = Incident Reporting	2
BR_15	Security Staff shall be able to receive and respond to security alerts generated by the system because rapid response is required to prevent potential threats.	Business Process = Alert Management	1
BR_16	Security Staff shall be able to review security logs and reports because monitoring historical security data helps identify risks and improve safety procedures.	Business Process = Security Monitoring	2
BR_17	Maintenance Staff shall be able to record equipment maintenance activities because airport systems and equipment require regular servicing.	Business Process = Maintenance Logging	1
BR_18	Maintenance Staff shall be able to schedule preventive maintenance tasks because regular inspections help avoid equipment failures.	Business Process = Preventive Maintenance	1
BR_19	Maintenance Staff shall be able to report infrastructure issues such as damaged facilities or equipment because operational safety depends on well-maintained infrastructure.	Business Process = Issue Reporting	1
BR_20	Maintenance Staff shall be able to track the status of maintenance requests because maintenance operations must be coordinated efficiently.	Business Process = Maintenance Tracking	2
BR_21	Maintenance Staff shall be able to identify, assess, track, and document maintenance issues and repairs because a complete maintenance lifecycle requires continuous monitoring and accurate reporting from detection to resolution.	Business Process = Issue Management	1
BR-22	Ground handling Staff shall view and manage the arrival schedule of aircraft, including assigned gates, parking stands, and estimated arrival times.	Business Process = Schedule Monitoring	2
BR-23	Ground handling Staff shall track, update, and manage baggage loading and unloading operations for each flight.	Business Process = Baggage Handling	1
BR-24	Ground handling Staff shall assign and monitor apron service tasks such as refueling coordination, catering delivery, and ground equipment allocation.	Business Process = Apron Management	1
BR-25	Ground handling Staff shall record and monitor aircraft turnaround activities to ensure readiness for departure.	Business Process = Departure Preparation	2
BR-26	Ground handling Staff shall share flight status updates and operational information with airline staff and airport authorities.	Business Process = Update Reporting	2
BR-27	Ground handling Staff shall confirm completion of ground services (baggage loading, catering, fueling readiness) before an aircraft departure.	Business Process = Confirmation Of Duty	1
BR-28	Airline Staff shall perform passenger check-in, verify identification, and confirm booking details.	Business Process = Passenger Verification	1

BR-29	Airline Staff shall generate and issue boarding passes to checked-in passengers.	Business Process = Pass Printing/Issuing	2
BR-30	Airline Staff shall assign, modify, or reallocate passenger seats for a flight.	Business Process = Seat Handling	3
BR-31	Airline Staff shall validate passenger tickets and confirm eligibility for boarding.	Business Process = Validation	1
BR-32	Airline Staff shall create, update, and view passenger manifests for their airline's flights.	Business Process = Monitoring Manifests	2
BR-33	Airline Staff shall manage the boarding process, including scanning boarding passes and marking passengers as boarded.	Business Process = Boarding Handling	1
BR-34	Finance Staff shall be able to generate invoices for airport services (landing fees, parking fees, passenger services) because financial transactions must be properly recorded and billed.	Business Process = Billing & Invoicing	1
BR-35	Finance Staff shall be able to generate financial reports (revenue, expenses, profit) because management requires financial insights.	Business Process = Financial Reporting	2
BR-36	HR Staff shall be able to manage employee records including personal details, roles, and departments because staff information must be maintained centrally.	Business Process = Employee Management	1
BR-37	HR Staff shall be able to manage employee schedules and shifts because airport operations require continuous staffing.	Business Process = Shift Scheduling	1
BR-38	Asset Management Staff shall be able to register and manage airport assets (vehicles, equipment, IT systems) because all assets must be tracked centrally.	Business Process = Asset Registration	1

3.2 Non-Functional Requirements

3.2.1 Product Requirements

Requirements which specify that the delivered product must behave in a particular way e.g. execution speed, reliability, etc.

Flight Information - The system must provide real-time flight information, including departure times, arrival times, delays, and gate numbers.

Passenger Check-in - Passengers must be able to check in online and receive a digital boarding pass.

Notifications - The system must send notifications for flight updates such as delays, cancellations, and gate changes.

Airport Navigation - The system must provide an interactive map to help passengers find gates, restaurants, and services inside the airport.

Airport Services - The system must allow passengers to access additional services such as parking reservations and lounge bookings.

3.2.1.1 Usability Requirements

1. The system interface shall be simple and user-friendly for airport staff and administrators.
2. The system shall provide clear navigation menus for accessing all major functions.
3. The system shall display error messages that clearly explain the problem and how to fix it.
4. New users shall be able to learn the basic functions within 30 minutes of training.

3.2.1.2 Performance Requirements

1. The system should respond quickly.
2. The system should support many users at the same time.
3. The system should update flight information in real time

3.2.1.3 Availability

1. The system shall be available 24 hours per day, 7 days per week.
2. Scheduled maintenance shall not exceed 2 hours per month.

3.2.1.4 Security

1. The system should require login.
2. The system should protect user data.
3. The system should use role based access.

3.2.1.5 Reliability

1. The system should work without crashes.
2. The system should save data automatically.
3. The system should backup data automatically.

3.2.1.6 Scalability

1. The system shall support future expansion to multiple airports if required.
2. The system shall allow increasing the number of users and flights without major system redesign.

3.2.2 Organizational Requirements

1. The system shall comply with the software development standards and documentation guidelines defined by the airport IT department.
2. The system shall be implemented using technologies approved by the airport organization's IT policies.
3. The system shall maintain system activity logs according to the airport authority's internal auditing procedures.
4. The system shall support regular data backup procedures defined by the airport IT department.
5. The system shall allow administrators to configure user roles and permissions according to airport organizational policies.
6. The system shall support staff training procedures by providing user manuals and system documentation.

The system shall follow the organization's data management policies for storing and maintaining passenger and flight information.

7. The system shall allow integration with the airport's internal information systems used by staff and management.

3.2.3 External Requirements

1. The system shall comply with international aviation regulations and standards applicable to airport operations.
2. The system shall comply with data protection laws regarding the storage and processing of passenger personal information.
3. The system shall support interoperability with airline information systems for exchanging flight data.
4. The system shall support integration with external security and immigration systems when required.
5. The system shall ensure secure communication with external airlines and booking systems.
6. The system shall comply with payment processing regulations when handling passenger payments.
7. The system shall support standard data exchange formats used in aviation systems.
8. The system shall ensure that passenger data is shared with authorized external systems only when permitted by law.

3.2.3.1 System Integration Requirements

The Airport Management System must be able to interact and exchange data with other external systems used within the airport ecosystem. These may include airline reservation systems, baggage handling systems, security monitoring systems, and the Airport Operational Database.

3.2.3.2 Legislative and Regulatory Requirements

The system must comply with international and national aviation regulations as well as data protection laws. Passenger information, flight data, and staff records must be processed and stored according to applicable privacy and security regulations.

3.2.3.3 Safety and Security Requirements

Due to the critical nature of airport operations, the system must implement strong security mechanisms. User authentication and authorization must be enforced based on defined user roles. Sensitive information such as passenger data, security logs, and operational details must be protected through secure communication protocols and data protection mechanisms.

4. User Scenarios/Use Cases

4.1 Use Cases 01-05 -> Moira Lako

UC Name	UC-01: Manage User Accounts
Summary	<i>Administrator creates, updates, deactivates user accounts and assigns roles/permissions.</i>
Dependency	<i>None</i>
Actors	Primary: Administrator
Preconditions	• <i>Administrator is authenticated; system is accessible; administrator has user-management permission.</i>
Description of the Main Sequence	• <i>Step 1: Administrator logs in.</i> • <i>Step 2: System authenticates credentials.</i> • <i>Step 3: Administrator opens User Management.</i> • <i>Step 4: System loads current user records.</i> • <i>Step 5: Administrator enters user data.</i> • <i>Step 6: System validates data.</i> • <i>Step 7: System checks for duplicates.</i> • <i>Step 8: Administrator assigns roles and permissions.</i> • <i>Step 9: System saves user account.</i> • <i>Step 10: System saves assigned roles/permissions.</i> • <i>Step 11: Confirmation message is displayed.</i>
Description of the Alternative Sequence	• <i>Step 1: Invalid user data entered → system shows error</i> • <i>Step 2: Duplicate user detected → system prevents creation</i> • <i>Step 4: Role/permission missing → system requests correction.</i> • <i>Step 4: Save failure → system displays failure message.</i>
Non functional requirements	<i>Secure authentication</i> <i>Role-based access control</i> <i>Fast response time</i>
Postconditions	<i>User account is successfully/created/updated/deactivated</i>

UC Name	<i>UC-02: Configure System Settings</i>
Summary	<i>Administrator views and modifies system configuration settings.</i>
Dependency	<i>UC-01</i>
Actors	<i>Primary: Administrator</i>
Preconditions	• <i>Administrator is logged in; administrator has configuration access.</i>
Description of the Main Sequence	• <i>Step 1: Administrator logs in.</i> • <i>Step 2: Open Configuration Settings.</i> • <i>Step 3: System loads current settings from repository.</i> • <i>Step 4: Administrator modifies parameters.</i> • <i>Step 5: System validates settings.</i> • <i>Step 6: Valid settings are applied to the system.</i> • <i>Step 7: Updated settings are saved.</i> • <i>Step 8: Confirmation message is shown.</i>
Description of the Alternative Sequence	• <i>Step 1: Invalid configuration → system rejects input</i> • <i>Step 2: Repository save failure → system keeps previous settings.</i>
Non functional requirements	<i>System stability</i> <i>Configuration validation</i>
Postconditions	<i>System settings are updated and stored successfully.</i>

UC Name	<i>UC-03: Monitor System Logs</i>
Summary	<i>Administrator searches, filters, views, and reviews system activity logs.</i>
Dependency	<i>None</i>
Actors	<i>Primary: Administrator</i>
Preconditions	• <i>Administrator is authenticated; logs exist or log repository is accessible.</i>
Description of the Main Sequence	• <i>Step 1: Administrator logs in.</i> • <i>Step 2: Open Logs Dashboard.</i> • <i>Step 3: System loads the dashboard.</i> • <i>Step 4: Administrator enters filter/search criteria.</i> • <i>Step 5: System retrieves matching logs.</i> • <i>Step 6: System displays log list.</i> • <i>Step 7: Administrator selects a log.</i> • <i>Step 8: System loads detailed log information.</i> • <i>Step 9: Administrator reviews activity logs.</i>
Description of the Alternative Sequence	• <i>Step 1: No logs found → system shows empty result</i> • <i>Step 2: Invalid filter → system asks for new criteria.</i> • <i>Step 3: Repository unavailable → system shows retrieval error.</i>
Non functional requirements	<i>High performance log retrieval Secure log storage</i>
Postconditions	<i>Logs are reviewed or no matching logs are reported.</i>

UC Name	<i>UC-04: Manage Airport Reference Data</i>
Summary	<i>Administrator manages airport reference data such as gates, stands, airport codes, and operational reference records.</i>
Dependency	<i>UC-02</i>
Actors	<i>Primary: Administrator</i>
Preconditions	Administrator is logged in; reference data module is available.
Description of the Main Sequence	<i>Step 1: Administrator logs in.</i> <i>Step 2: Accesses Reference Data Module.</i> <i>Step 3: System loads current reference data.</i> <i>Step 4: Administrator adds, updates, or deletes data.</i> <i>Step 5: System validates data.</i> <i>Step 6: Valid changes are applied.</i> <i>Step 7: Updated reference data is saved.</i> <i>Step 8: Confirmation message is shown.</i>
Description of the Alternative Sequence	<i>Step 1: Invalid data → system shows error</i> <i>Step 2: Duplicate reference value → system rejects change.</i> <i>Step 3: Delete blocked due to dependency → system displays warning.</i>
Non functional requirements	<i>Data integrity Validation rules</i>
Postconditions	<i>Reference data is updated successfully.</i>

UC Name	<i>UC-05: Assign Gates to Flights</i>
Summary	<i>Operations Staff assigns available gates to scheduled flights while avoiding conflicts.</i>
Dependency	<i>UC-06</i>
Actors	<i>Primary: Operations Staff</i>
Preconditions	• <i>Flight schedule exists; gates exist; system can access flight and gate repositories.</i>
Description of the Main Sequence	• <i>Step 1: Operations Staff views flight schedule.</i> • <i>Step 2: System loads scheduled flights.</i> • <i>Step 3: Operations Staff selects a flight.</i> • <i>Step 4: System retrieves flight details.</i> • <i>Step 5: System retrieves available gates.</i> • <i>Step 6: System validates gate assignment.</i> • <i>Step 7: Operations Staff confirms selected gate.</i> • <i>Step 8: System creates assignment.</i> • <i>Step 9: Flight gate status is updated.</i> • <i>Step 10: Assignment is saved.</i> • <i>Step 11: Confirmation message is shown.</i>
Description of the Alternative Sequence	• <i>Step 1: Gate unavailable → system prompts another selection</i> • <i>Step 2: Assignment conflict → system displays alternative gates.</i> • <i>Step 3: No gates available → system informs staff.</i> • <i>Step 4: Save failure → assignment is not confirmed.</i>
Non functional requirements	<i>Real-time updates Conflict prevention</i>
Postconditions	<i>Gate is assigned to the flight and stored successfully.</i>

4.2 Use Cases 06-10 -> Tea Metani

UC Name	UC-06: Manage Flight Schedule
Summary	<i>Operations Staff creates and updates flight schedules.</i>
Dependency	<i>None</i>
Actors	Primary: <i>Operations Staff</i>
Preconditions	• <i>Staff logged in</i>
Description of the Main Sequence	• <i>Open scheduling module</i> • <i>Create/update flight details</i> • <i>Save schedule</i>
Description of the Alternative Sequence	• <i>Invalid schedule → rejected</i>
Non functional requirements	<i>High availability; Data accuracy</i>
Postconditions	<i>Flight schedule updated</i>

UC Name	<i>UC-07: Assign Aircraft Stands</i>
Summary	<i>Operations Staff assigns stands to aircraft.</i>
Dependency	<i>UC-06</i>
Actors	<i>Operations Staff</i>
Preconditions	• <i>Aircraft and schedule exist</i>
Description of the Main Sequence	• <i>Select flight/aircraft</i> • <i>Assign stand</i> • <i>Confirm assignment</i>
Description of the Alternative Sequence	• <i>Stand conflict → alert</i>
Non functional requirements	<i>Conflict detection; Real-time updates</i>
Postconditions	<i>Stand assigned</i>

UC Name	<i>UC-08: Monitor Flight Status</i>
Summary	<i>Operations Staff monitors flight status.</i>
Dependency	<i>UC-06</i>
Actors	<i>Operations Staff</i>
Preconditions	• <i>Flights exist in system</i>
Description of the Main Sequence	• <i>Open monitoring dashboard</i> • <i>View flight status</i> • <i>Track updates</i>
Description of the Alternative Sequence	• <i>Data unavailable → notification</i>
Non functional requirements	<i>Real-time data; High performance</i>
Postconditions	<i>Flight status monitored</i>

UC Name	<i>UC-09: Update Operational Records</i>
Summary	<i>Operations Staff updates operational records.</i>
Dependency	<i>UC-05, UC-07</i>
Actors	<i>Operations Staff</i>
Preconditions	• <i>Operational data exists</i>
Description of the Main Sequence	• <i>Select record</i> • <i>Modify details</i> • <i>Save updates</i>
Description of the Alternative Sequence	• <i>Invalid input → error</i>
Non functional requirements	<i>Data consistency; Audit logging</i>
Postconditions	<i>Records updated</i>

UC Name	<i>UC-10: Detect Scheduling Conflicts</i>
Summary	<i>Operations Staff detects conflicts.</i>
Dependency	<i>UC-05, UC-07</i>
Actors	<i>Operations Staff</i>
Preconditions	• <i>Scheduling data exists</i>
Description of the Main Sequence	• <i>Run conflict detection</i> • <i>System analyzes schedules</i> • <i>Display conflicts</i>
Description of the Alternative Sequence	• <i>No conflicts → confirmation</i>
Non functional requirements	<i>Fast processing; Accurate detection</i>
Postconditions	<i>Conflicts identified</i>

4.3 Use Cases 11-15 -> Stelina Spaho

UC Name	<i>UC-11:Monitor Surveillance Camera</i>
Summary	<i>Security staff monitors live camera feeds to detect suspicious activities and ensure airport safety.</i>
Dependency	<i>UC-01 and UC-02</i>
Actors	Primary: Security Staff Secondary: Surveillance System
Preconditions	• Security staff is logged into the system • Cameras are active and connected
Description of the Main Sequence	• Step 1: Security staff logs into the system • Step 2: User opens surveillance monitoring dashboard • Step 3: System displays live camera feeds • Step 4: Security staff selects specific camera views • Step 5: Security staff monitors activities in real time • Step 6: Suspicious activity is identified
Description of the Alternative Sequence	• Step 1: Camera feed is unavailable • Step 2: System displays error message • Step 3: Staff reports issue to maintenance
Non functional requirements	<i>Real-time video streaming High system availability (24/7) Secure access to surveillance feeds</i>

UC Name	<i>UC-12 – Control Access to Restricted Areas</i>
Summary	<i>Security staff verifies and controls access to restricted airport zones.</i>
Dependency	<i>UC-01</i>
Actors	<i>Primary Actor: Security Staff</i> <i>Secondary Actor: Staff Member / Passenger</i>
Preconditions	• <i>Security staff is logged in</i> • <i>Access control system is active</i>
Description of the Main Sequence	• <i>Step 1: Person requests access to restricted area</i> • <i>Step 2: Security staff scans ID or access badge</i> • <i>Step 3: System verifies credentials</i> • <i>Step 4: System grants or denies access</i> • <i>Step 5: Security staff allows or blocks entry</i>
Description of the Alternative Sequence	• <i>Step 1: Invalid or expired credentials detected</i> • <i>Step 2: System denies access</i> • <i>Step 3: Security staff reports incident</i>
Non functional requirements	<i>High accuracy detection</i> <i>Fast processing</i> <i>Compliance with safety regulations</i>
Postconditions	<i>Only authorized individuals gain access</i> <i>Access attempt is logged</i>

UC Name	<i>UC-13 – Perform Security Screening</i>
Summary	<i>Security staff screens passengers and staff before entering secure areas.</i>
Dependency	<i>None</i>
Actors	<i>Primary Actor: Security Staff</i> <i>Secondary Actor: Passenger / Staff</i>
Preconditions	• <i>Screening equipment is operational</i> • <i>Passenger/staff is present</i>
Description of the Main Sequence	• <i>Step 1: Passenger arrives at screening checkpoint</i> • <i>Step 2: Security staff checks identification</i> • <i>Step 3: Passenger places belongings on scanner</i> • <i>Step 4: Security staff monitors scanning process</i> • <i>Step 5: Screening results are displayed</i> • <i>Step 6: Passenger is cleared</i>
Description of the Alternative Sequence	• <i>Step 1: Suspicious item detected</i> • <i>Step 2: Security staff performs manual inspection</i> • <i>Step 3: Item is confiscated or escalated</i>
Non functional requirements	<i>Fast response time</i> <i>Secure authentication</i> <i>High reliability</i>
Postconditions	<i>Passenger is either cleared or flagged</i> <i>Screening result is recorded</i>

UC Name	<i>UC-14– Record and Report Security Incident</i>
Summary	<i>Security staff records and reports incidents for investigation and auditing.</i>
Dependency	<i>UC-11</i>
Actors	<i>Primary Actor: Security Staff</i> <i>Secondary Actor: System Administrator</i>
Preconditions	• <i>Security staff is logged in</i> • <i>Incident has occurred</i>
Description of the Main Sequence	• <i>Step 1: Security staff identifies incident</i> • <i>Step 2: Staff opens incident reporting module</i> • <i>Step 3: Staff enters incident details</i> • <i>Step 4: Staff submits report</i> • <i>Step 5: System stores report in database</i>
Description of the Alternative Sequence	• <i>Step 1: Incomplete information entered</i> • <i>Step 2: System prompts user to complete fields</i>
Non functional requirements	<i>Data integrity</i> <i>Secure storage of reports</i> <i>Audit logging</i>
Postconditions	<i>Incident is recorded in system</i> <i>Report is available for review</i>

UC Name	<i>UC-15– Respond to Security Alerts</i>
Summary	<i>Security staff receives and responds to alerts generated by the system.</i>
Dependency	<i>UC-11 and UC-12</i>
Actors	<i>Primary Actor: Security Staff</i> <i>Secondary Actor: System</i>
Preconditions	• <i>System is operational</i> • <i>Alert mechanism is active</i>
Description of the Main Sequence	• <i>Step 1: System detects abnormal activity</i> • <i>Step 2: System generates alert</i> • <i>Step 3: Security staff receives notification</i> • <i>Step 4: Staff reviews alert details</i> • <i>Step 5: Staff takes appropriate action</i> • <i>Step 6: Staff marks alert as resolved</i>
Description of the Alternative Sequence	• <i>Step 1: False alert detected</i> • <i>Step 2: Staff dismisses alert</i>
Non functional requirements	<i>Real-time alert delivery</i> <i>High reliability</i> <i>Secure communication</i>
Postconditions	<i>Alert is handled and resolved</i> <i>Action is logged in the system</i>

4.4 Use Cases 16-20 -> Yeldos Rakhymzhanov

UC Name	<i>UC-16: Review Security Logs and Reports</i>
Summary	<i>Security staff review security logs and reports to monitor risks and improve safety procedures.</i>
Dependency	<i>UC-03, UC-14, UC-15</i>
Actors	<i>Primary: Security Staff</i>
Preconditions	• <i>Security staff is authenticated</i> • <i>Security logs exist in the system</i>
Description of the Main Sequence	• <i>Step 1: Security staff logs into the system</i> • <i>Step 2: Navigates to security logs and reports module</i> • <i>Step 3: Applies filters (date, type, location)</i> • <i>Step 4: Views logs and reports</i> • <i>Step 5: Optionally exports reports</i>
Description of the Alternative Sequence	• <i>Step 1: No filters applied → system shows recent logs</i> • <i>Step 2: No data found → system displays message</i>
Non functional requirements	• <i>Data confidentiality and integrity</i> • <i>Fast query response time</i> • <i>Secure access control</i>
Postconditions	<i>Security logs and reports are reviewed</i>

UC Name	<i>UC-17: Record Maintenance Activities</i>
Summary	<i>Maintenance staff record completed maintenance activities for equipment.</i>
Dependency	<i>UC-21</i>
Actors	<i>Primary: Maintenance Staff</i>
Preconditions	• <i>Maintenance staff is authenticated</i> • <i>Equipment exists in the system</i>
Description of the Main Sequence	• <i>Step 1: Maintenance staff logs into the system</i> • <i>Step 2: Navigates to maintenance logging module</i> • <i>Step 3: Selects equipment</i> • <i>Step 4: Enters maintenance details</i> • <i>Step 5: Submits the record</i>
Description of the Alternative Sequence	• <i>Step 1: Equipment not found → system prompts to add equipment</i> • <i>Step 2: Missing required fields → system shows error</i>
Non functional requirements	• <i>Data accuracy</i> • <i>System reliability</i> • <i>Quick data entry response</i>
Postconditions	<i>Maintenance activity is recorded</i>

UC Name	<i>UC-18: Schedule Preventive Maintenance</i>
Summary	<i>Maintenance staff schedule preventive maintenance tasks to avoid equipment failure.</i>
Dependency	<i>UC-21</i>
Actors	<i>Primary: Maintenance Staff</i>
Preconditions	• <i>Maintenance staff is authenticated</i> • <i>Equipment list is available</i>
Description of the Main Sequence	• <i>Step 1: Maintenance staff logs into the system</i> • <i>Step 2: Navigates to preventive maintenance module</i> • <i>Step 3: Selects equipment</i> • <i>Step 4: Defines schedule (date, frequency)</i> • <i>Step 5: Saves schedule</i>
Description of the Alternative Sequence	• <i>Step 1: Invalid date entered → system shows error</i> • <i>Step 2: Equipment not available → system prompts registration</i>
Non functional requirements	• <i>Scheduling accuracy</i> • <i>System availability</i> • <i>Notification support</i>
Postconditions	<i>Preventive maintenance schedule is created</i>

UC Name	<i>UC-19: Report Infrastructure Issues</i>
Summary	<i>Maintenance staff report infrastructure or equipment issues.</i>
Dependency	<i>None</i>
Actors	<i>Primary: Maintenance Staff</i>
Preconditions	• <i>Maintenance staff is authenticated</i>
Description of the Main Sequence	• <i>Step 1: Maintenance staff logs into the system</i> • <i>Step 2: Navigates to issue reporting module</i> • <i>Step 3: Enters issue details</i> • <i>Step 4: Attaches supporting files (optional)</i> • <i>Step 5: Submits the report</i>
Description of the Alternative Sequence	• <i>Step 1: Missing required information → system shows error</i> • <i>Step 2: File upload fails → system prompts retry</i>
Non functional requirements	• <i>Data integrity</i> • <i>File upload reliability</i> • <i>System usability</i>
Postconditions	<i>Issue report is submitted and stored</i>

UC Name	<i>UC-20: Track Maintenance Requests</i>
Summary	<i>Maintenance staff track and monitor maintenance request status.</i>
Dependency	<i>UC-19, UC-17</i>
Actors	<i>Primary: Maintenance Staff</i>
Preconditions	• <i>Maintenance staff is authenticated</i> • <i>Maintenance requests exist</i>
Description of the Main Sequence	• <i>Step 1: Maintenance staff logs into the system</i> • <i>Step 2: Navigates to maintenance tracking module</i> • <i>Step 3: Views list of requests</i> • <i>Step 4: Selects a request</i> • <i>Step 5: Views or updates status</i>
Description of the Alternative Sequence	• <i>Step 1: No requests available → system shows empty list</i> • <i>Step 2: Request not found → system shows error</i>
Non functional requirements	• <i>Real-time updates</i> • <i>System performance</i> • <i>Data consistency</i>
Postconditions	<i>Maintenance request status is viewed or updated</i>

4.5 Use Cases 21-25 -> Fatih Gökteke

UC Name	UC-21 — Access Asset Information
Summary	<i>Maintenance Staff accesses and reviews information about airport assets and equipment stored in the system. The user can search for an asset, view its details, and check its current status and maintenance history. This helps maintenance operations be managed more efficiently and accurately.</i>
Dependency	<i>None</i>
Actors	<i>Primary: Maintenance Staff</i>
Preconditions	● <i>Maintenance Staff is authenticated. Asset records exist in the system.</i>
Description of the Main Sequence	● <i>Step 1: Maintenance Staff opens the Asset Module.</i> ● <i>Step 2: System retrieves the asset list.</i> ● <i>Step 3: Maintenance Staff searches or selects an asset.</i> ● <i>Step 4: System displays asset details.</i> ● <i>Step 5: Maintenance Staff reviews asset information.</i>
Description of the Alternative Sequence	● <i>Step 1: Asset not found.</i> ● <i>Step 2: System displays “Asset not found” error message.</i>
Non functional requirements	● <i>Fast access</i> ● <i>Data accuracy</i>
Postconditions	<i>Asset information displayed successfully.</i>

UC Name	<i>UC-22 — Update Maintenance Report</i>
Summary	<i>Maintenance Staff updates maintenance reports related to repair and maintenance activities. The system allows staff to open a maintenance request, modify report details, and save the updated information.</i>
Dependency	<i>UC-20, UC-21</i>
Actors	<i>Primary: Maintenance Staff</i>
Preconditions	• <i>Maintenance Staff is authenticated. Maintenance requests and reports exist in the system.</i>
Description of the Main Sequence	• <i>Step 1: Maintenance Staff opens the Maintenance Report Module.</i> • <i>Step 2: System retrieves maintenance request list.</i> • <i>Step 3: Maintenance Staff selects a maintenance request.</i> • <i>Step 4: System opens the related maintenance report.</i> • <i>Step 5: Maintenance Staff enters or modifies report details.</i> • <i>Step 6: System validates the report data.</i> • <i>Step 7: System saves the updated maintenance report.</i>
Description of the Alternative Sequence	• <i>Step 1: Maintenance request not found.</i> • <i>Step 2: System shows “Request not found” message.</i> • <i>Step 3: Missing or invalid report data entered.</i> • <i>Step 4: System shows validation error.</i>
Non functional requirements	• <i>Reliable saving</i> • <i>Data consistency</i>
Postconditions	<i>Report Updated</i>

UC Name	UC-23 — Manage Arrival Schedule
Summary	<i>Ground Handling Staff views and manages the arrival schedule of aircraft in the airport system. The system displays arrival flight details including assigned gates, parking stands, and estimated arrival times. This helps ground handling operations prepare required services before aircraft arrival.</i>
Dependency	<i>UC-05,UC-06,UC-07</i>
Actors	Primary: <i>Ground Handling Staff</i>
Preconditions	● <i>Ground Handling Staff is authenticated. Flight schedules, gate assignments, and stand assignments exist in the system.</i>
Description of the Main Sequence	● <i>Step 1: Ground Handling Staff opens the Arrival Schedule Module.</i> ● <i>Step 2: System retrieves scheduled arrival flights.</i> ● <i>Step 3: System displays flight details, assigned gates, parking stands, and estimated arrival times.</i> ● <i>Step 4: Ground Handling Staff selects a flight.</i> ● <i>Step 5: Ground Handling Staff reviews or updates arrival schedule information.</i> ● <i>Step 6: System saves updated schedule information.</i>
Description of the Alternative Sequence	● <i>Step 1: No scheduled arrival flights found.</i> ● <i>Step 2: System displays an empty schedule message.</i> ● <i>Step 3: Selected flight not found.</i> ● <i>Step 4: System displays “Flight not found” error message.</i>
Non functional requirements	● <i>Real-time updates</i> ● <i>Conflict prevention</i>
Postconditions	<i>Arrival schedule information is displayed or updated successfully.</i>

UC Name	UC-24 — Manage Baggage Operations
Summary	<i>Ground Handling Staff manages baggage loading and unloading operations for selected flights. The system allows staff to view flight information, update baggage status, and confirm baggage operation progress. This helps ensure baggage handling is tracked accurately and completed on time.</i>
Dependency	UC-23
Actors	Primary: <i>Ground Handling Staff</i>
Preconditions	• <i>Ground Handling Staff is authenticated. Flight schedule and baggage status records exist in the system.</i>
Description of the Main Sequence	• <i>Step 1: Ground Handling Staff opens the Baggage Operations Module.</i> • <i>Step 2: System displays the flight list.</i> • <i>Step 3: Ground Handling Staff selects a flight.</i> • <i>Step 4: System displays baggage loading/unloading details.</i> • <i>Step 5: Ground Handling Staff updates baggage status.</i> • <i>Step 6: System validates the updated baggage information.</i> • <i>Step 7: Ground Handling Staff confirms the update.</i> • <i>Step 8: System saves the baggage status.</i>
Description of the Alternative Sequence	• <i>Step 1: No baggage record is found for the selected flight.</i> • <i>Step 2: System displays an error message.</i> • <i>Step 3: Invalid baggage status is entered.</i> • <i>Step 4: System shows validation error and asks staff to correct the data.</i>
Non functional	• <i>Real-time tracking</i>

requirements	• <i>Accuracy</i>
Postconditions	<i>Baggage status updated</i>

UC Name	<i>UC-25 — Manage Apron Services</i>
Summary	• <i>Ground Handling Staff assigns and monitors apron service tasks for selected flights. These tasks include refueling coordination, catering delivery, and ground equipment allocation. The system helps staff check resource availability and track task progress during ground operations.</i>
Dependency	<i>UC-23,UC-24</i>
Actors	<i>Primary:</i> <i>Ground Handling Staff</i>
Preconditions	• <i>Ground Handling Staff is authenticated. Flight schedule and resource records exist in the system.</i>
Description of the Main Sequence	• <i>Step 1: Ground Handling Staff opens the Apron Services Module.</i> • <i>Step 2: System displays available flights.</i> • <i>Step 3: Ground Handling Staff selects a flight.</i> • <i>Step 4: System displays apron service tasks for the selected flight.</i> • <i>Step 5: Ground Handling Staff assigns required resources.</i> • <i>Step 6: System checks resource availability.</i> • <i>Step 7: Ground Handling Staff monitors task progress.</i> • <i>Step 8: System saves task status updates.</i>
Description of the Alternative Sequence	• <i>Step 1: Selected resource is unavailable.</i> • <i>Step 2: System displays “Resource unavailable” message.</i>

	• <i>Step 3: Ground Handling Staff chooses another resource.</i>
Non functional requirements	• <i>Real-time monitoring</i> • <i>Resource management</i>
Postconditions	<i>Apron service tasks are assigned and tracked successfully.</i>

4.6 Use Cases 26-30 -> Leart Mukaj

UC Name	<i>UC-26: Baggage Loading</i>
Summary	<i>Ground handling staff tracks, updates and manages baggage loading</i>
Dependency	<i>None</i>
Actors	<i>Primary: Ground Staff</i>
Preconditions	• <i>Plane is on ground</i> • <i>Staff member is logged in</i>
Description of the Main Sequence	• <i>Step 1: Staff is logged in</i> • <i>Step 2: Staff receives luggage from check-in and marks as loading</i> • <i>Step 3: After luggage is moved with luggage car, staff members scan each tag as its being put on board to mark as loaded</i>
Description of the Alternative Sequence	• <i>Luggage does not belong to flight → System alerts staff to relocate luggage</i> • <i>Luggage fails scan → System sends error report and asks for re-scan</i>
Non functional requirements	<i>Errors handled quickly</i> <i>Data updated in real time</i>
Postconditions	<i>Luggage is successfully loaded into cargo and marked as such in system</i>

UC Name	<i>UC-27: Final confirmation</i>
Summary	<i>Ground handling staff shall reconfirm that all processes have been fully completed to avoid</i>
Dependency	<i>UC-23, 24, 25, 26</i>
Actors	<i>Primary: Ground Staff</i>
Preconditions	• <i>Apron services, turnaround process and luggage handling has been completed</i> • <i>Staff is logged in</i>
Description of the Main Sequence	• <i>Step 1: Staff re-confirms completion of turnaround process</i> • <i>Step 2: Staff re-confirms completion of apron services</i> • <i>Step 3: Staff re-confirms completion of luggage handling</i>
Description of the Alternative Sequence	• <i>Failure to re-confirm marks plane as unfit for departure</i> • <i>Issues found during re-confirmation are addressed</i>
Non functional requirements	<i>Re-confirmation made easy to do to prevent avoidance</i>
Postconditions	<i>Plane is fully cleared for departure</i>

UC Name	<i>UC-28: Passenger Check-in</i>
Summary	<i>Staff confirms check-in, identification and booking</i>
Dependency	<i>None</i>
Actors	<i>Primary:</i> <i>Airline staff</i>
Preconditions	• <i>Staff member is logged in</i>
Description of the Main Sequence	• <i>Step 1: Staff is logged in</i> • <i>Step 2: Staff scans ticket</i> • <i>Step 3: Staff scans id</i> • <i>Step 4: System outputs if details match</i> • <i>Step 5: System shows booking details</i>
Description of the Alternative Sequence	• <i>Failure of information matching</i>
Non functional requirements	<i>Data updated in real time</i>
Postconditions	<i>None</i>

UC Name	<i>UC-29: Boarding pass</i>
Summary	<i>Staff generates and issues boarding passes</i>
Dependency	<i>None</i>
Actors	<i>Primary: Airline Staff</i>
Preconditions	• <i>Staff member is logged in</i> • <i>Passenger is at check-in</i>
Description of the Main Sequence	• <i>Step 1: Staff receives id and booking information</i> • <i>Step 2: Staff request printing of boarding pass</i> • <i>Step 3: Staff hands pass to passenger</i>
Description of the Alternative Sequence	
Non functional requirements	<i>Re-confirmation made easy to do to prevent avoidance</i>
Postconditions	<i>None</i>

UC Name	<i>UC-30: Seat handling</i>
Summary	<i>Airline staff shall assign, modify and reallocate seats for a flight</i>
Dependency	<i>None</i>
Actors	<i>Primary: Airline Staff</i>
Preconditions	• <i>Staff logged in</i>
Description of the Main Sequence	• <i>Step 1: Staff sees request from passenger for seat relocation</i> • <i>Step 2: Staff checks seat availability in system</i> • <i>Step 3: Staff accepts or denies relocation</i>
Description of the Alternative Sequence	• <i>Staff receives last minute seat assignment, checks empty seats and assigns</i>
Non functional requirements	<i>Re-confirmation made easy to do to prevent avoidance</i>
Postconditions	<i>None</i>

4.6 Use Cases 31-35 -> Erisa Meta



UC Name	Assign / Modify Passenger Seats
Summary	Airline staff shall assign, modify, or reallocate passenger seats for a flight.
Dependency	Flight schedule must exist; Passenger bookings must be present.
Actors	Airline Staff
Preconditions	Flight and passenger bookings exist.
Description of the Main Sequence:	• <u>Step 1</u>: Staff selects flight and views passenger list. • <u>Step 2</u>: Staff assigns, modifies, or reallocates seats. • <u>Step 3</u>: System updates seat assignments.
Description of the Alternative Sequence	• Step 1: Seat already occupied → system notifies staff and prompts for a different seat.
Non-functional Requirements	• <u>System</u> must process seat assignments within 2 seconds per passenger. • Only authorized staff can modify seats.



UC Name	UC-31 Validate Passenger Ticket
Summary	Airline staff shall validate passenger tickets and confirm eligibility for boarding.
Dependency	Valid ticket database; Boarding eligibility rules.
Actors	Airline Staff
Preconditions	Passenger holds a valid ticket.
Description of the Main Sequence:	• <u>Step 1</u>: Staff scans or inputs ticket information. • <u>Step 2</u>: System verifies ticket validity. • <u>Step 3</u>: Staff approves or rejects boarding.
Description of the Alternative Sequence	Step 1: Ticket invalid or expired → boarding denied; system alerts staff.
Non-functional Requirements	• <u>Verification</u> must occur in under 1 second. • System logs all validation attempts for audit purposes.





UC Name	UC-32 Manage Passenger Manifest
Summary	Airline staff shall create, update, and view passenger manifests for their airline's flights.
Dependency	Flight schedule and passenger booking data must exist.
Actors	Airline Staff
Preconditions	Flight exists.
Description of the Main Sequence:	• <u>Step 1</u>: Select flight. • <u>Step 2</u>: Add, update, or remove passengers. • <u>Step 3</u>: System saves and displays the manifest.
Description of the Alternative Sequence	Step 1: Missing or inconsistent data → system alerts staff and prevents saving.
Non-functional Requirements	• <u>Manifest</u> updates must be saved immediately. • Only authorized staff can modify manifests.



UC Name	UC-33 Manage Boarding Process
Summary	Airline staff shall manage the boarding process, including scanning boarding passes and marking passengers as boarded.
Dependency	Valid tickets; Boarding procedures must be defined.
Actors	Airline Staff
Preconditions	Boarding started; passengers have valid tickets.
Description of the Main Sequence:	• <u>Step 1</u>: Scan boarding passes. • <u>Step 2</u>: Verify passenger eligibility. • <u>Step 3</u>: Mark passengers as boarded.
Description of the Alternative Sequence	Step 1: Invalid or duplicate pass → system alerts staff; passenger denied boarding.
Non-functional Requirements	• <u>Boarding</u> process must handle at least 100 passengers per 5 minutes. • System logs all boarding events.





UC Name	UC-34 Generate Invoice
Summary	Finance staff shall generate invoices for landing fees, parking fees, and passenger services.
Dependency	Recorded airport services; Billing rules must be defined.
Actors	Finance Staff
Preconditions	Services recorded for billing.
Description of the Main Sequence:	• <u>Step 1</u>: Select services to invoice. • <u>Step 2</u>: Calculate fees. • <u>Step 3</u>: Generate invoice.
Description of the Alternative Sequence	Missing or incomplete service data → system alerts staff; invoice not generated.
Non-functional Requirements	• <u>Invoice</u> generation must complete in under 2 seconds. • Only authorized finance staff can generate invoices. 



UC Name	UC-35 <u>Generate Financial Reports</u>
Summary	Finance staff shall generate revenue, expense, and profit reports.
Dependency	Complete financial data; Reporting criteria must be available.
Actors	Finance Staff
Preconditions	Financial data available and up-to-date.
Description of the Main Sequence:	• <u>Step 1</u>: Select report type and period. • <u>Step 2</u>: System compiles data and generates report. • <u>Step 3</u>: Staff reviews and exports report.
Description of the Alternative Sequence	• Incomplete or inconsistent data → system alerts staff; report not generated.
Non-functional Requirements	• <u>Reports</u> must be generated in under 5 seconds. • System ensures data integrity and confidentiality. 

4.7 Use Cases 36-38 -> Moira Lako & Stelina Spaho

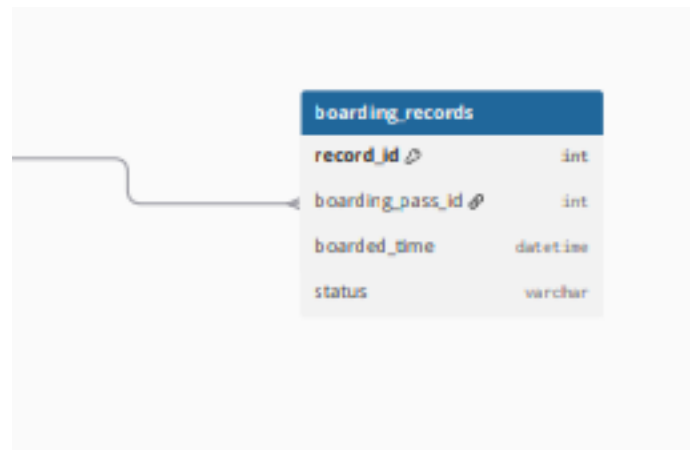
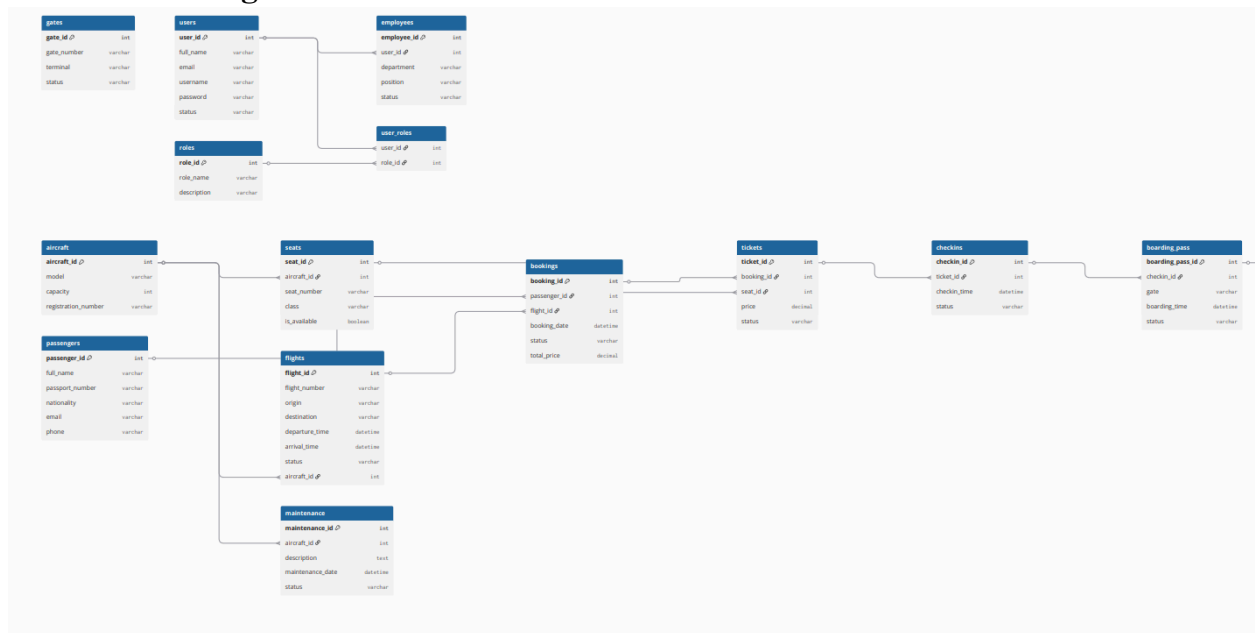
UC Name	<i>UC-36: Manage Employee Records</i>
Summary	<i>HR Staff manages employee information, employment status, department assignments, roles, certifications, and contact details to maintain an accurate workforce database.</i>
Dependency	<i>UC-01</i>
Actors	Primary: HR Staff
Preconditions	• <i>HR Staff is authenticated</i> • <i>Employee database is available</i>
Description of the Main Sequence	• <i>Step 1: HR Staff opens Employee Management Module</i> • <i>Step 2: System displays employee records and search options</i> • <i>Step 3: HR Staff creates a new employee record or selects an existing employee</i> • <i>Step 4: HR Staff enters or updates employee information</i> • <i>Step 5: System validates mandatory fields and duplicate records</i> • <i>Step 6: HR Staff reviews and confirms changes</i> • <i>Step 7: System updates the employee database and records the modification in the audit log</i>
Description of the Alternative Sequence	• <i>Step 1: Duplicate employee record detected → System displays warning and prevents creation</i> • <i>Step 2: Mandatory information missing → System highlights required fields</i> • <i>Step3: Invalid role assignment → System rejects update</i>
Non functional requirements	<i>Secure access control Data consistency and integrity Audit trail for record modification</i>
Postconditions	<i>Employee record is created, updated, or archived successfully</i>

UC Name	<i>UC-37:Manage Employee Schedules and Shifts</i>
Summary	<i>HR staff manages employee schedules and shifts to ensure continuous airport operations.</i>
Dependency	<i>UC-01 and UC-02</i>
Actors	Primary: HR Staff Secondary: Scheduling System
Preconditions	• <i>HR staff is logged into the system</i> • <i>Employee records exist in the system</i>
Description of the Main Sequence	• <i>Step 1: HR staff logs into the system</i> • <i>Step 2: User opens scheduling module</i> • <i>Step 3: System displays current schedules and shifts</i> • <i>Step 4: HR staff selects an employee or group of employees</i> • <i>Step 5: HR staff assigns or updates shifts and schedules</i> • <i>Step 6: System checks for conflicts or overlaps</i> • <i>Step 7: System saves the schedule</i>
Description of the Alternative Sequence	• <i>Step 1: Schedule conflict is detected</i> • <i>Step 2: System displays conflict warning</i> • <i>Step 3: HR staff modifies the schedule</i> • <i>Step 4: System revalidates and saves</i>
Non functional requirements	<i>Real-time schedule updates</i> <i>High system availability</i> <i>Conflict detection and resolution</i>
Postconditions	<i>Updated employee schedules are stored and available for operational planning</i>

UC Name	<i>UC-38: Manage Airport Assets</i>
Summary	<i>Asset management staff registers and manages airport assets such as vehicles, equipment, and IT systems to ensure centralized tracking.</i>
Dependency	<i>UC-01 and UC-02</i>
Actors	Primary: Asset Management Staff Secondary: Asset Management System
Preconditions	• <i>Asset management staff is logged into the system</i> • <i>Asset database is available</i>
Description of the Main Sequence	• <i>Step 1: Asset management staff logs into the system</i> • <i>Step 2: User navigates to asset management module</i> • <i>Step 3: System displays list of registered assets</i> • <i>Step 4: Staff adds a new asset or selects an existing one</i> • <i>Step 5: Staff enters or updates asset details (type, status, location)</i> • <i>Step 6: System validates the information</i> • <i>Step 7: System saves and updates the asset record</i>
Description of the Alternative Sequence	• <i>Step 1: Duplicate or invalid asset data is entered</i> • <i>Step 2: System displays error message</i> • <i>Step 3: Staff corrects the data and resubmits</i>
Non functional requirements	<i>Accurate asset tracking</i> <i>Secure access and role-based permissions</i> <i>Scalable asset database</i>
Postconditions	<i>Asset inventory is updated and available for maintenance, operational, and management purposes</i>

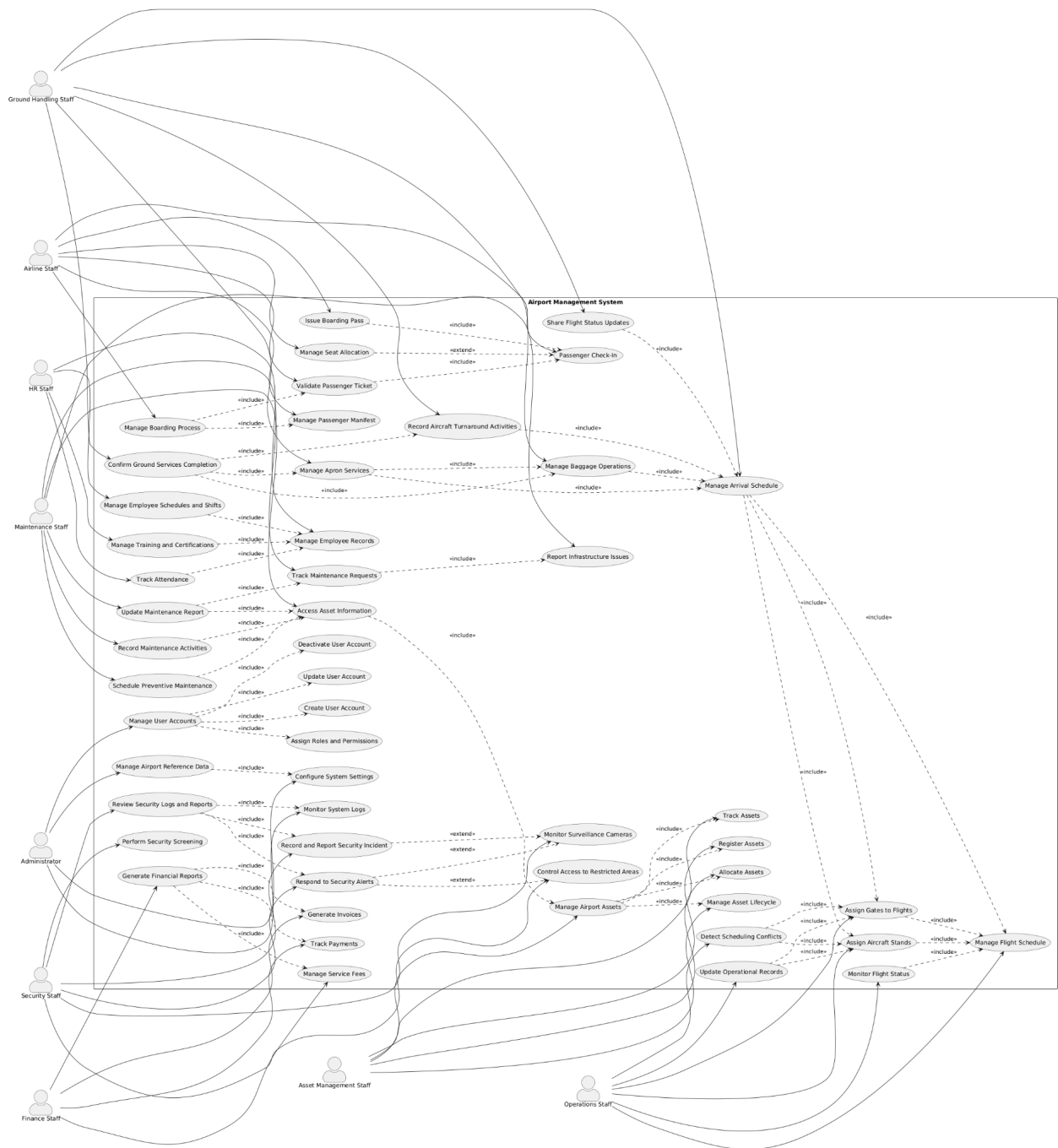
5. Diagrams

5.1 ER Diagram



By: Erisa Meta

5.2 Use Case Diagram

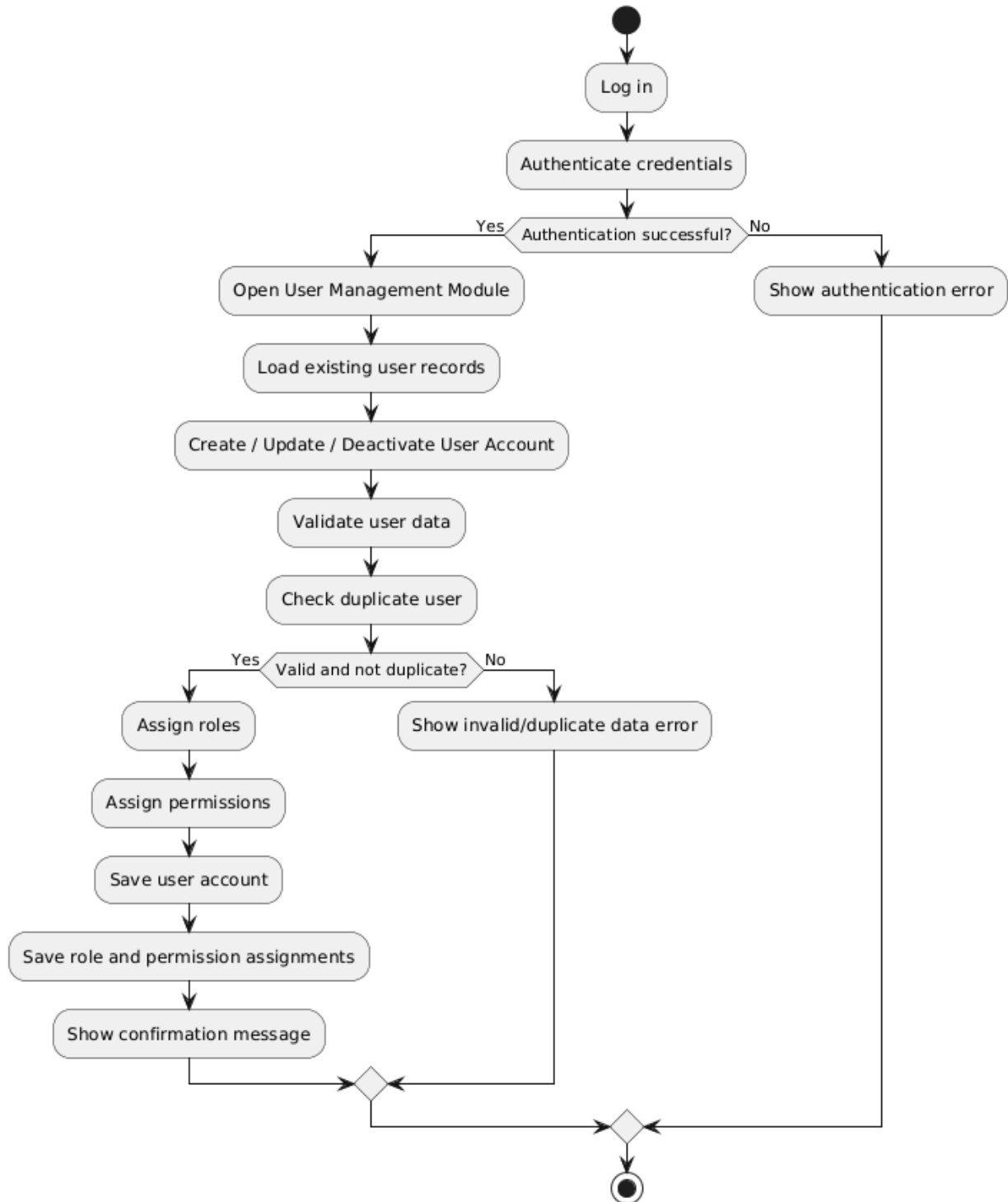


By: Moira Lako

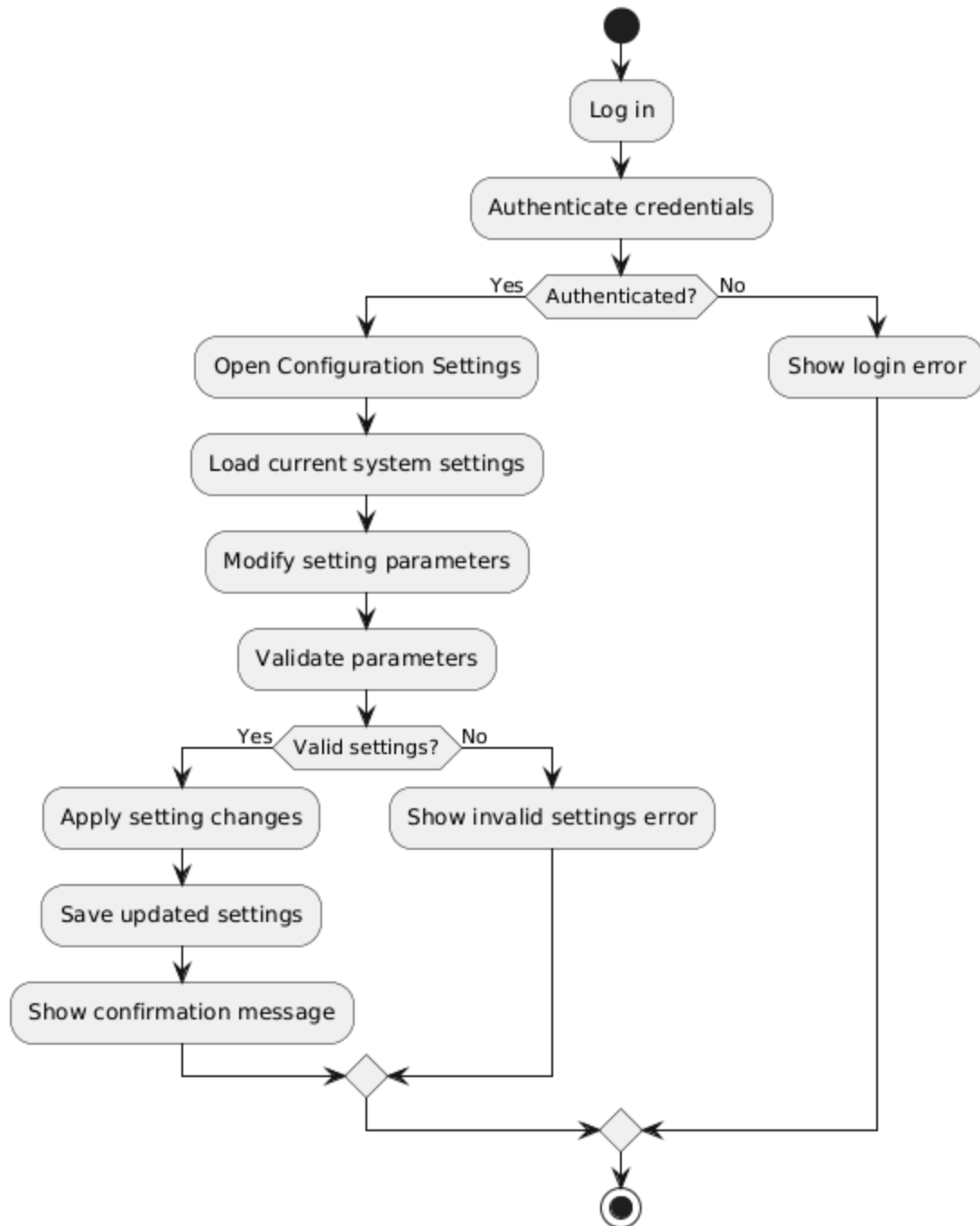
5.3 Activity Diagrams

5.3.1 Activity Diagrams UC 01-05 -> Moira Lako

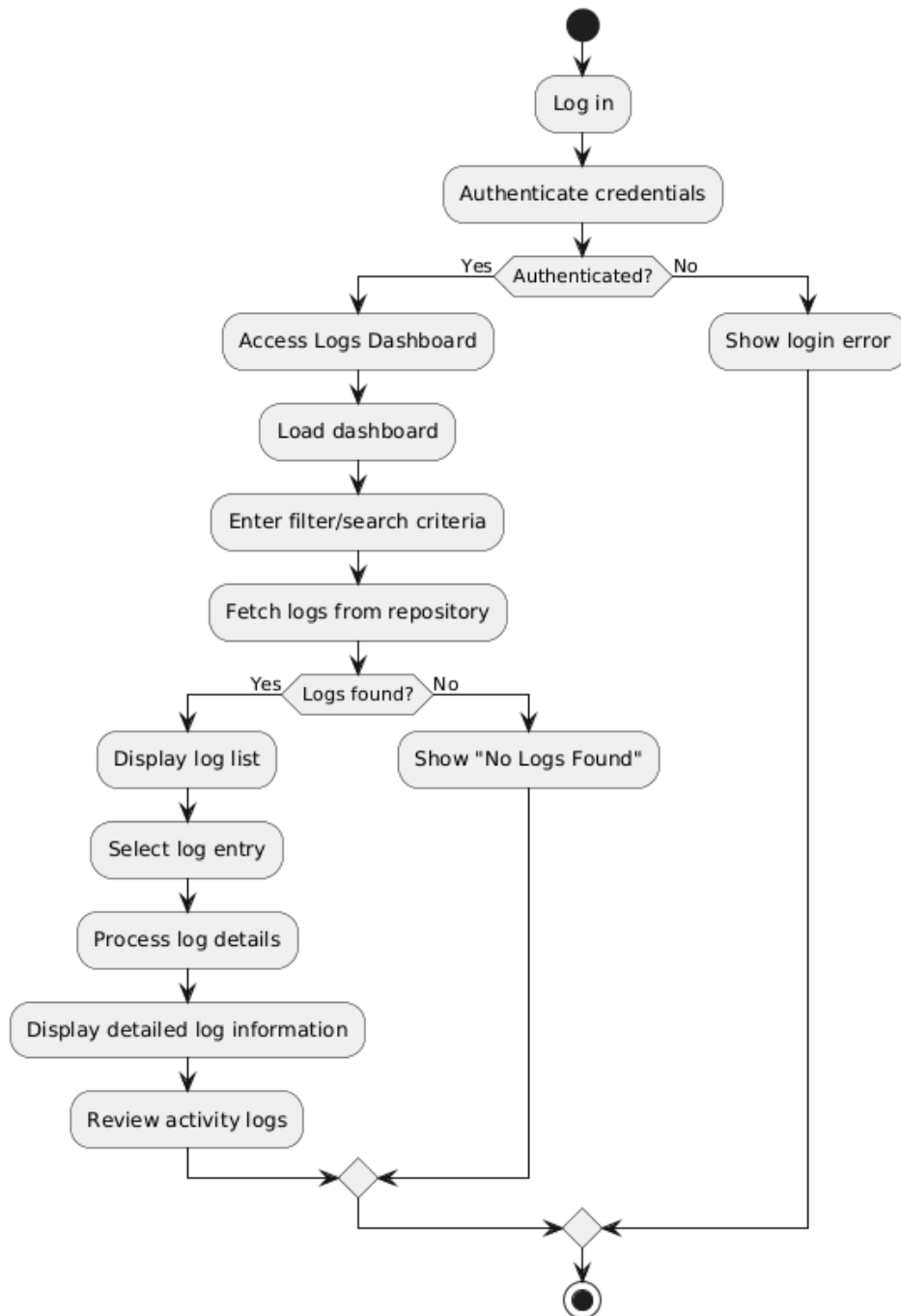
UC-01: Manage User Accounts



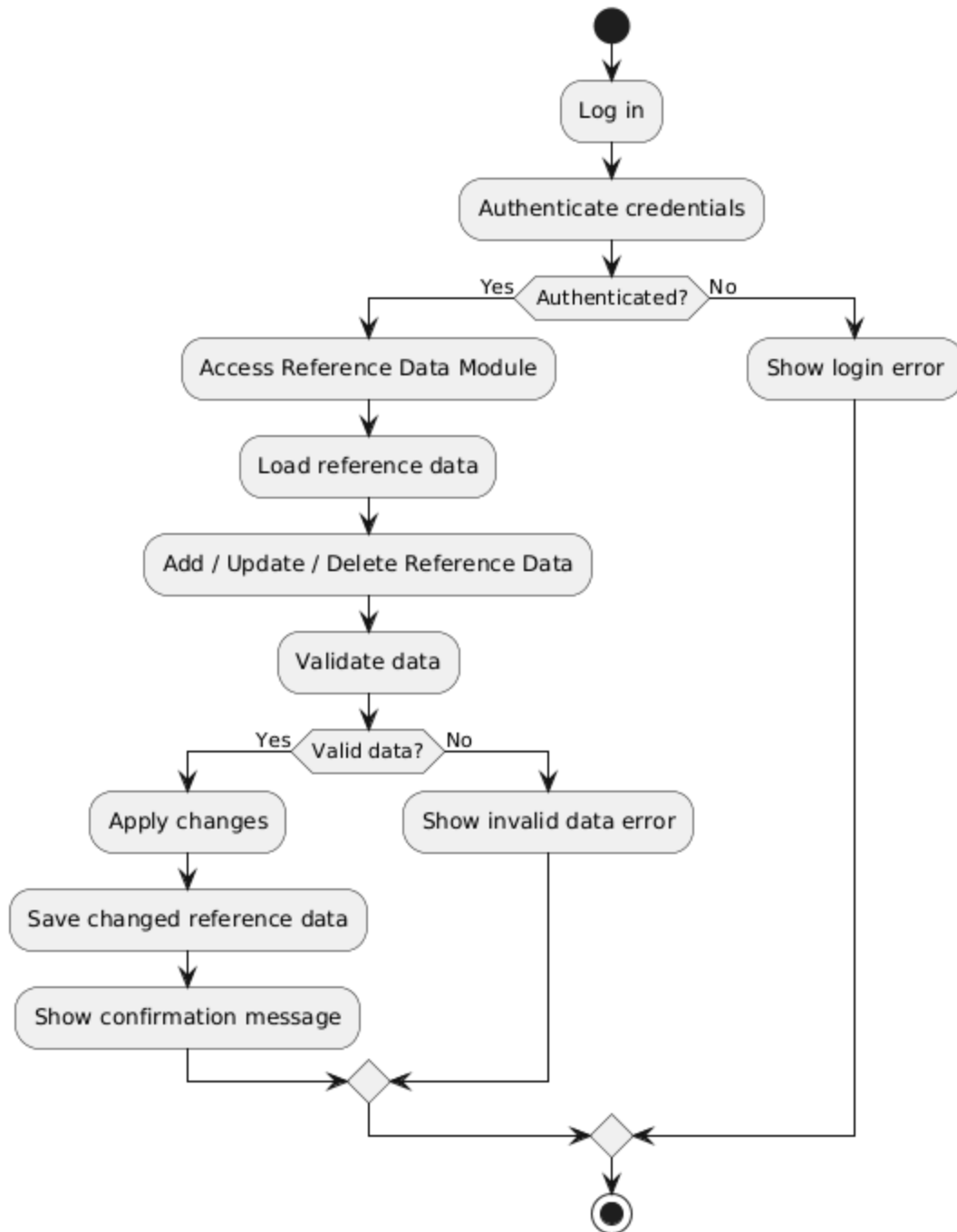
UC-02: Configure System Settings



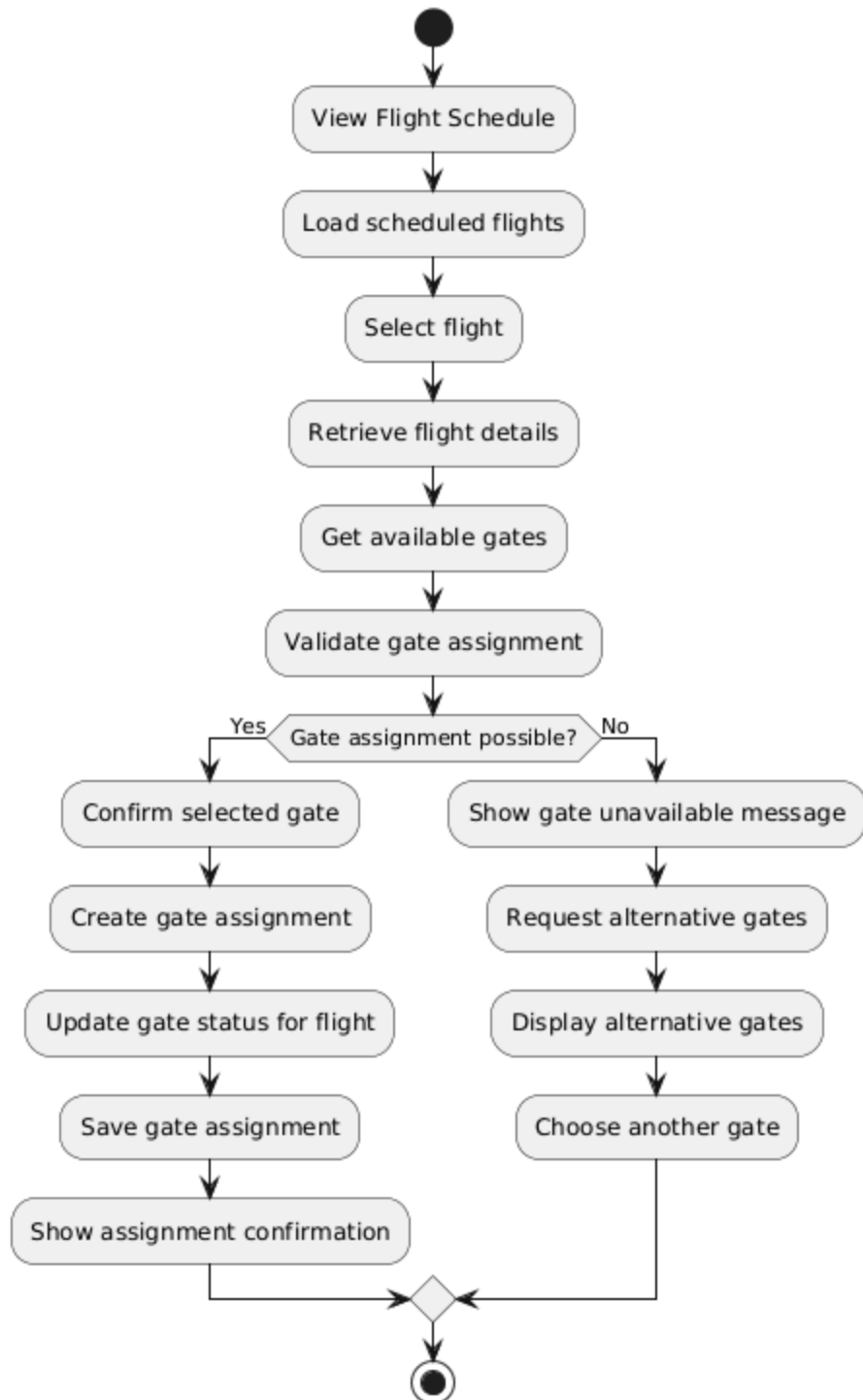
UC-03: Monitor System Logs



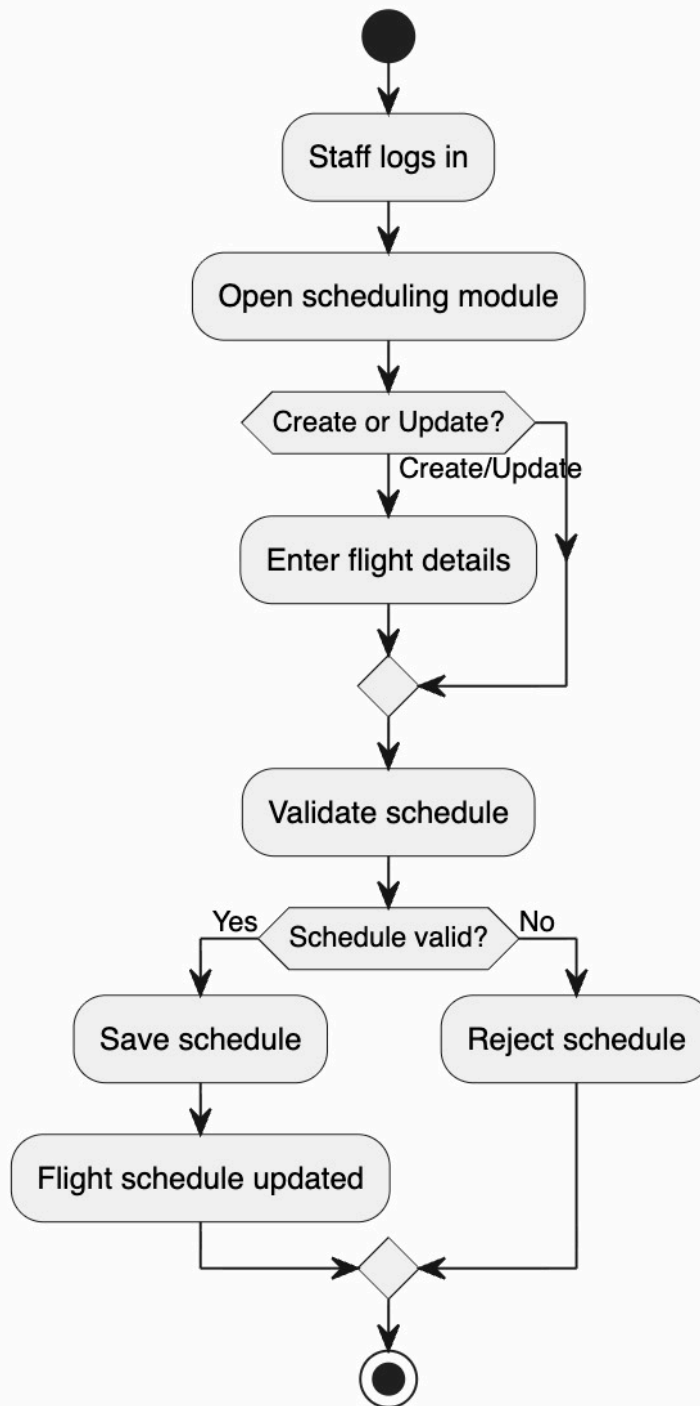
UC-04: Manage Airport Reference Data



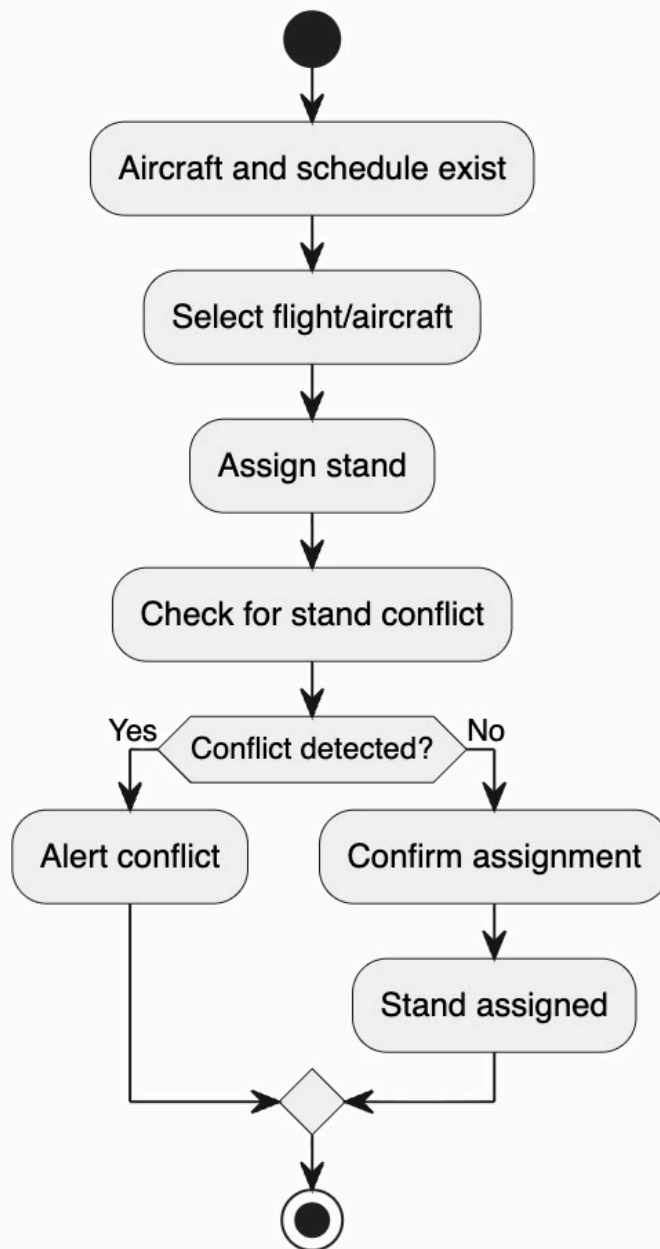
UC-05: Assign Gates to Flights



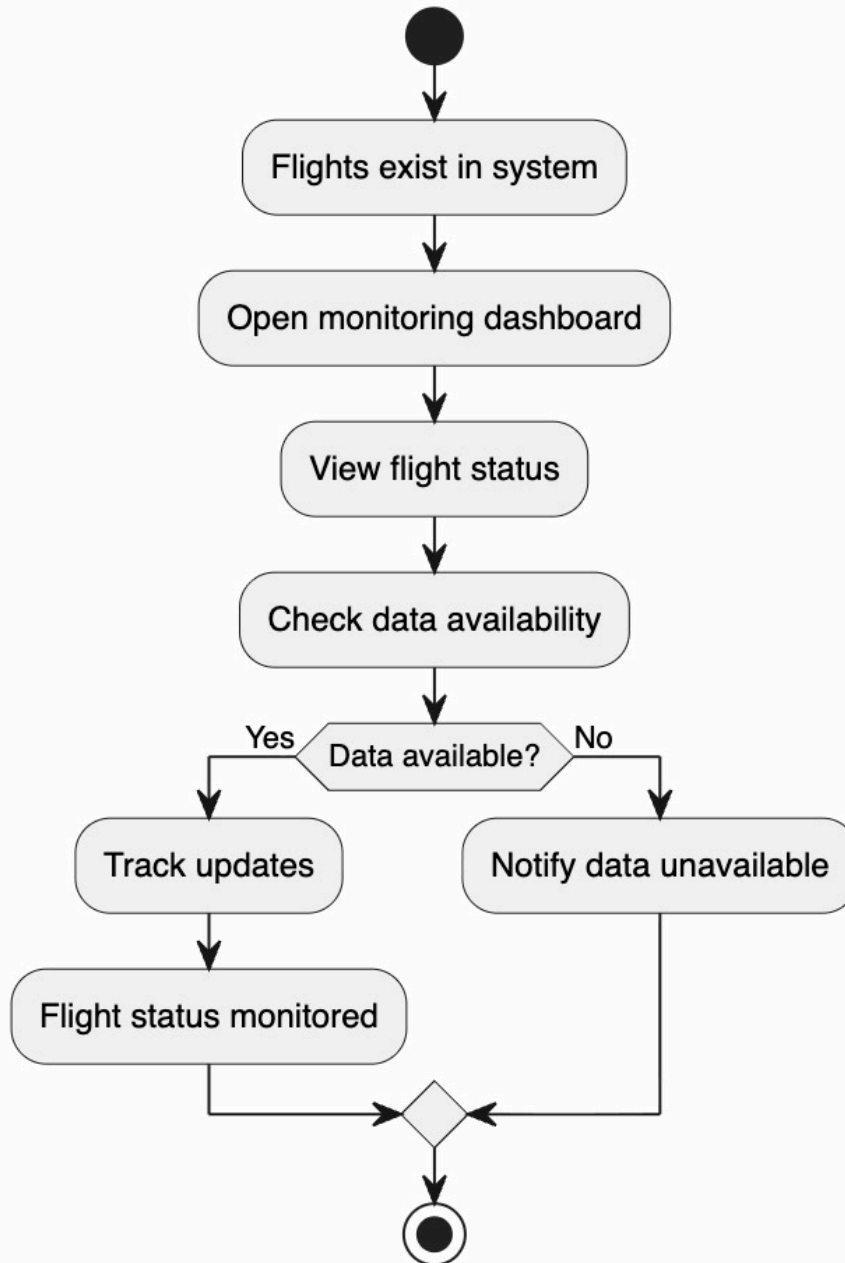
UC-06: Manage Flight Schedule - Activity Diagram



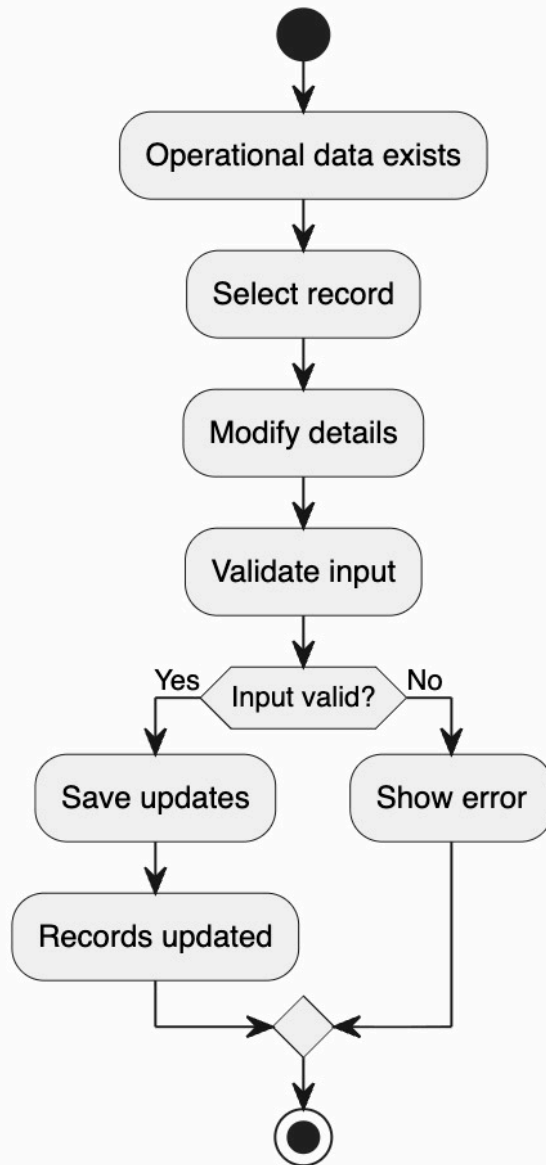
UC-07: Assign Aircraft Stands - Activity Diagram



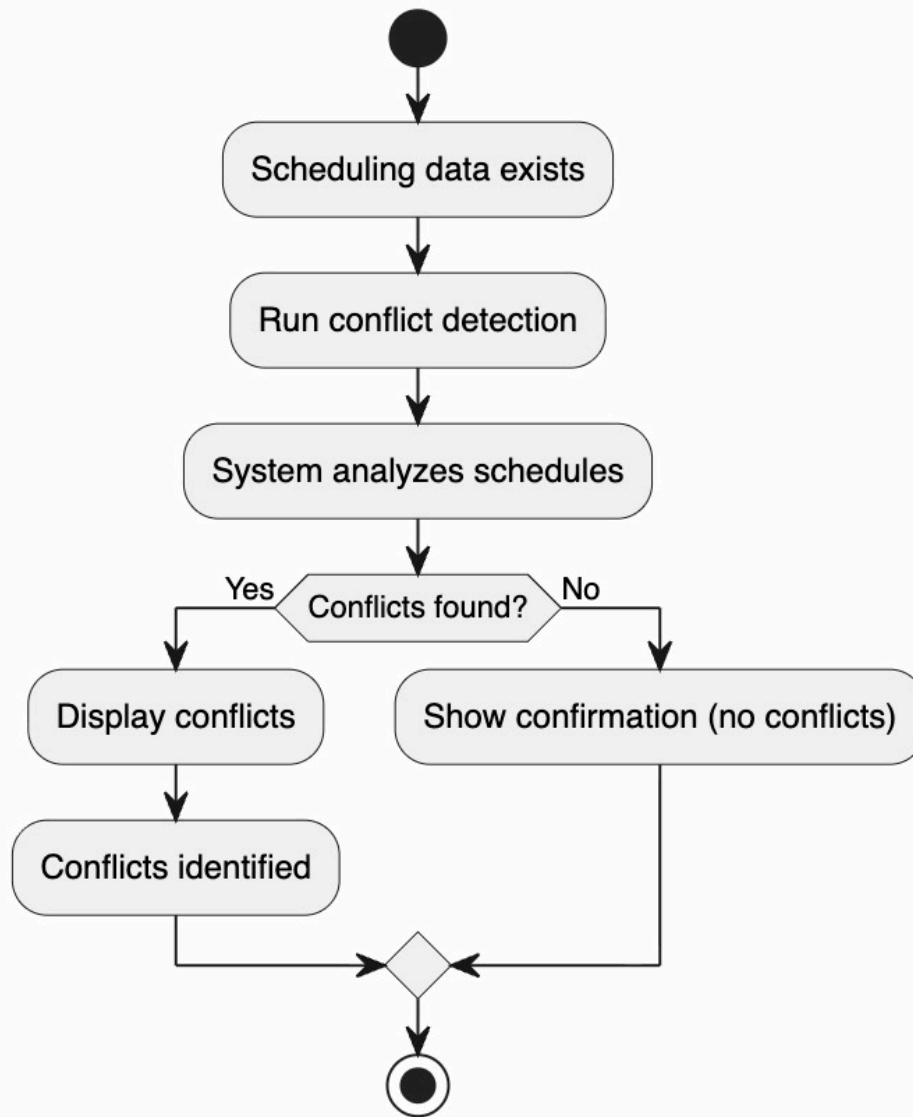
UC-08: Monitor Flight Status - Activity Diagram



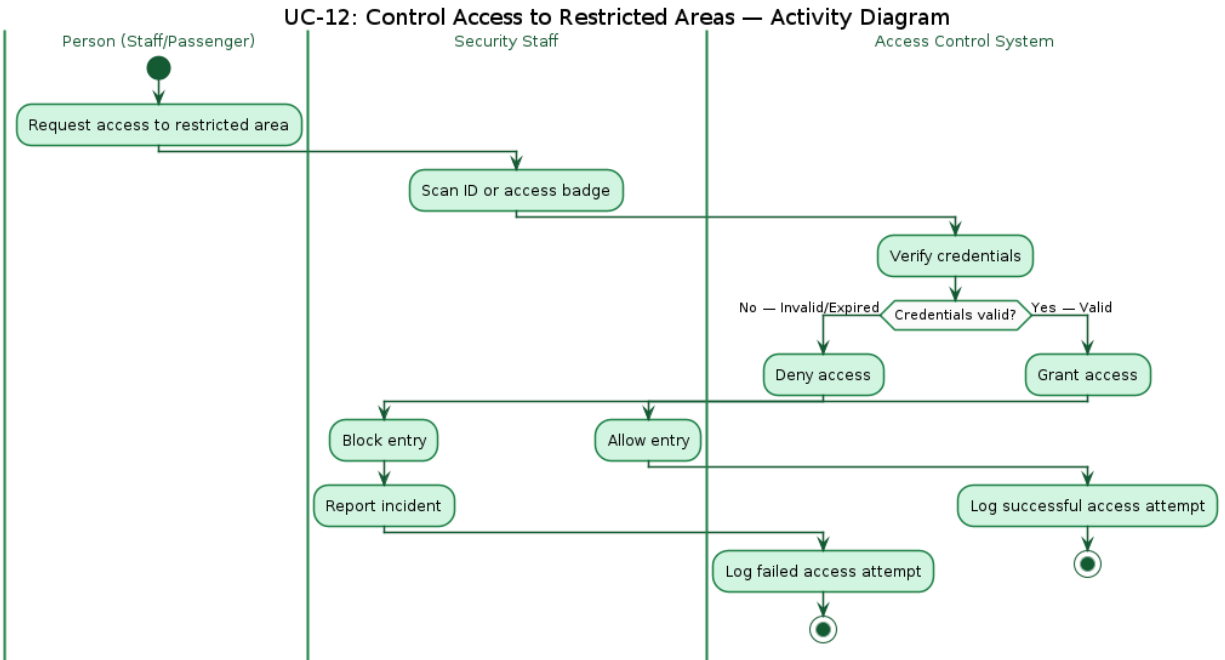
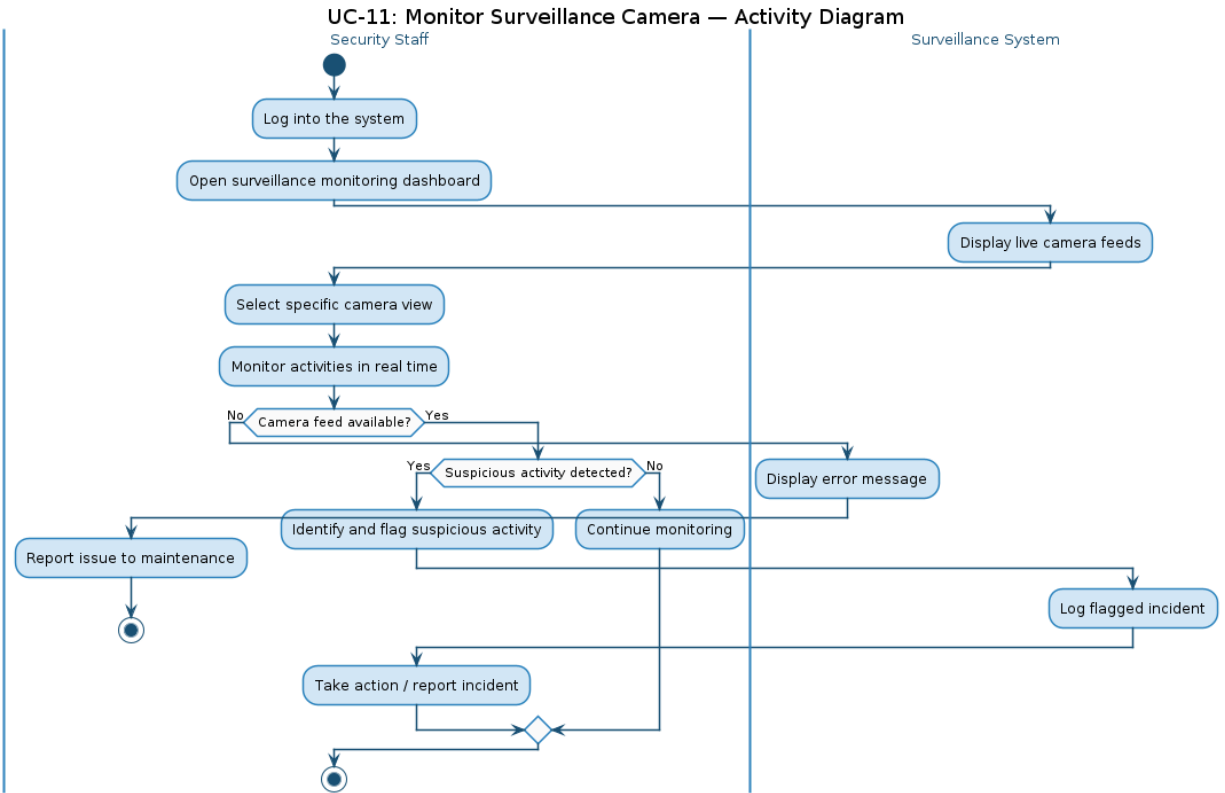
UC-09: Update Operational Records - Activity Diagram



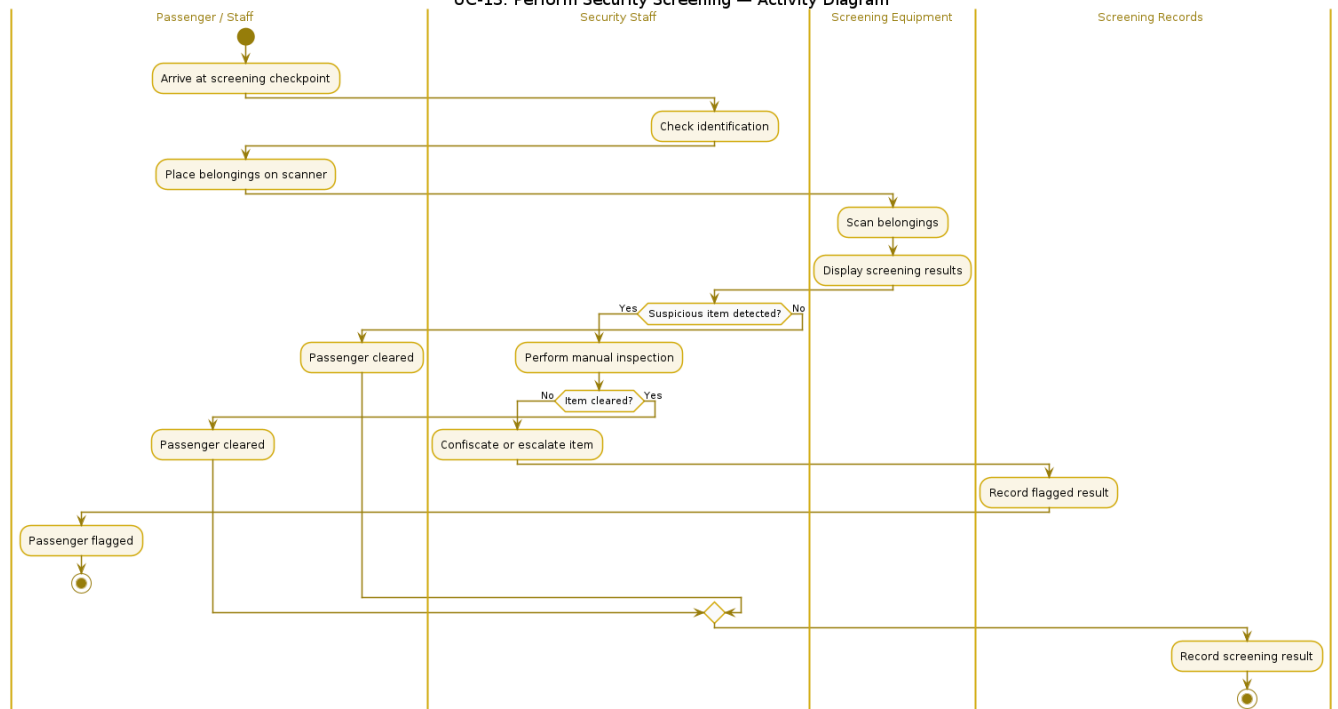
UC-10: Detect Scheduling Conflicts - Activity Diagram



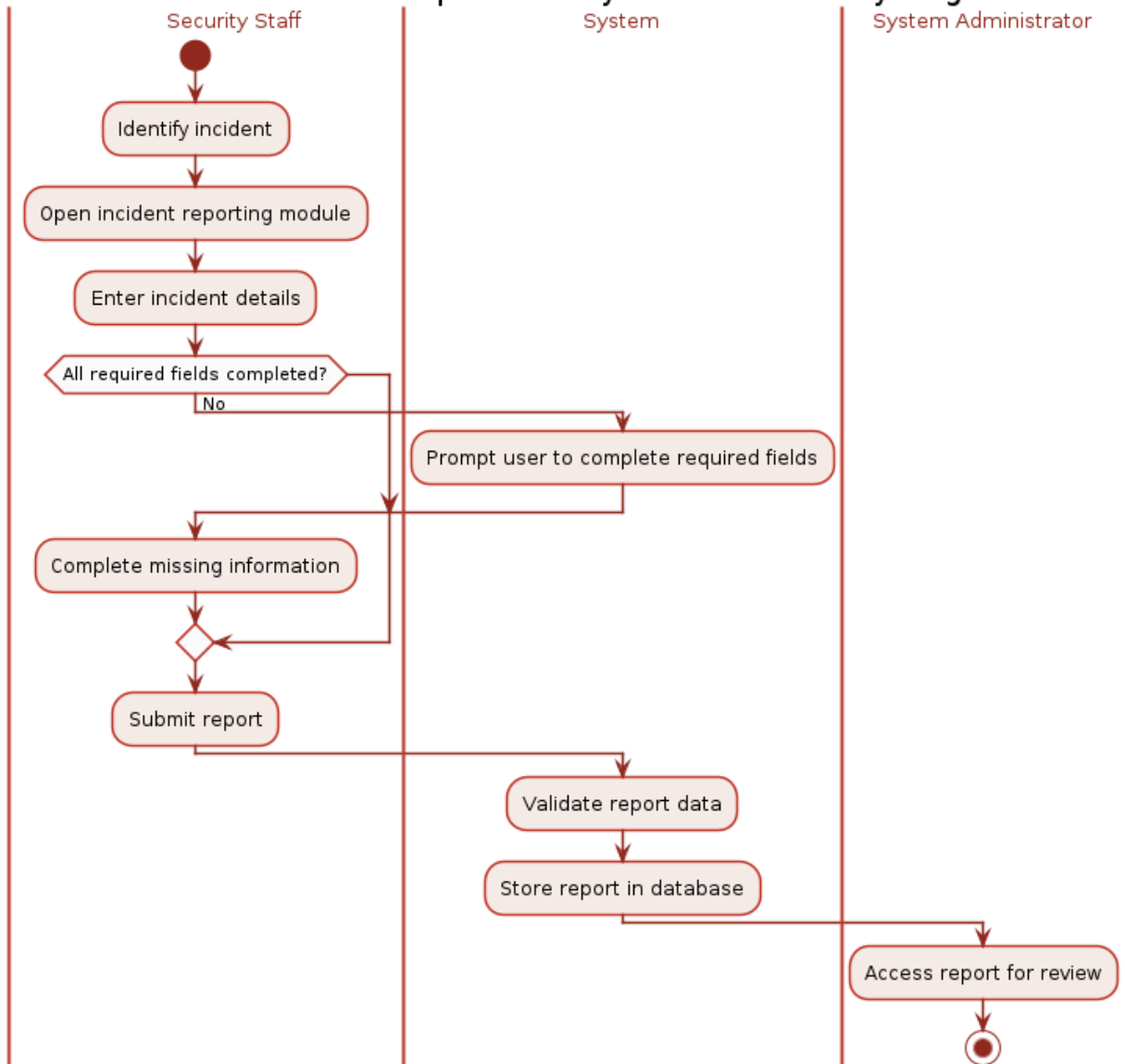
5.3.3 Activity Diagrams UC 11-15 -> Stelina Spaho



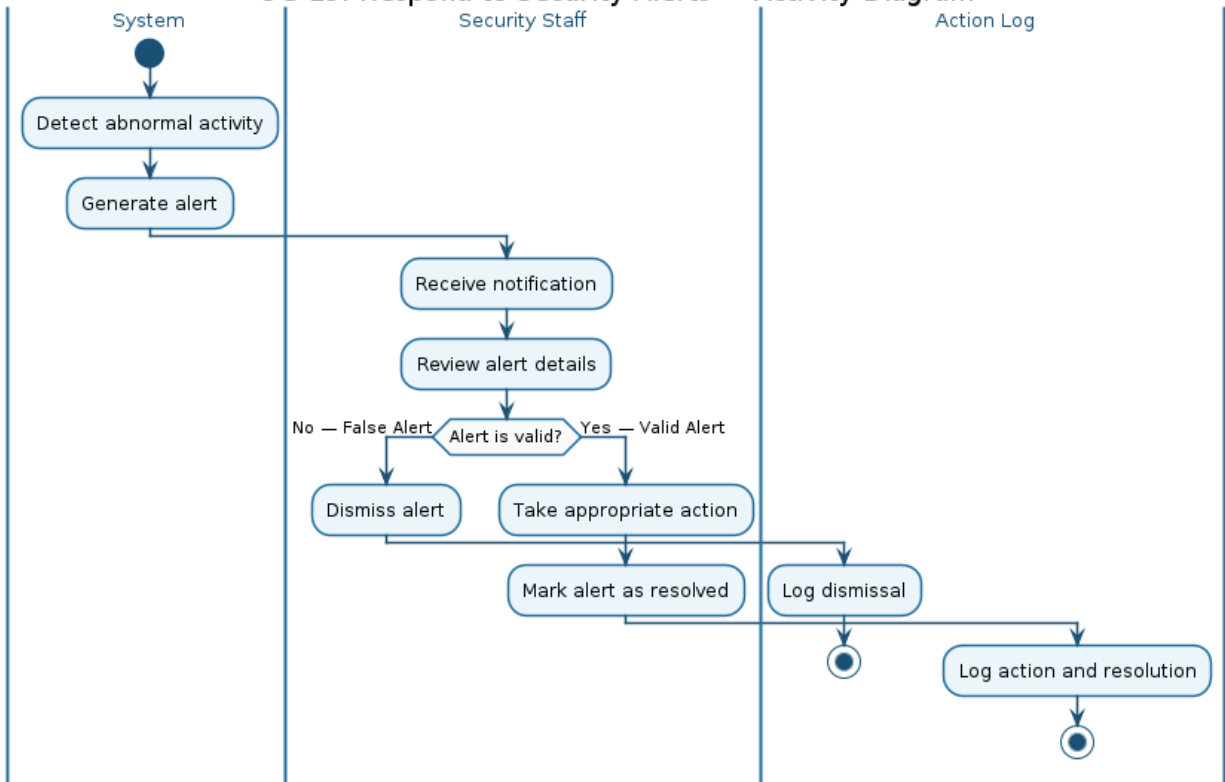
UC-13: Perform Security Screening — Activity Diagram



UC-14: Record and Report Security Incident — Activity Diagram

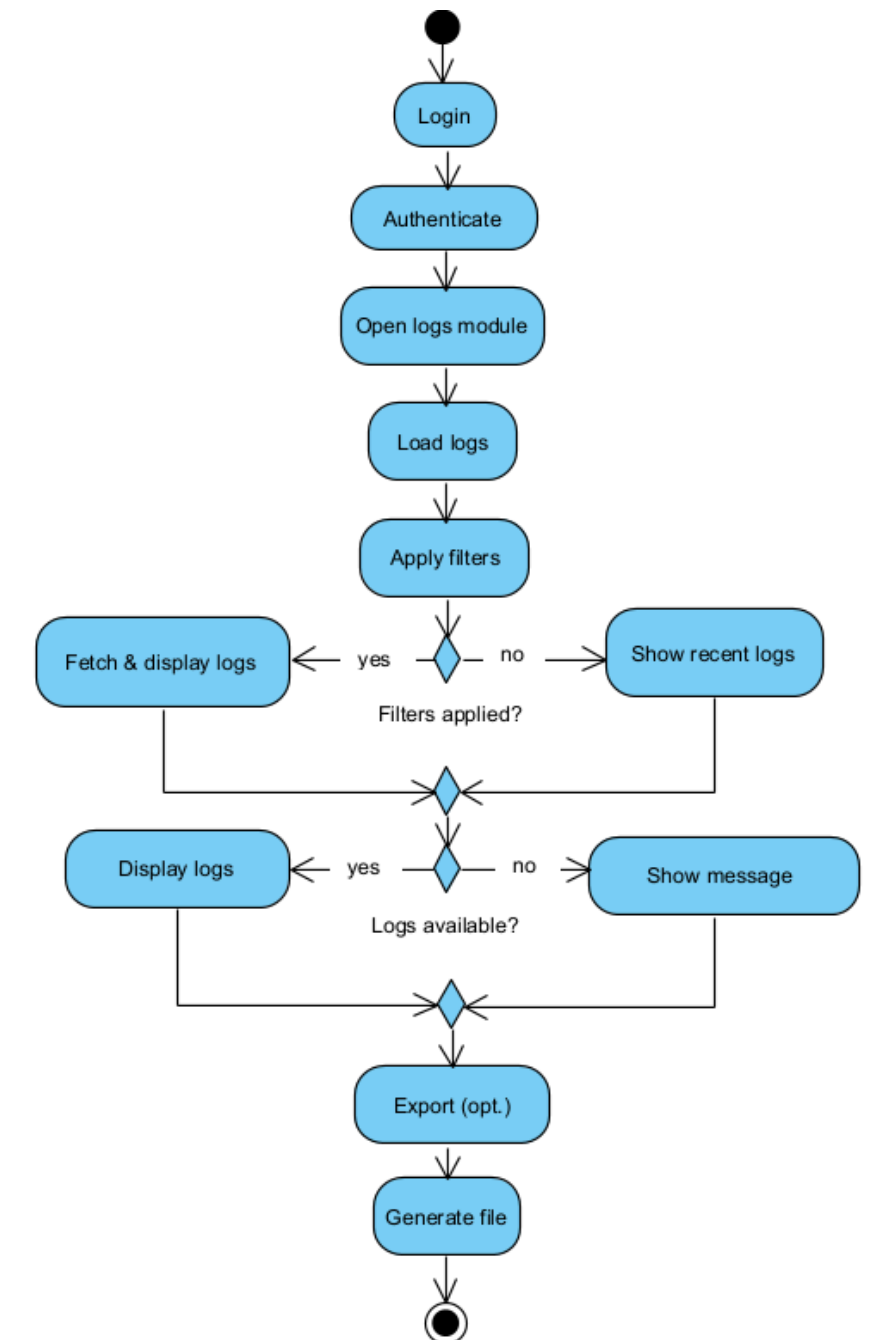


UC-15: Respond to Security Alerts — Activity Diagram

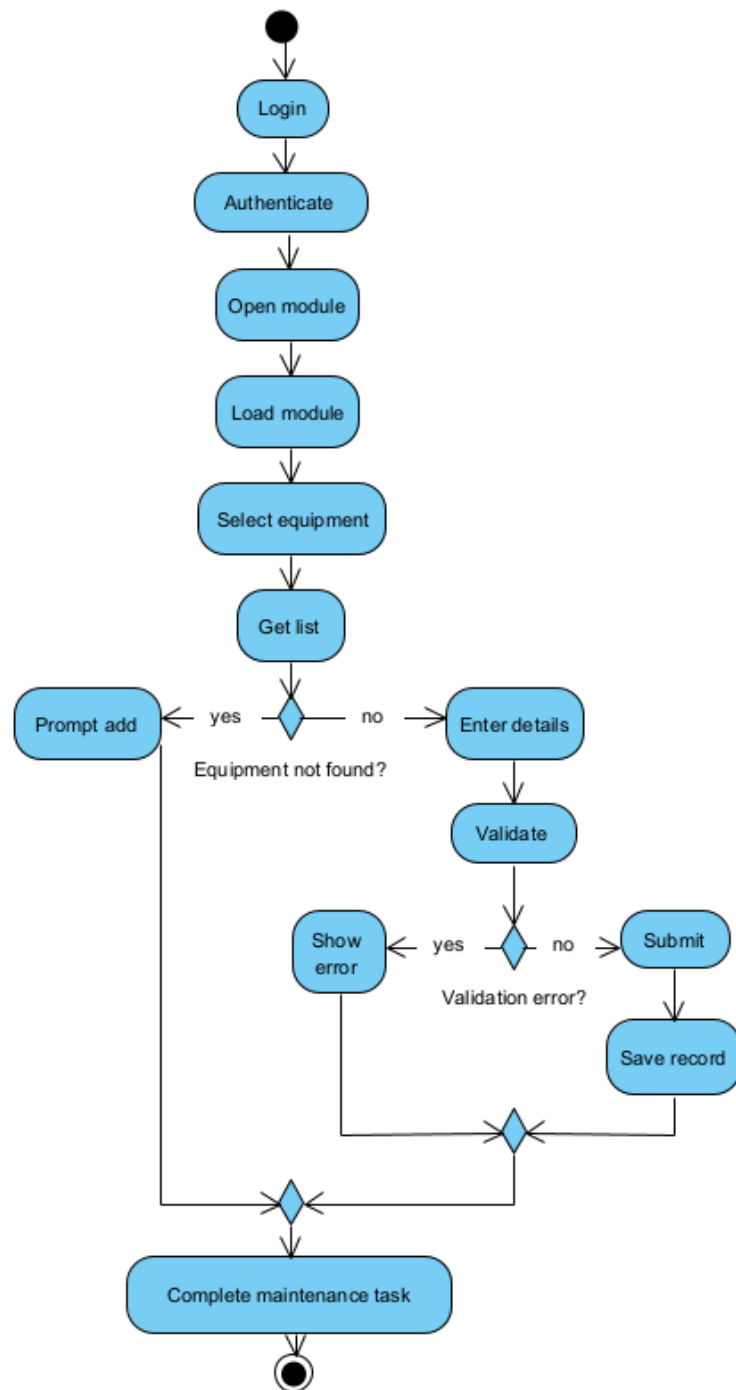


5.3.4 Activity Diagrams UC 16-20 -> Yeldos Rakhymzhanov

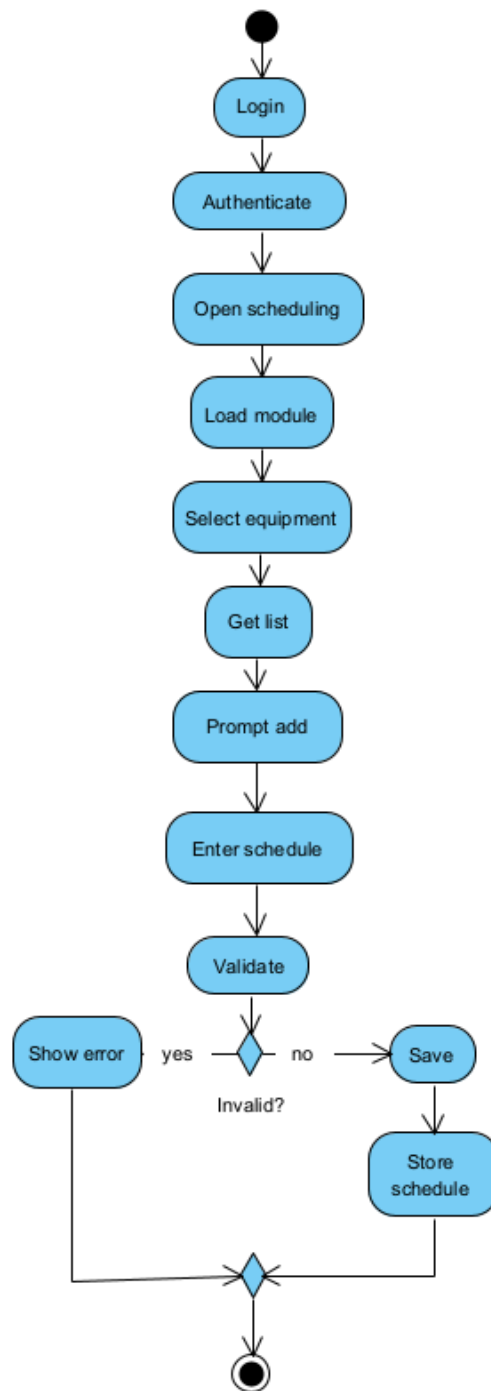
UC-16: Review Security Logs and Reports



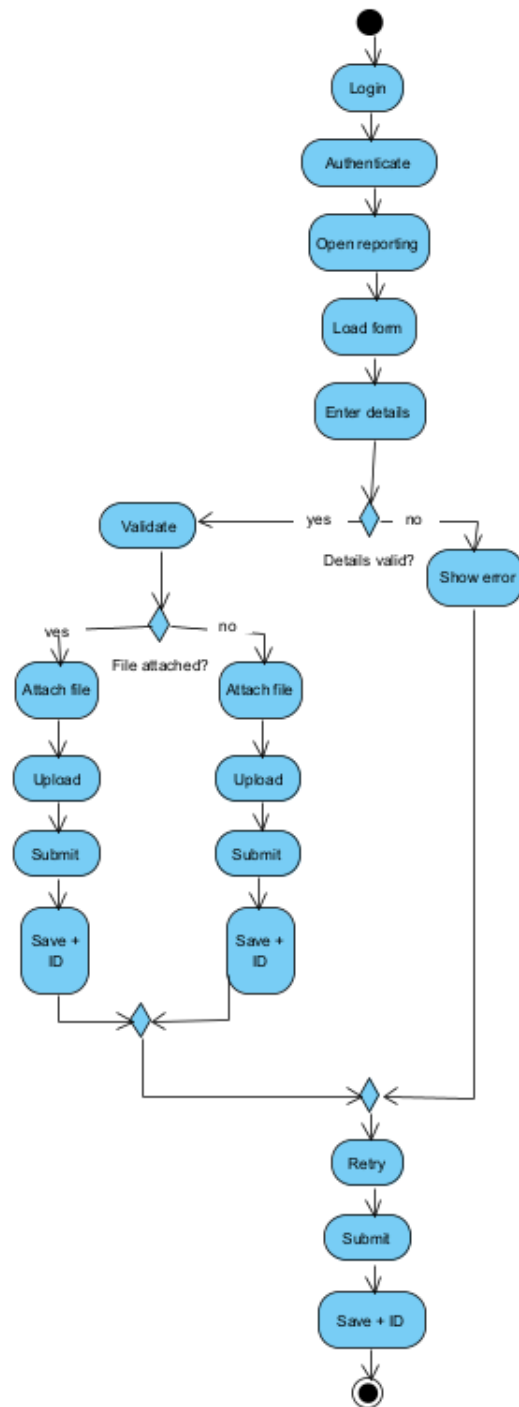
UC-17: Record Maintenance Activities



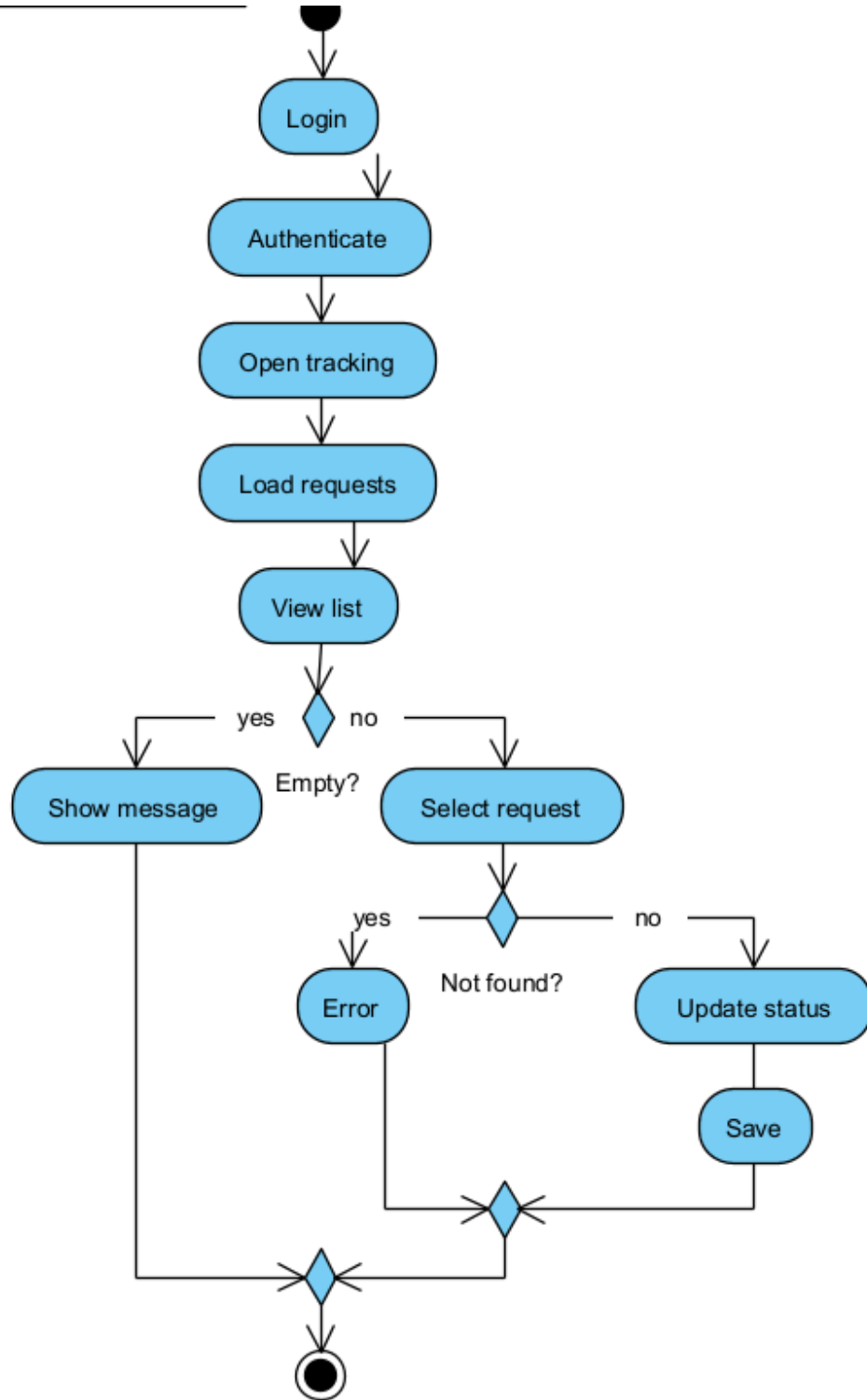
UC-18: Schedule Preventive Maintenance



UC-19: Report Infrastructure Issues

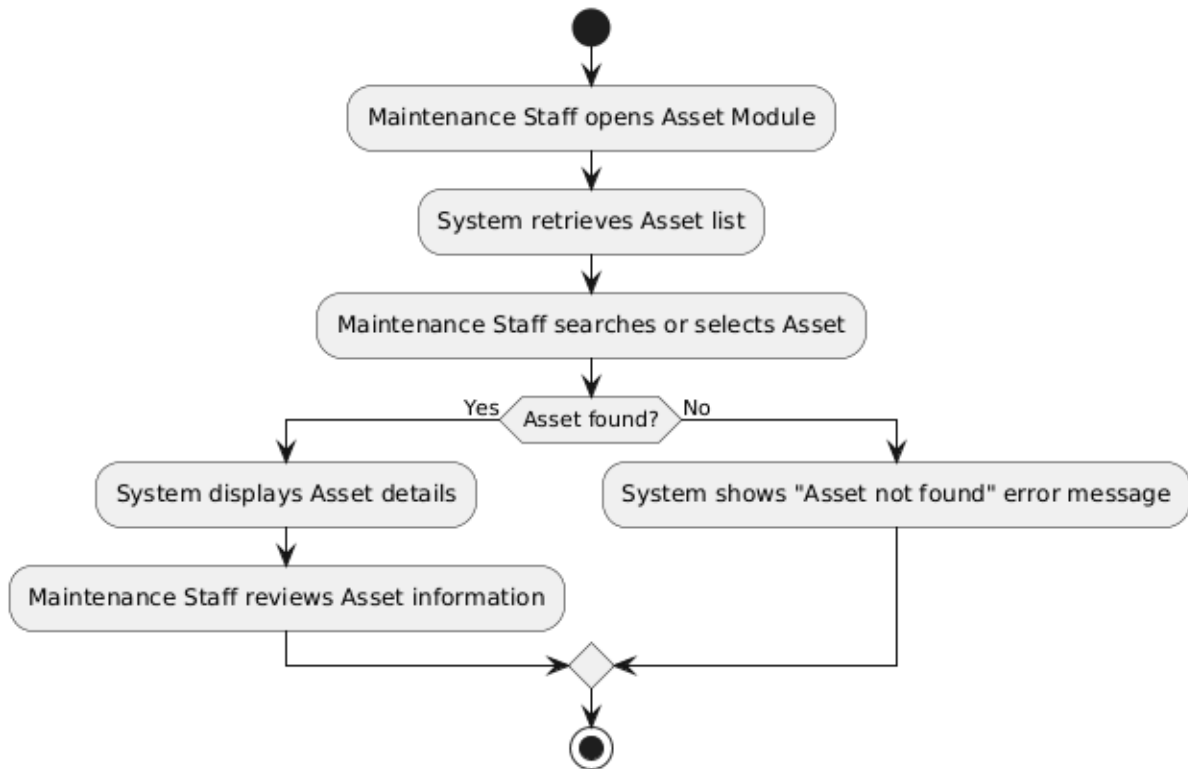


UC-20: Track Maintenance Requests

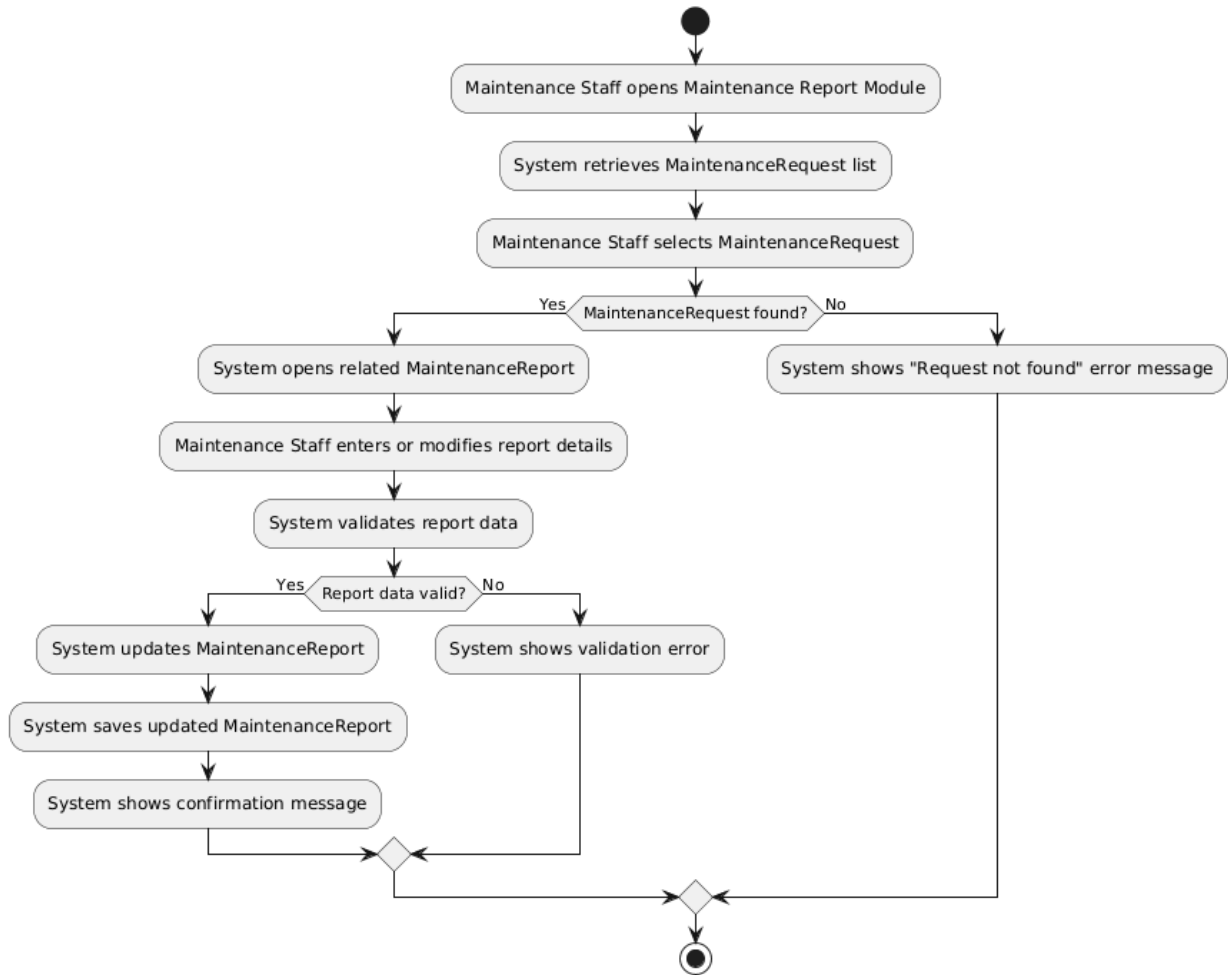


5.3.5 Activity Diagrams UC 21-25 -> Fatih Gökteke

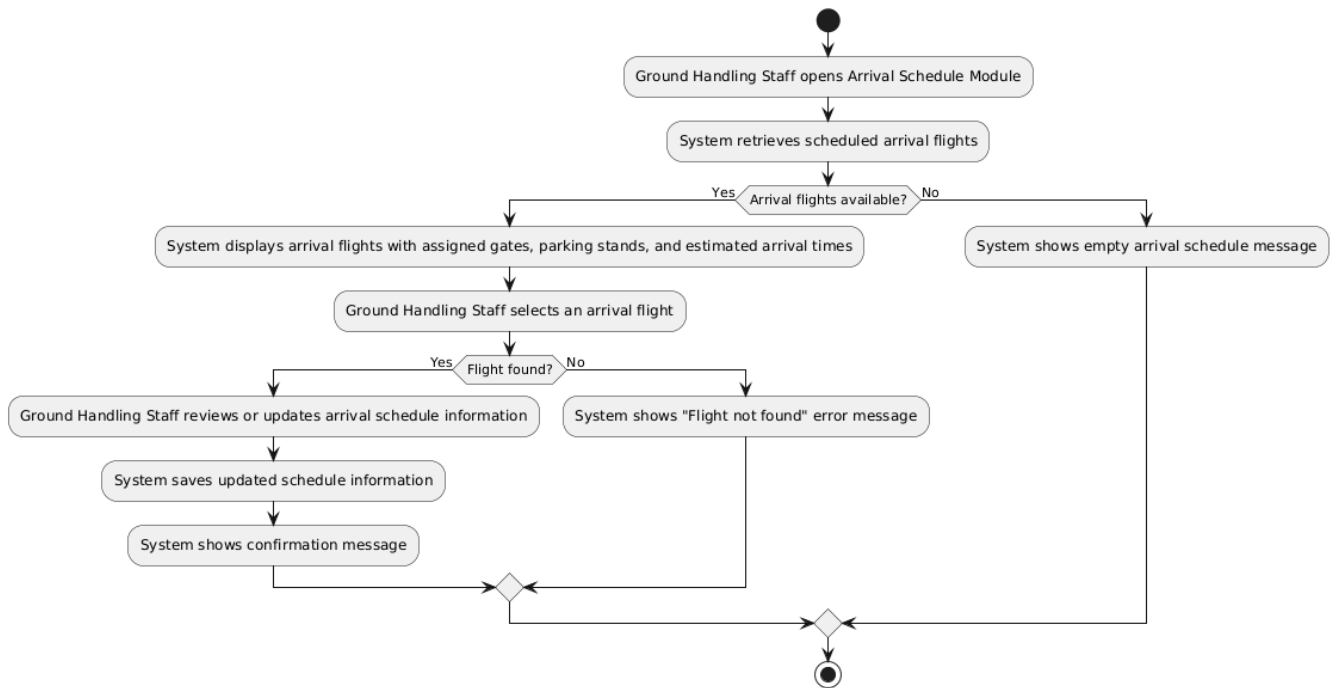
UC-21: Access Asset Information - Activity Diagram



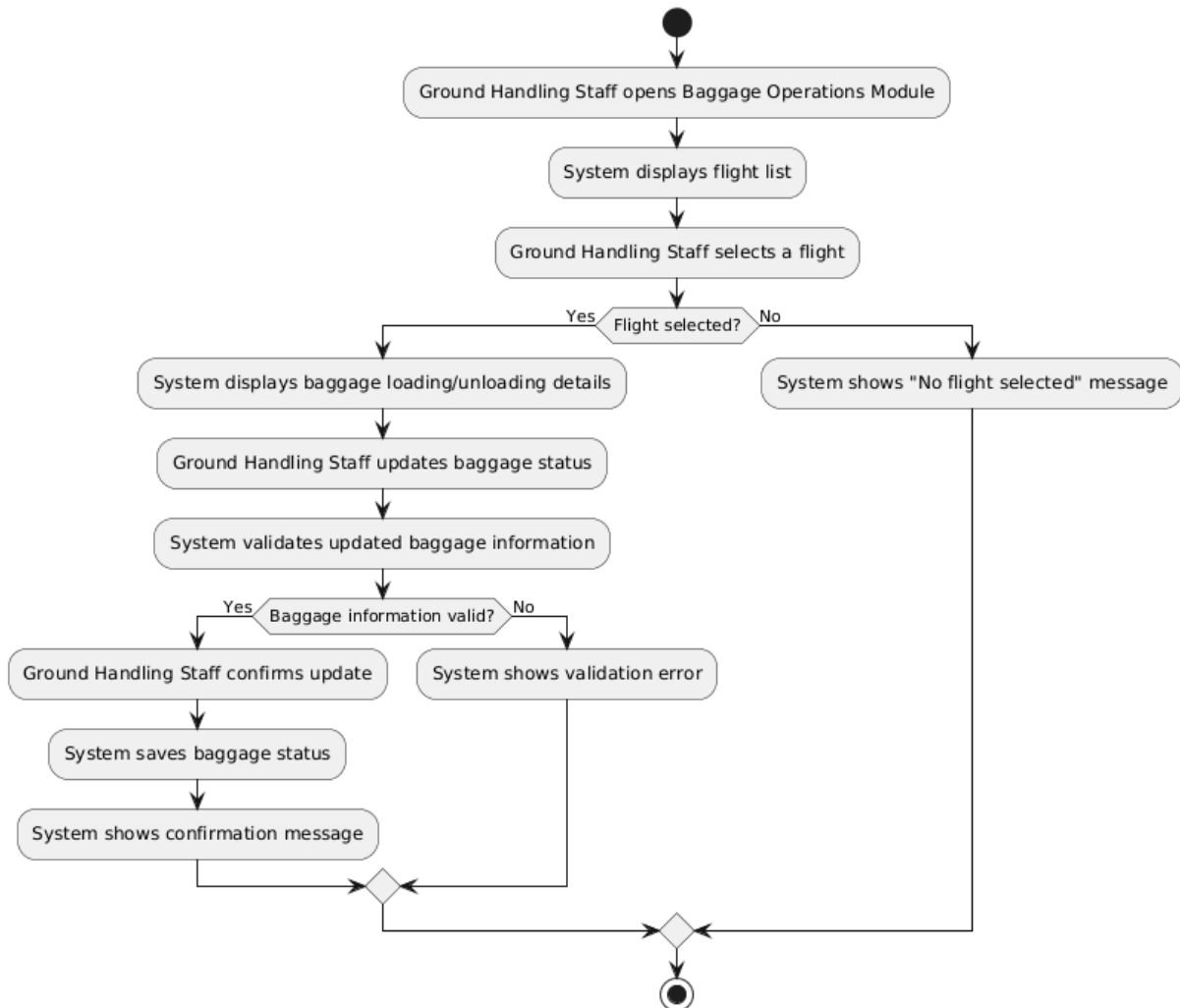
UC-22: Update Maintenance Report - Activity Diagram



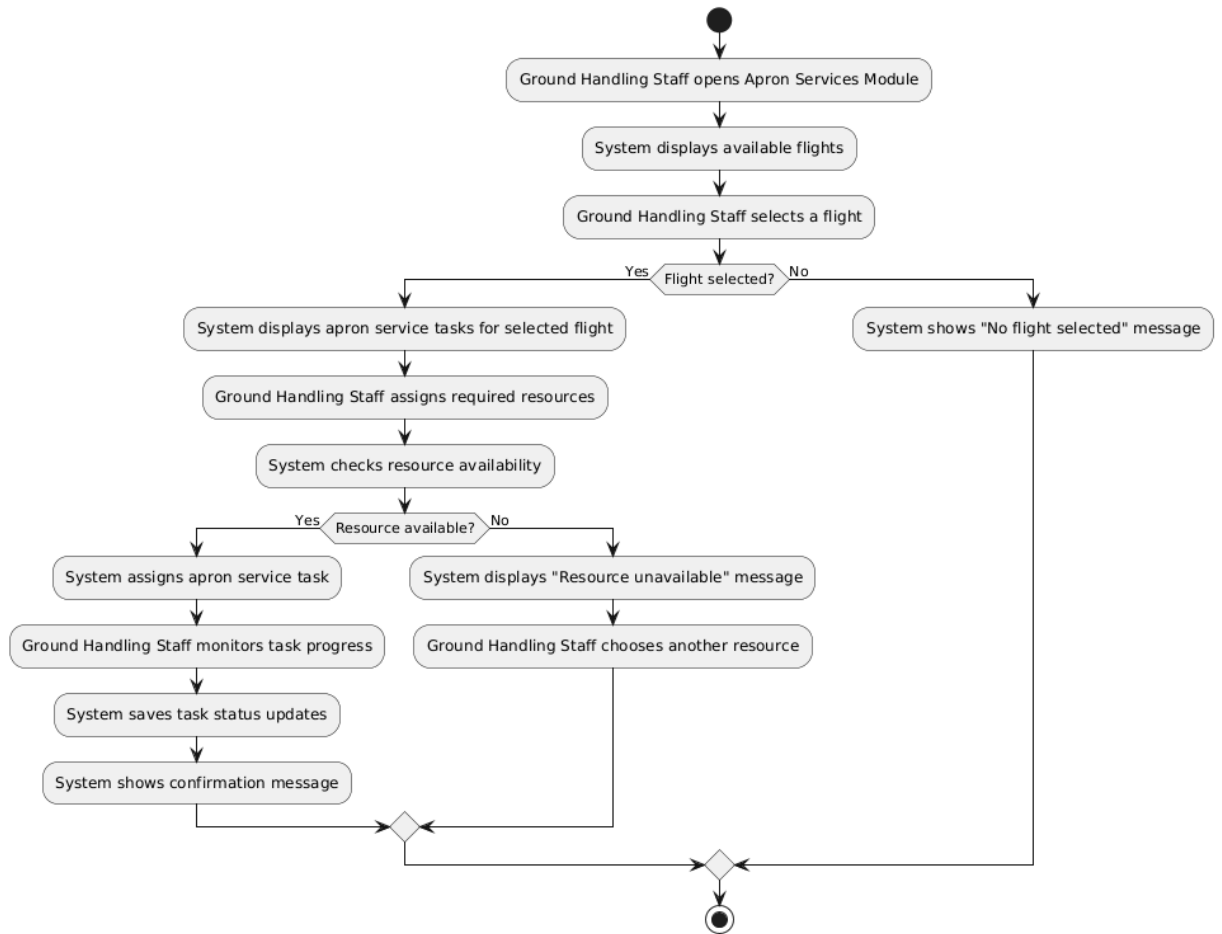
UC-23: Manage Arrival Schedule - Activity Diagram



UC-24: Manage Baggage Operations - Activity Diagram

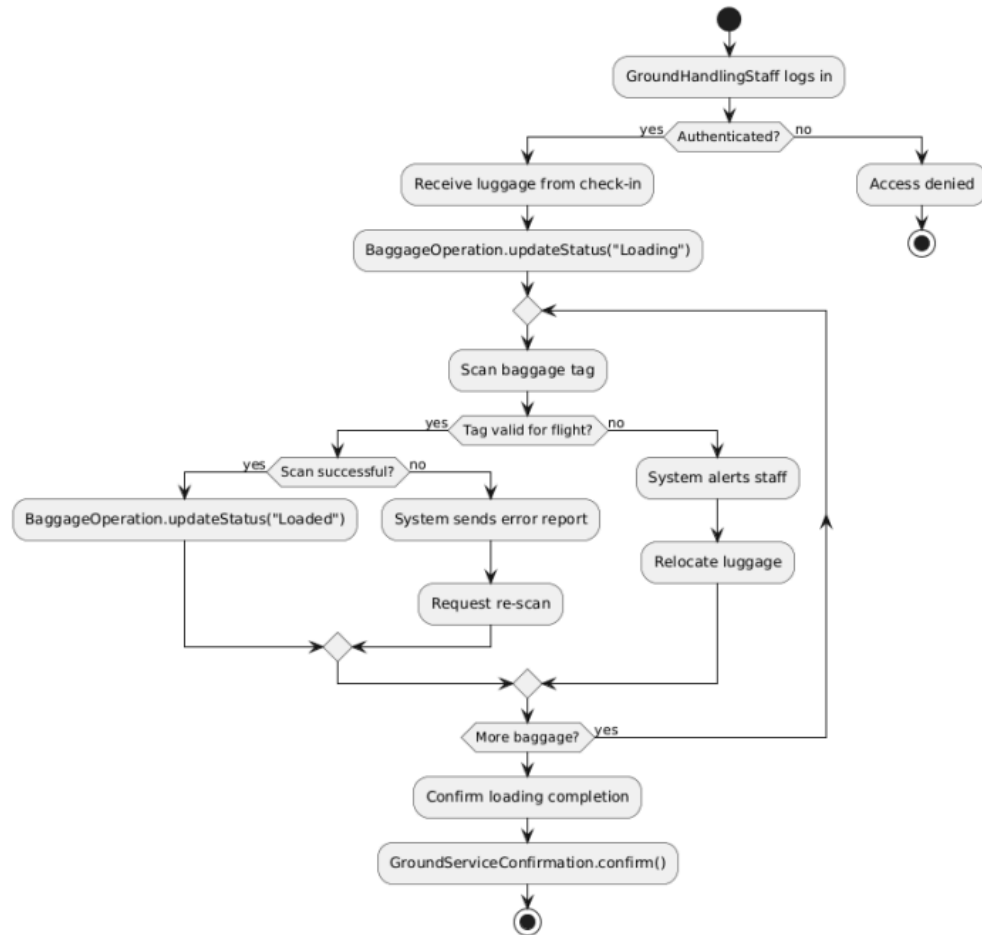


UC-25: Manage Apron Services - Activity Diagram

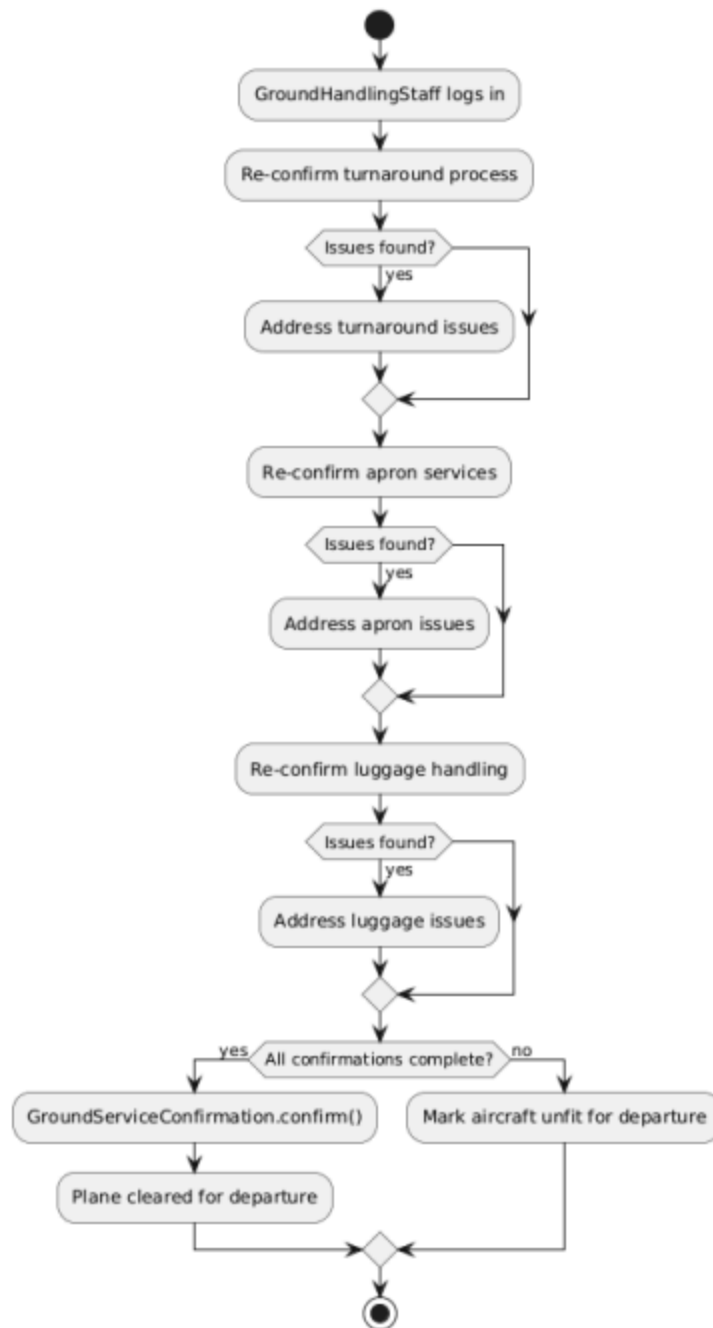


5.3.6 Activity Diagrams UC 26-30 -> Leart Mukaj

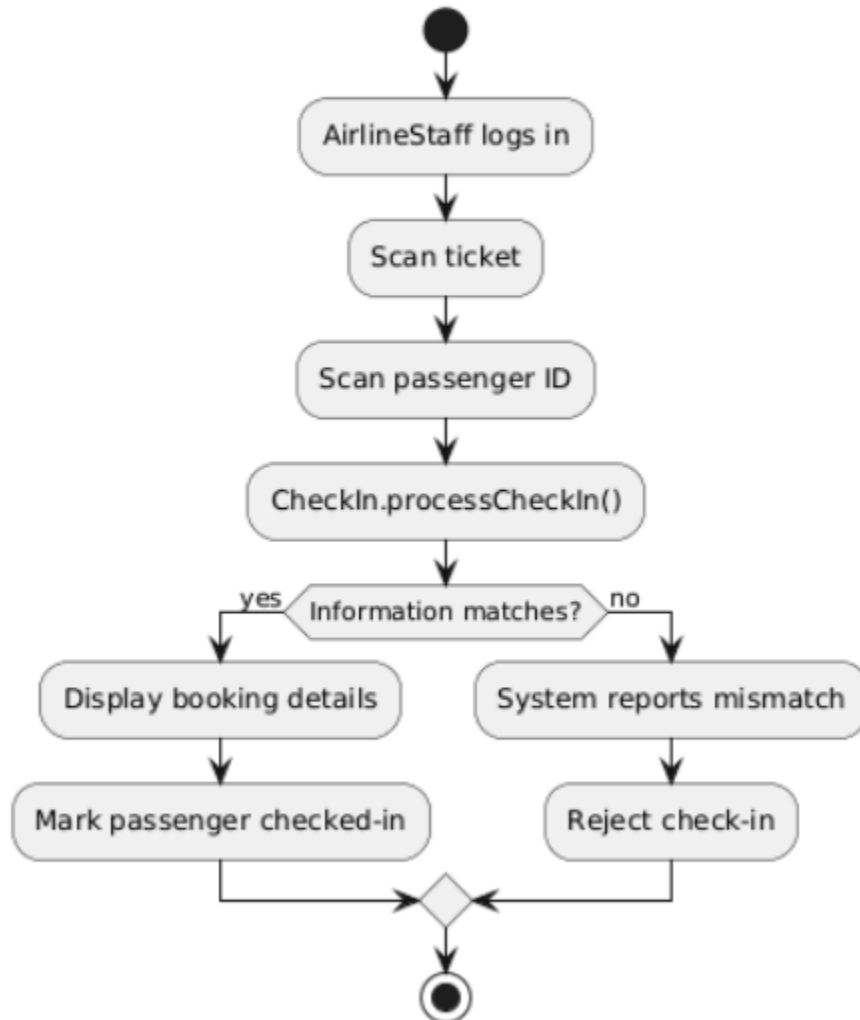
UC-26 Baggage Loading - Activity Diagram



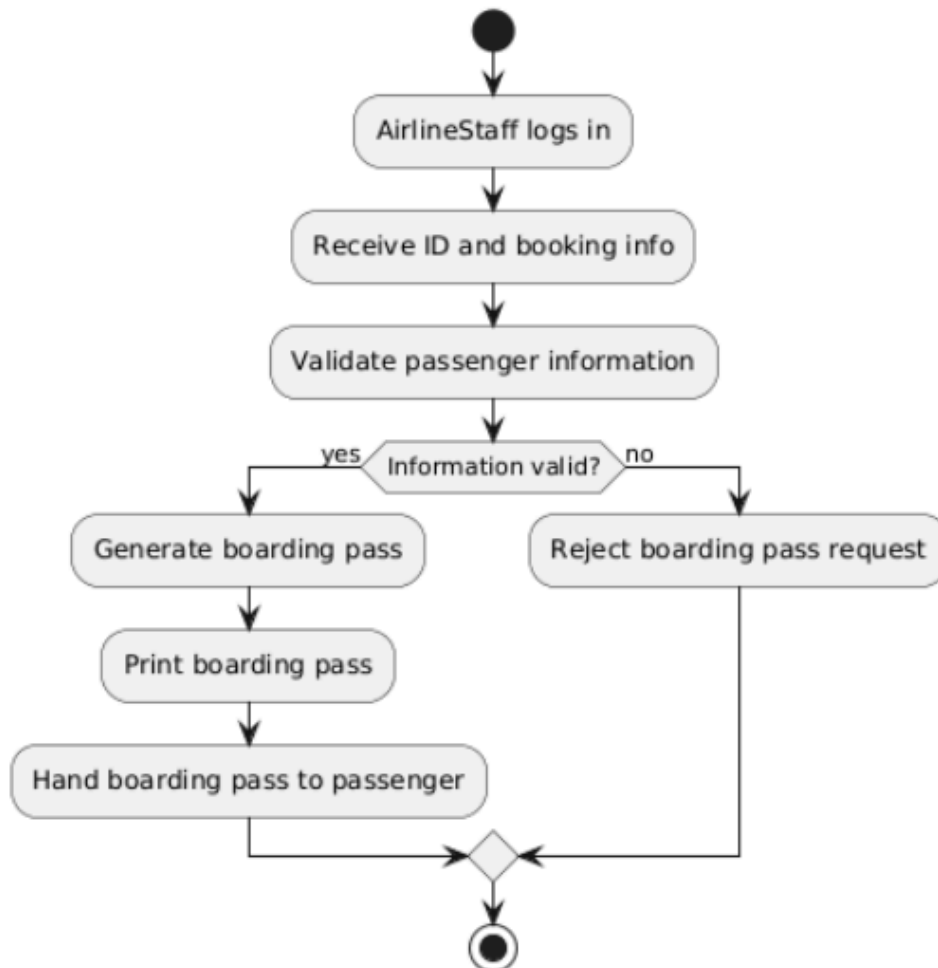
UC-27 Final Confirmation - Activity Diagram



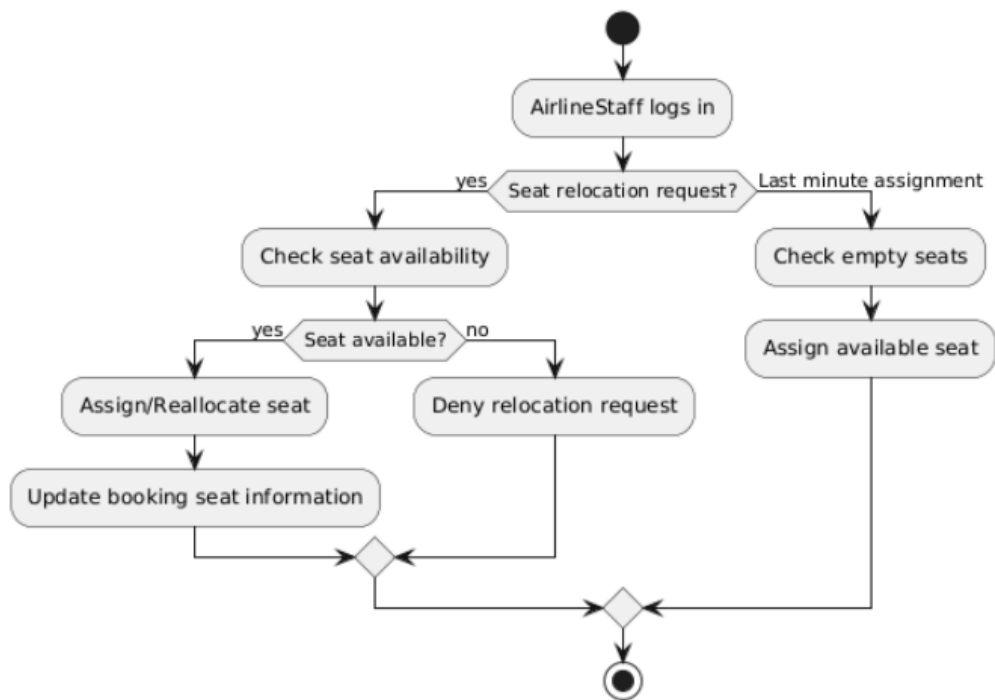
UC-28 Passenger Check-in - Activity Diagram



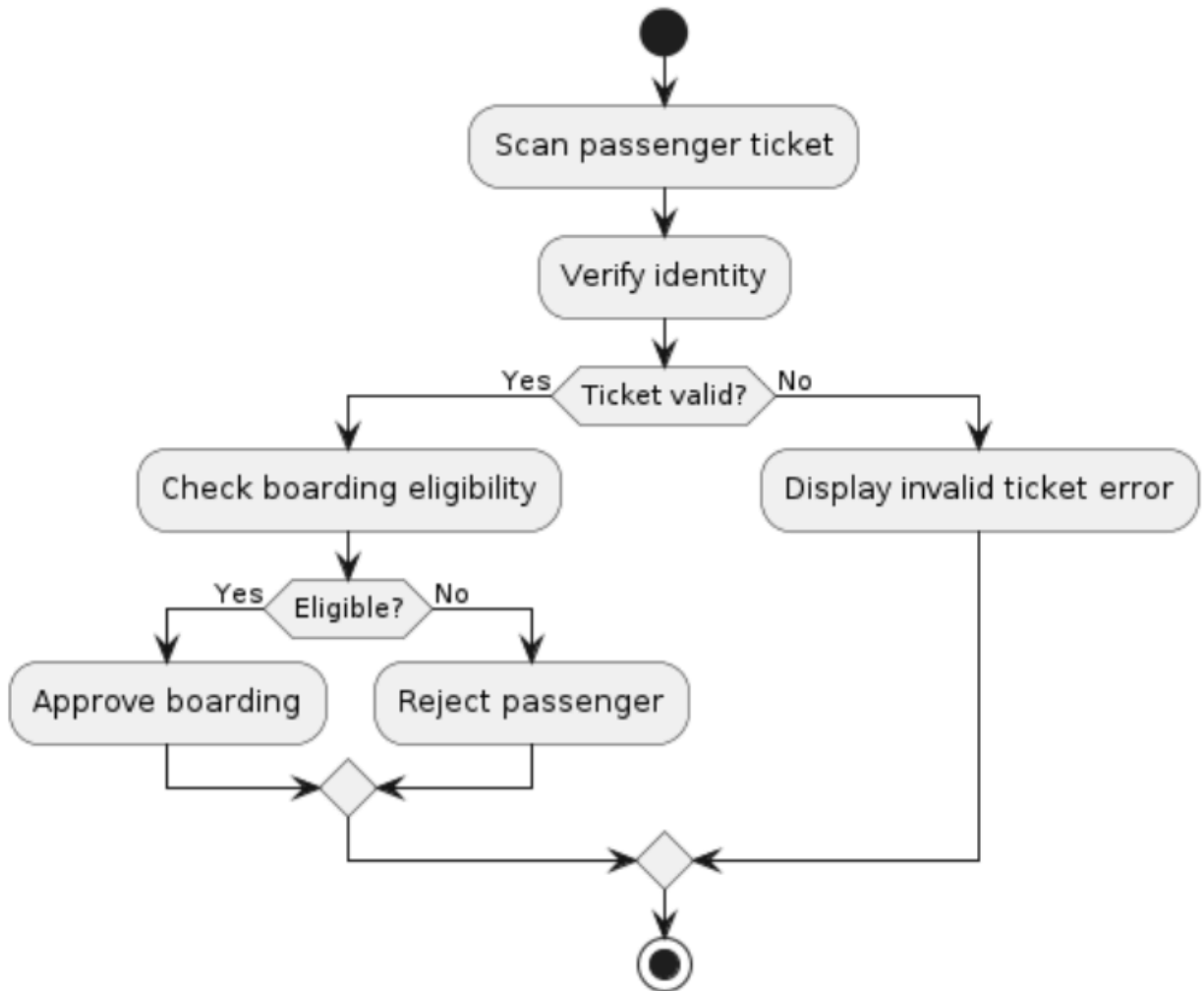
UC-29 Boarding Pass - Activity Diagram



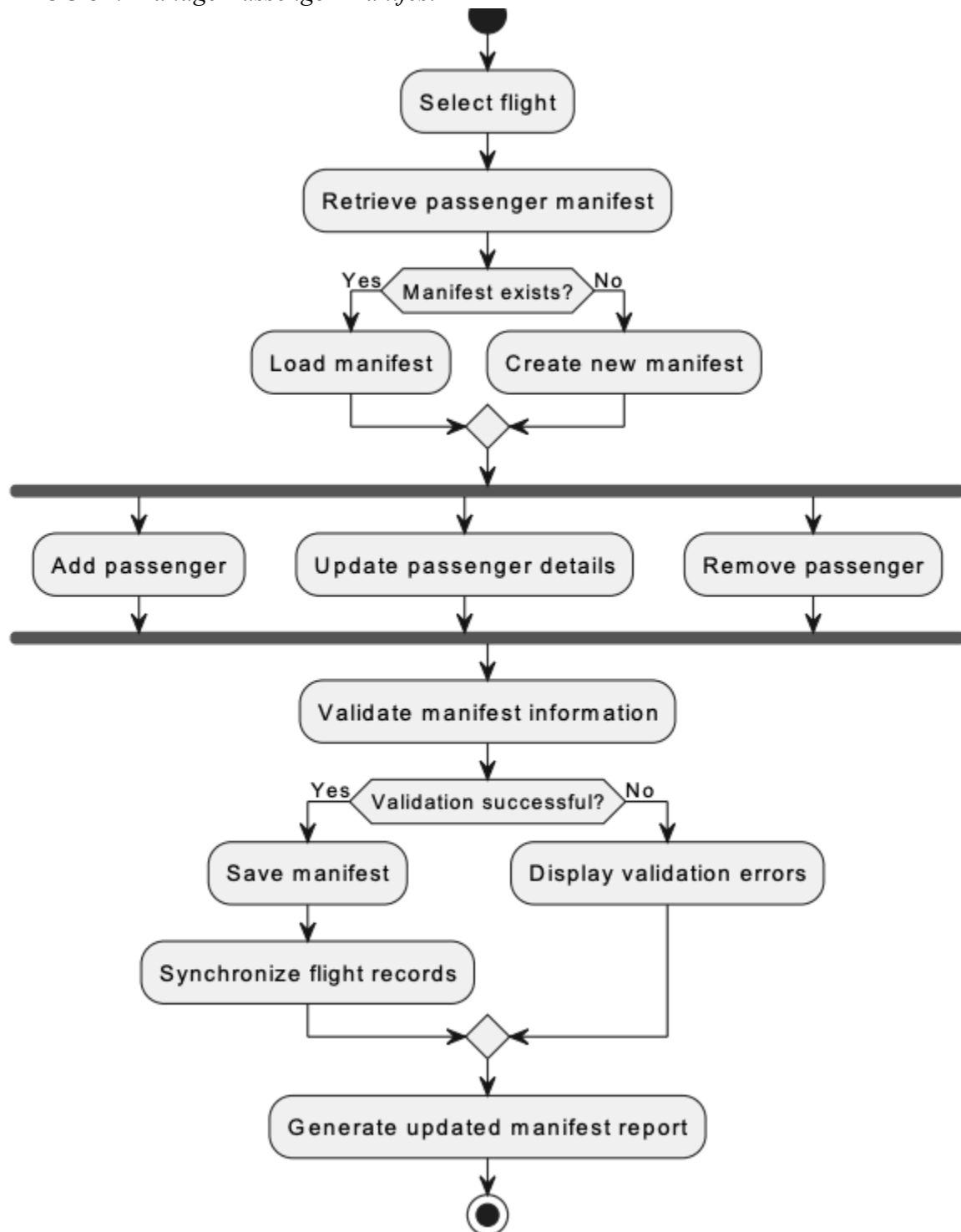
UC-30 Seat Handling - Activity Diagram



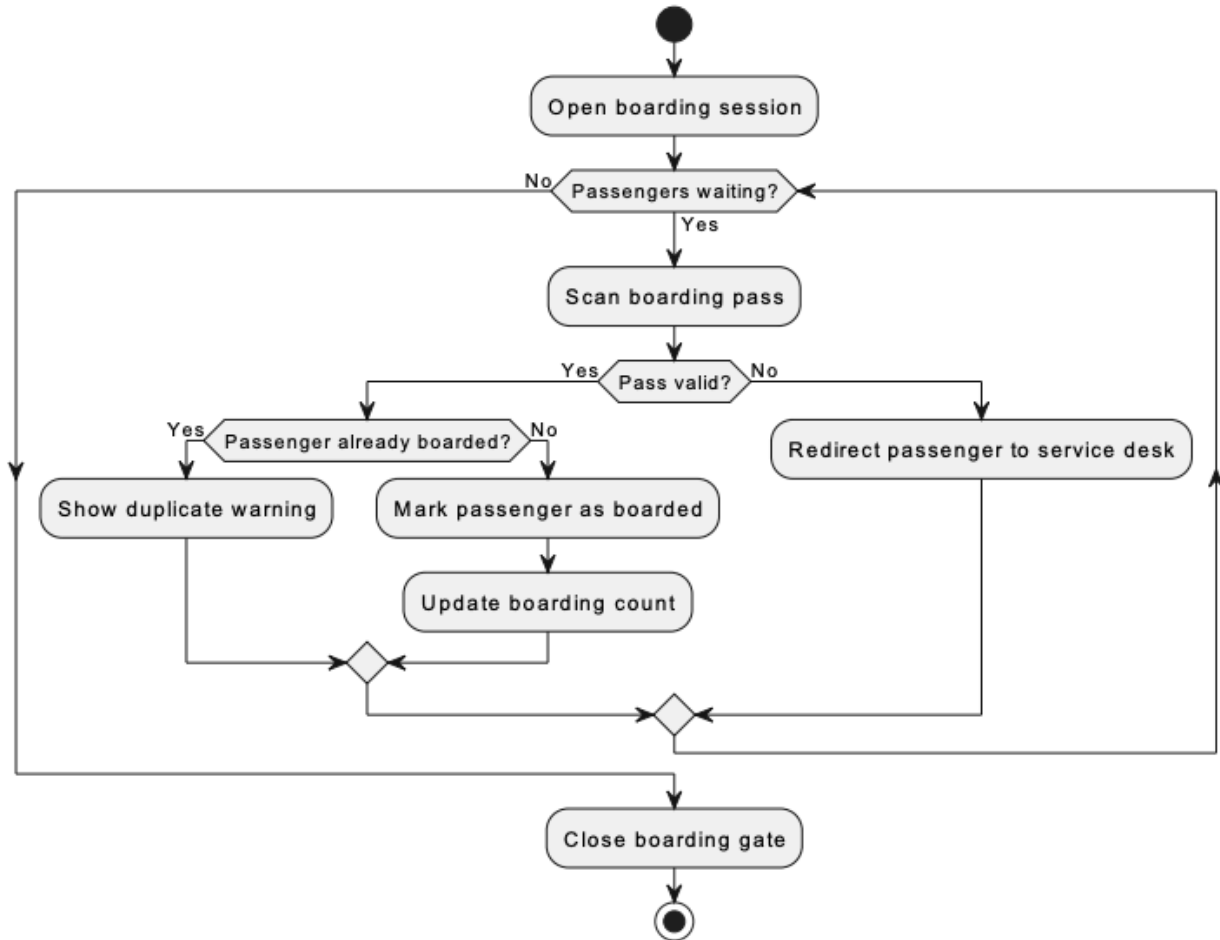
BR-31 Activity Diagram - Ticket Validation



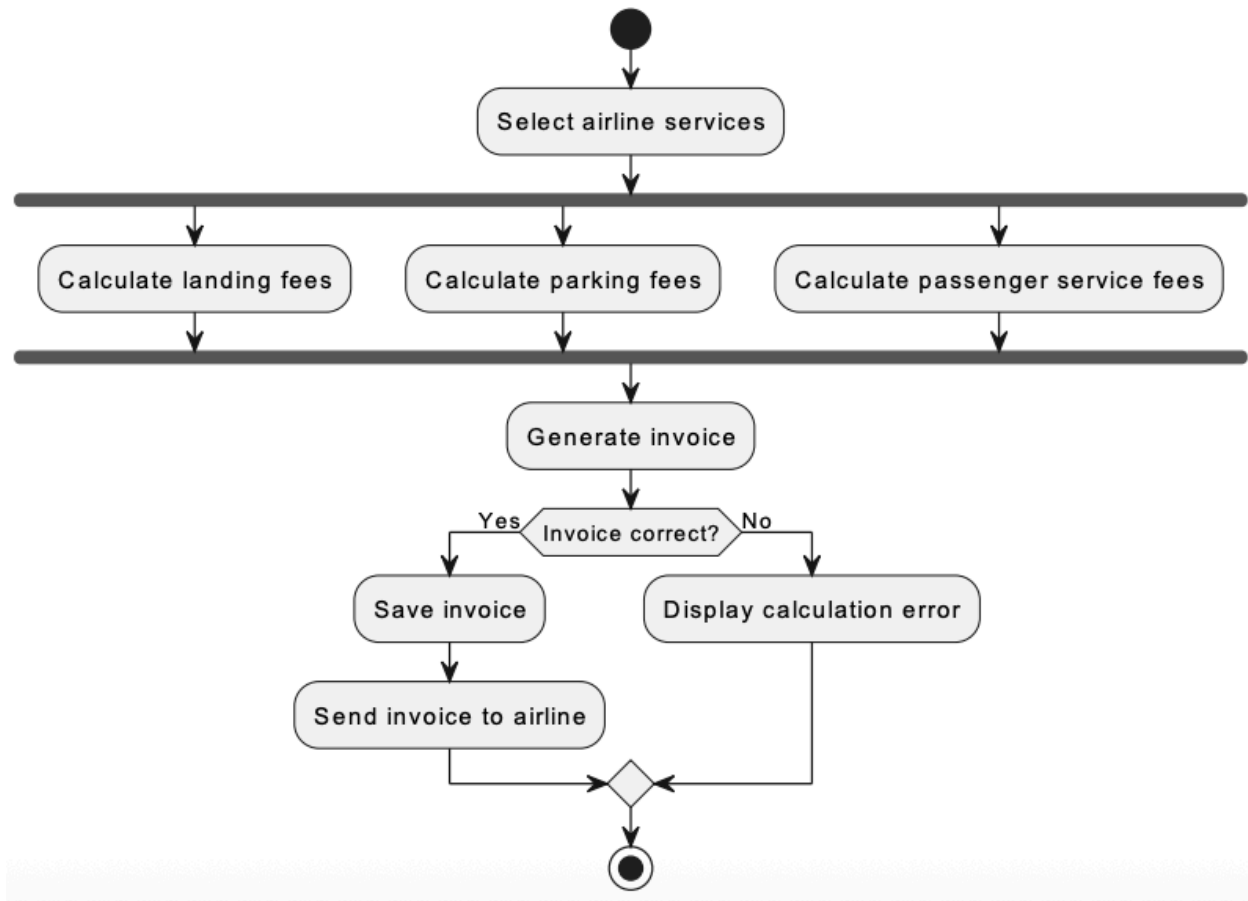
UC-32: Manage Passenger Manifest



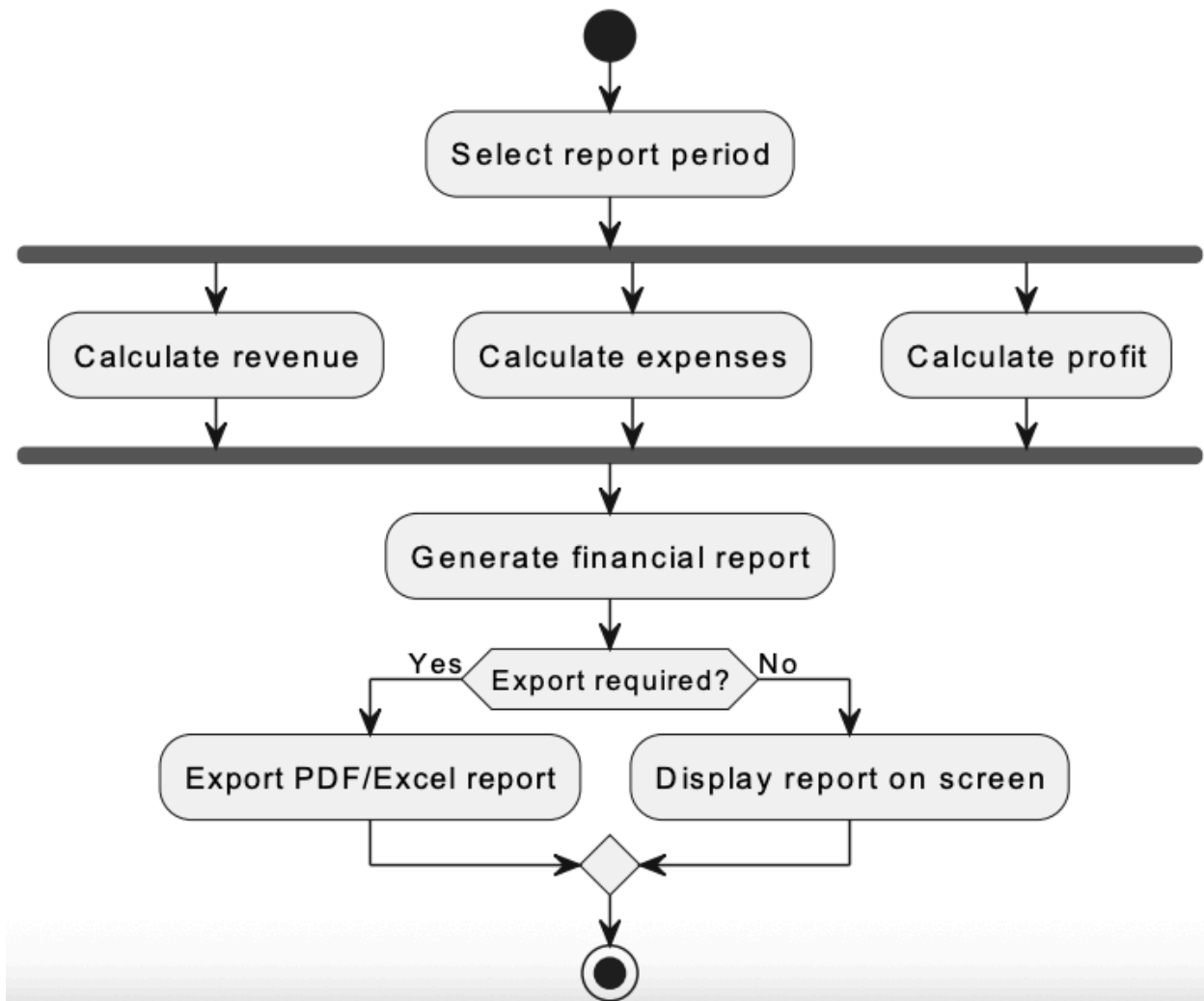
BR-33 Activity Diagram - Boarding Process



BR-34 Activity Diagram - Invoice Generation

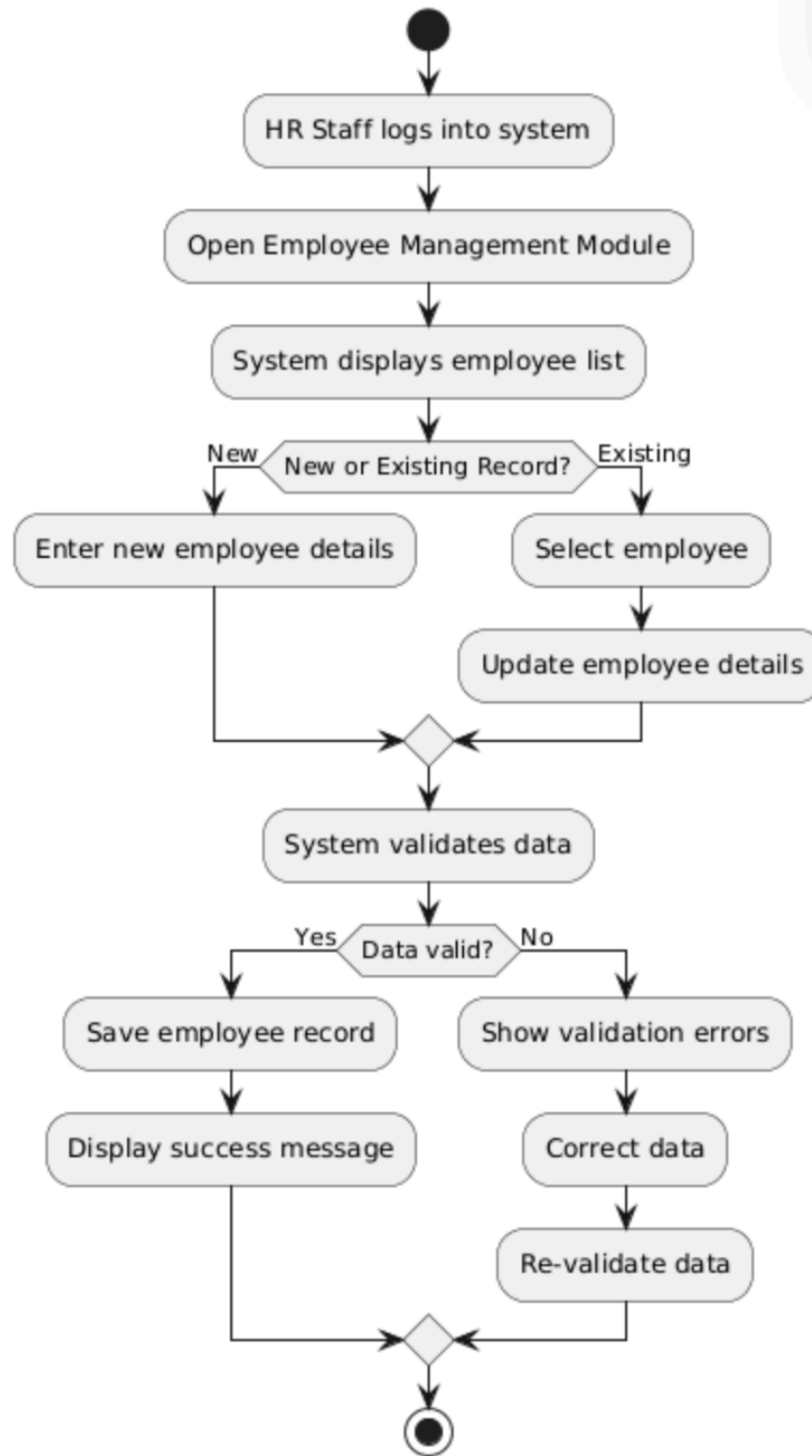


BR-35 Activity Diagram - Financial Reporting

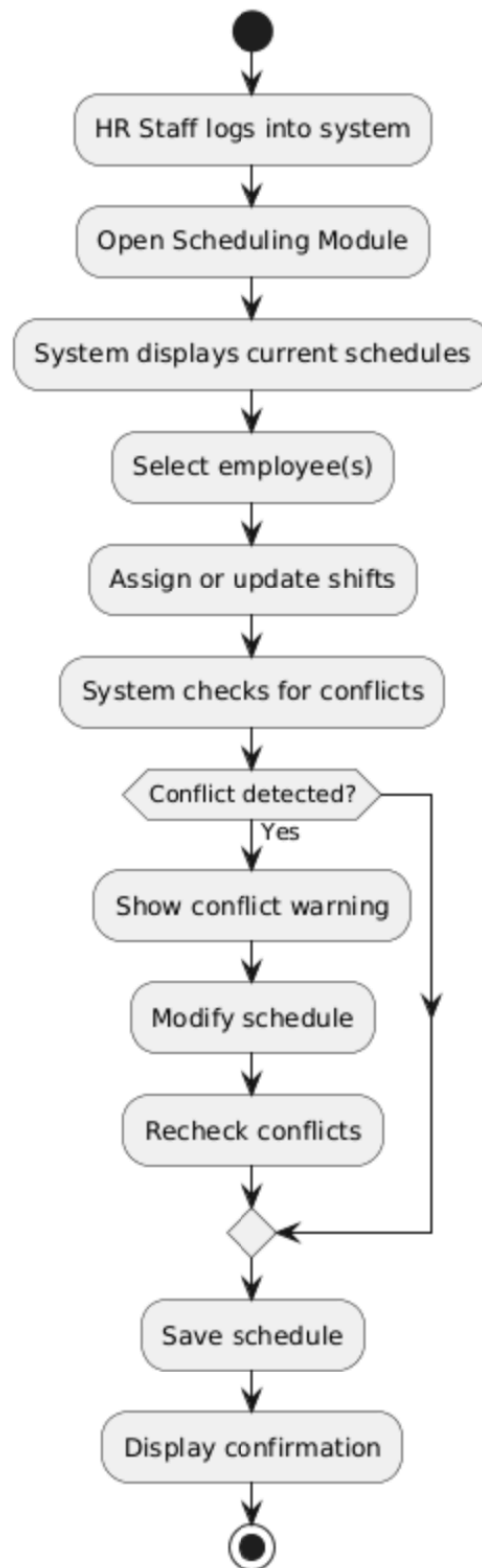


5.3.7 Activity Diagrams UC 36-38 -> Stelina Spaho

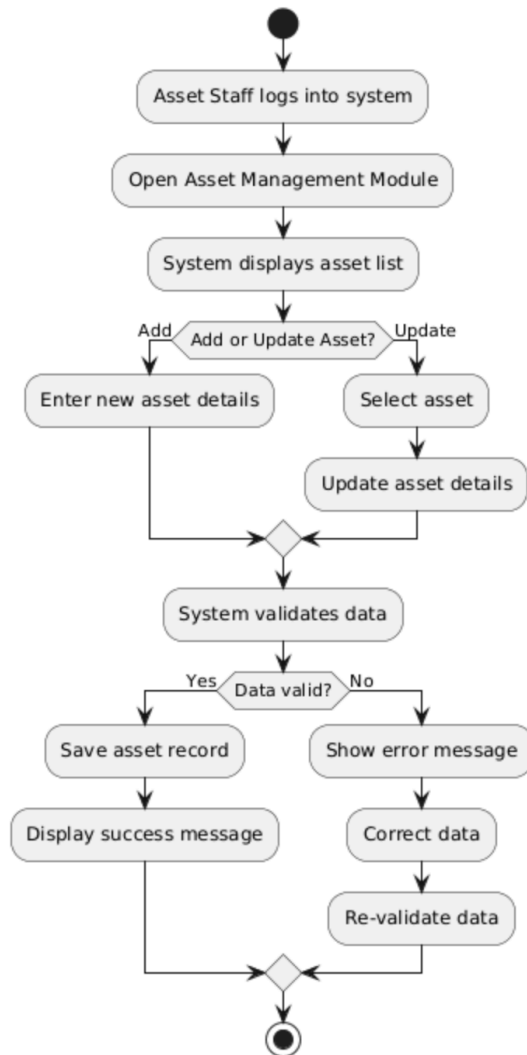
UC-36: Manage Employee Records



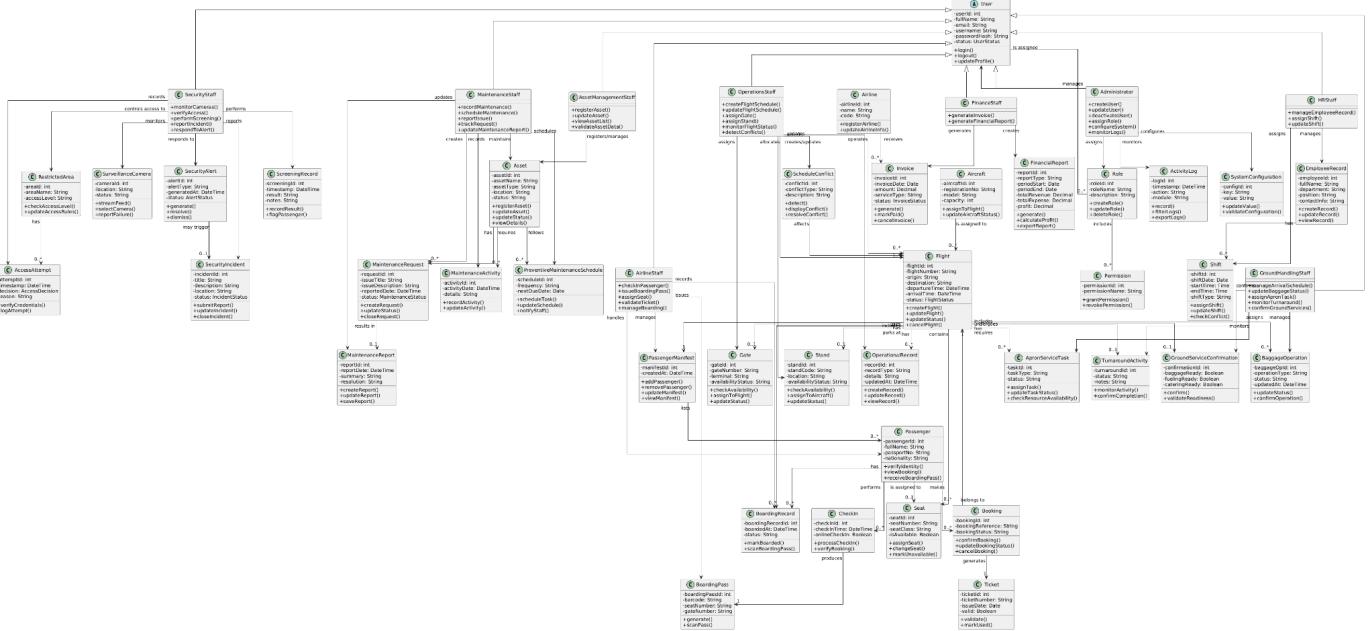
UC-37: Manage Employee Schedules and Shifts



UC-38: Manage Airport Assets



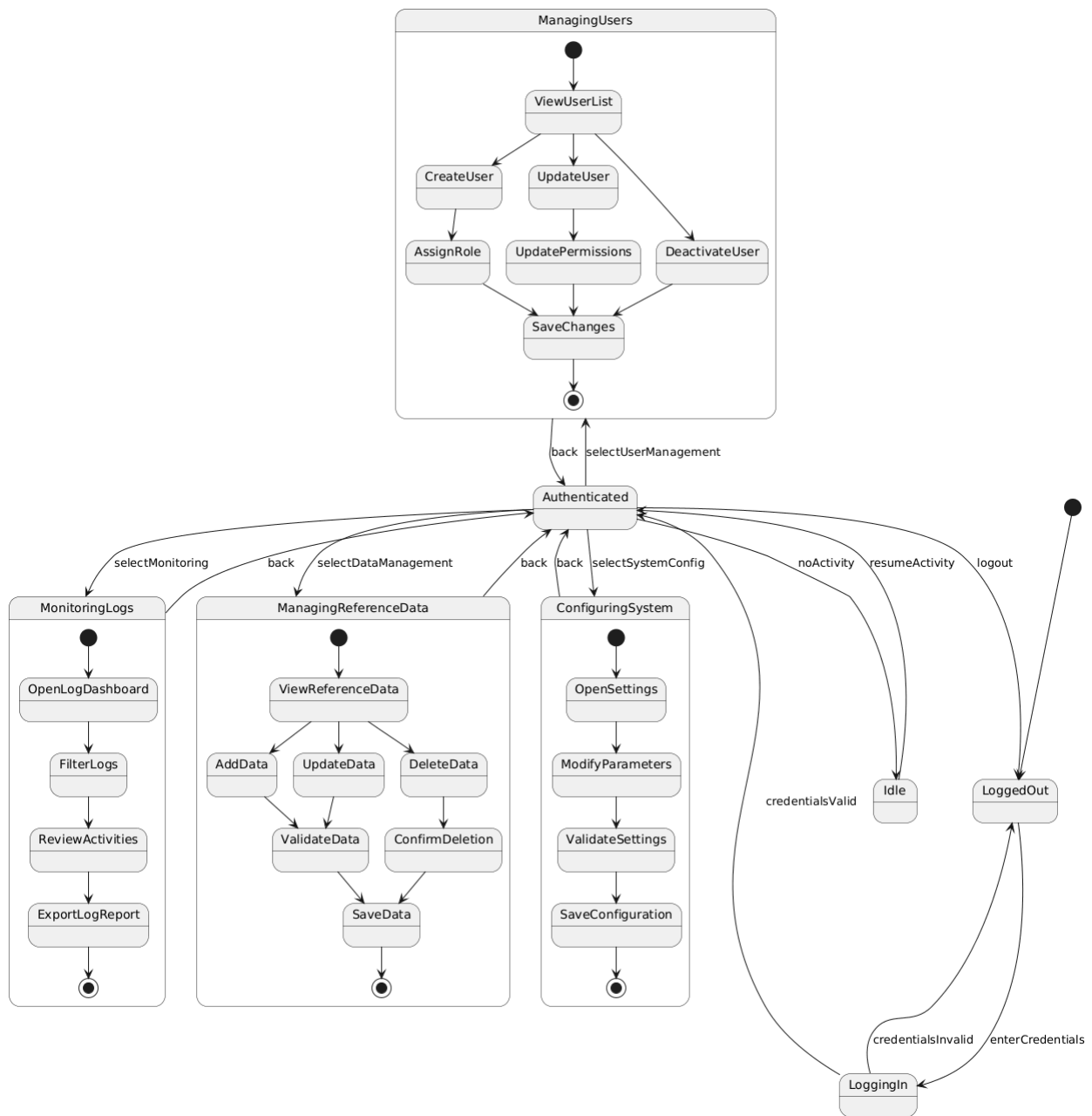
5.4 Class Diagram



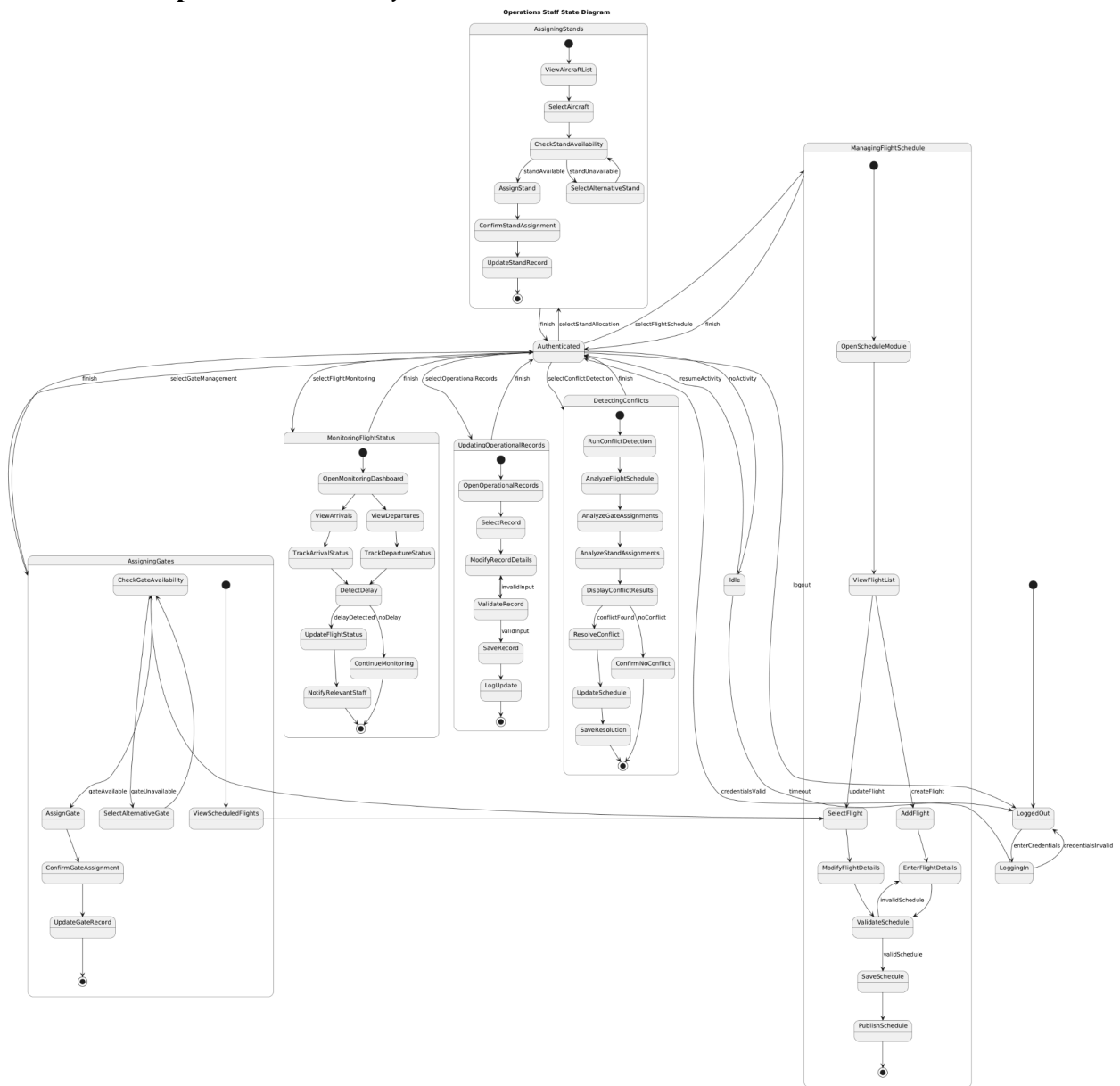
By: Moira Lako

5.5 State Diagrams

5.5.1 Administrator - *by Moira Lako*

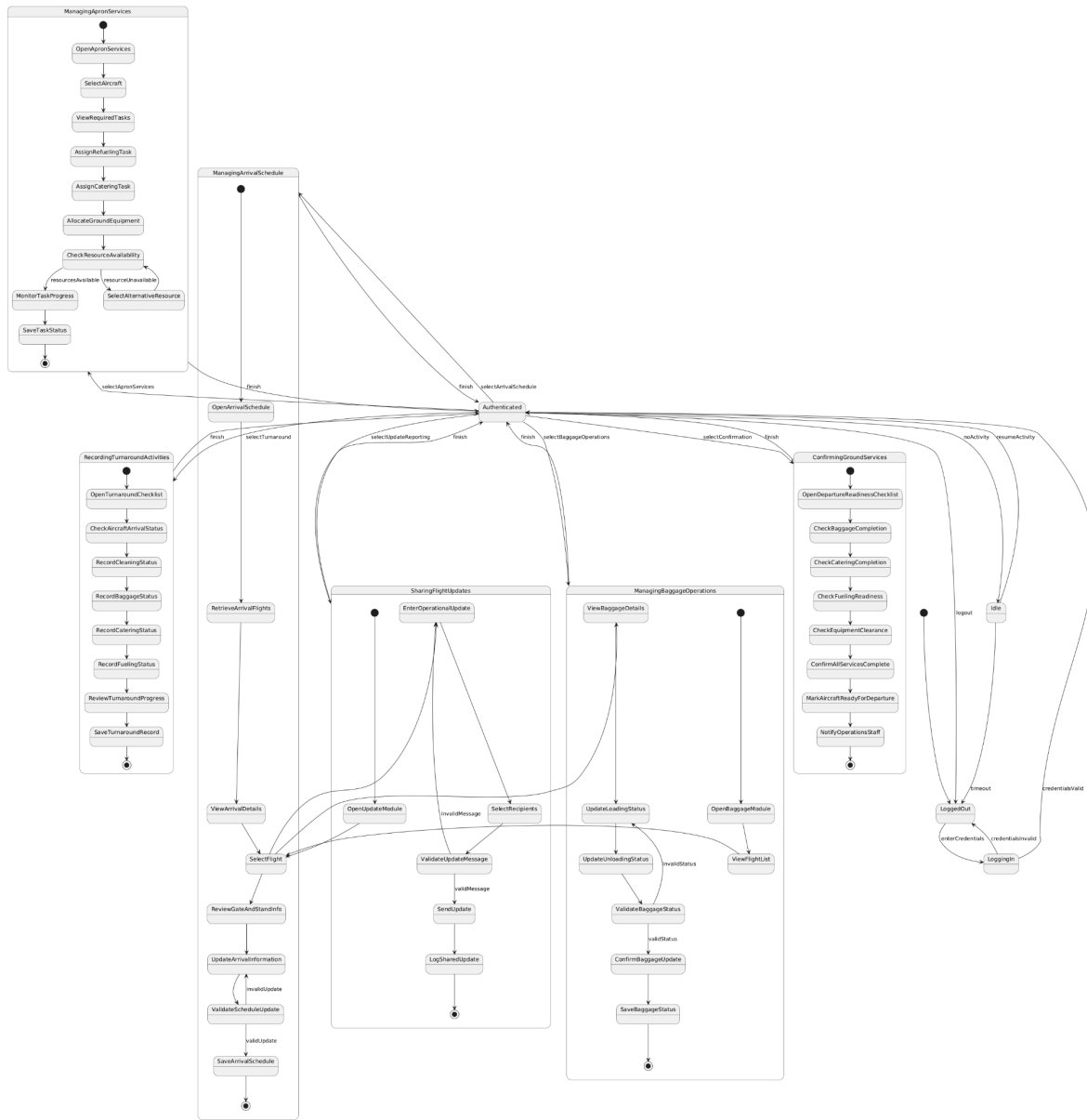


5.5.2 Operations Staff - *by Moira Lako*

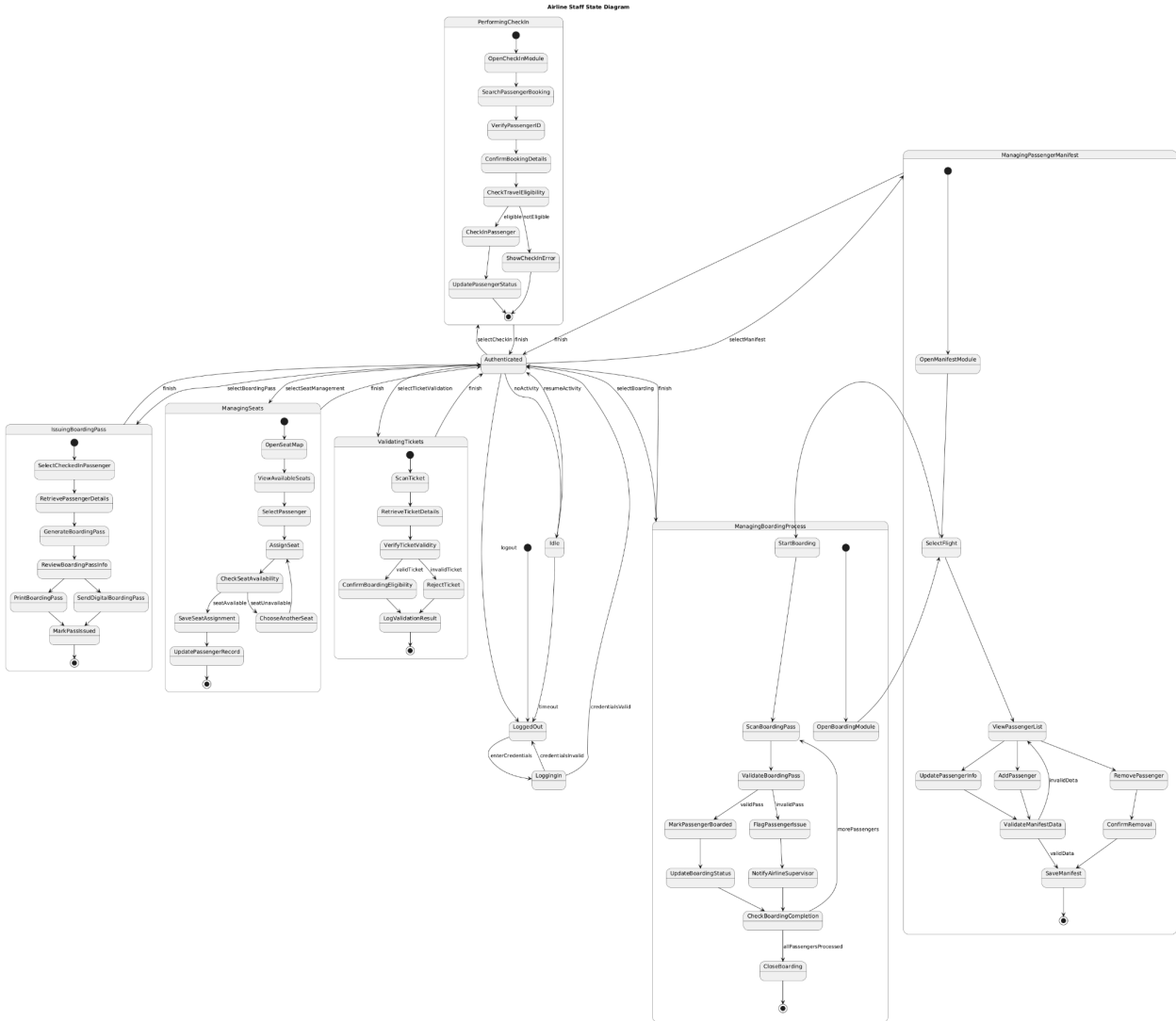


5.5.3 Ground Handling Staff - by Moira Lako

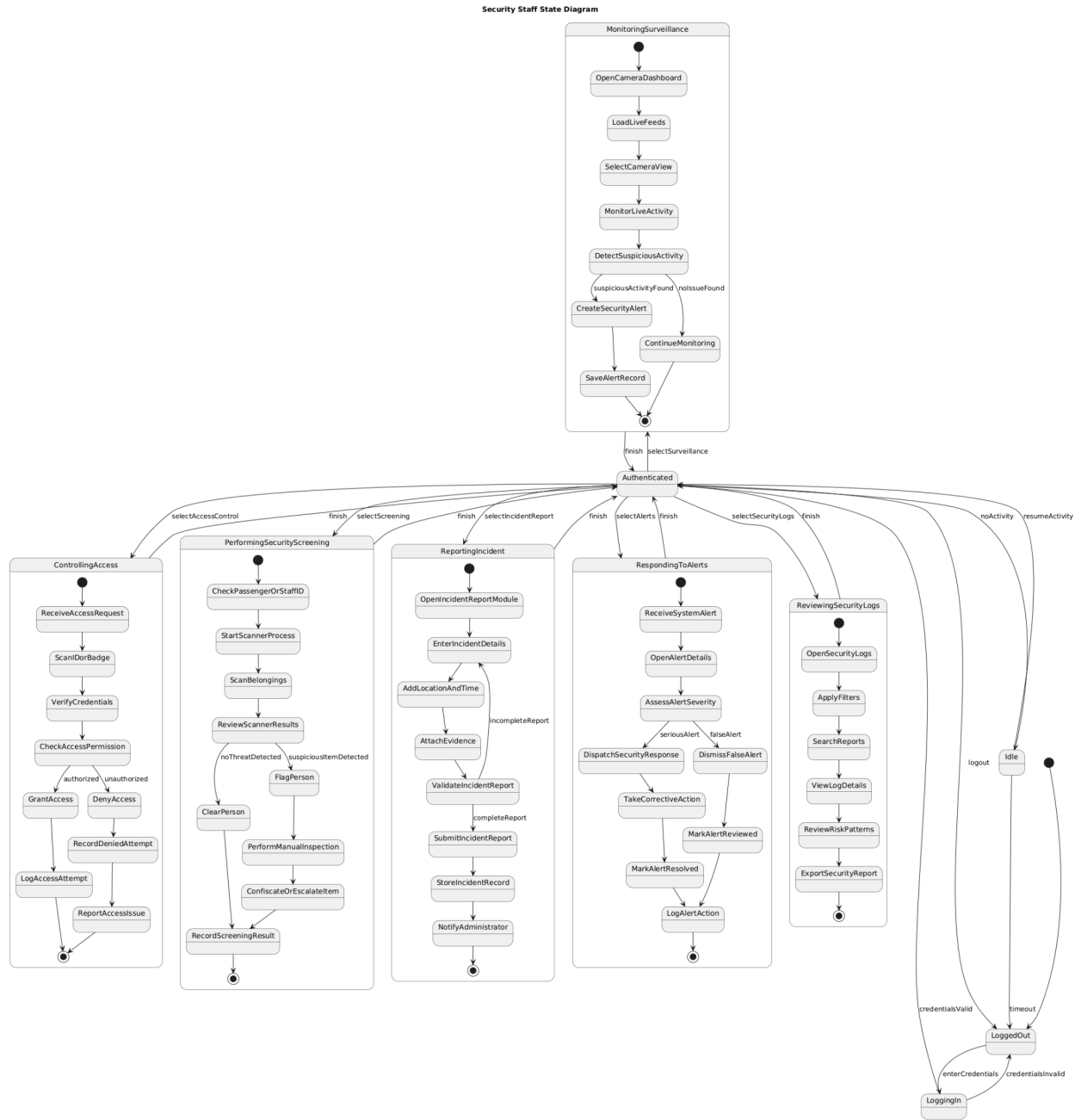
Ground Handling Staff State Diagram



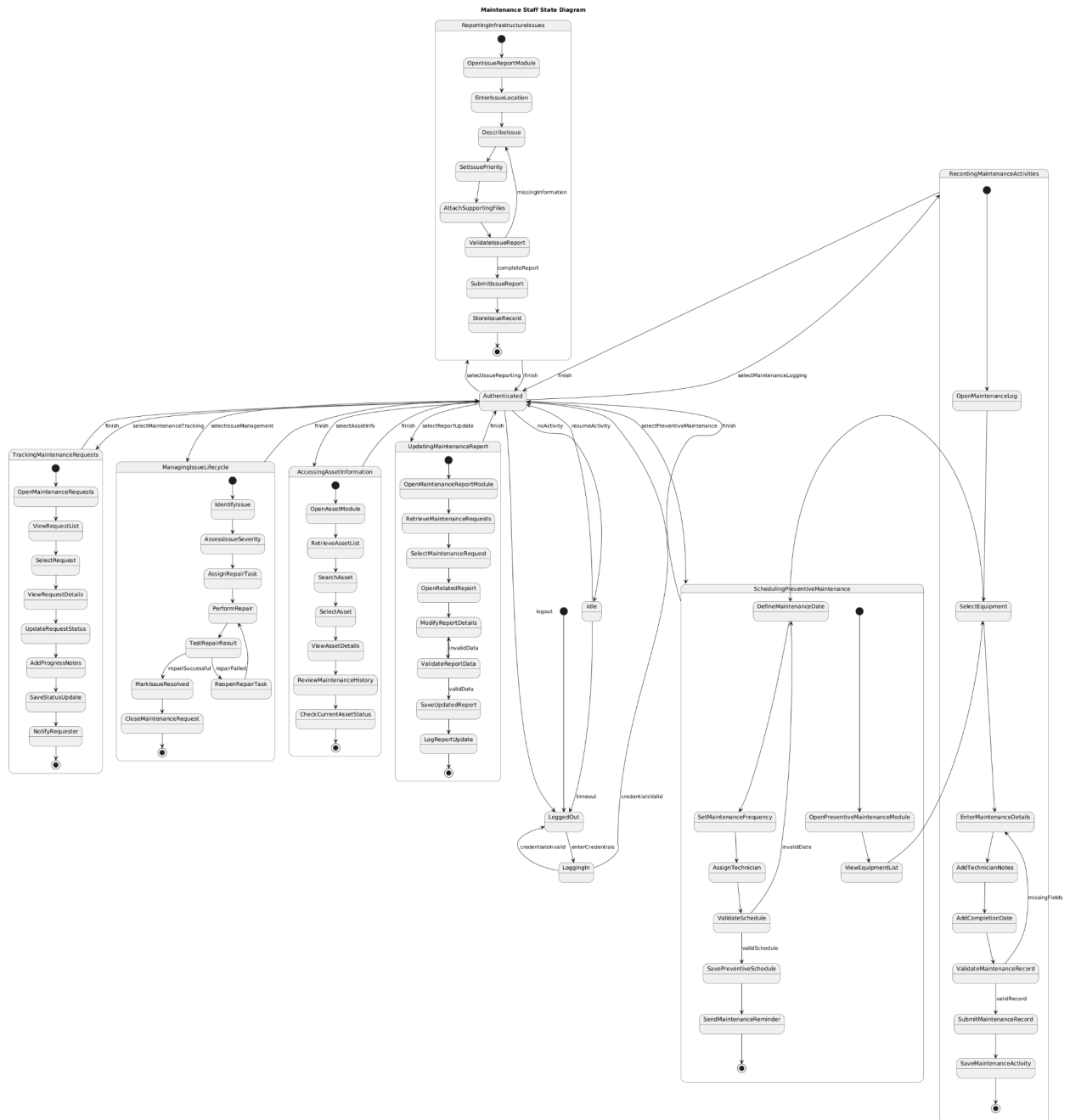
5.5.4 Airline Staff - *by Moira Lako*



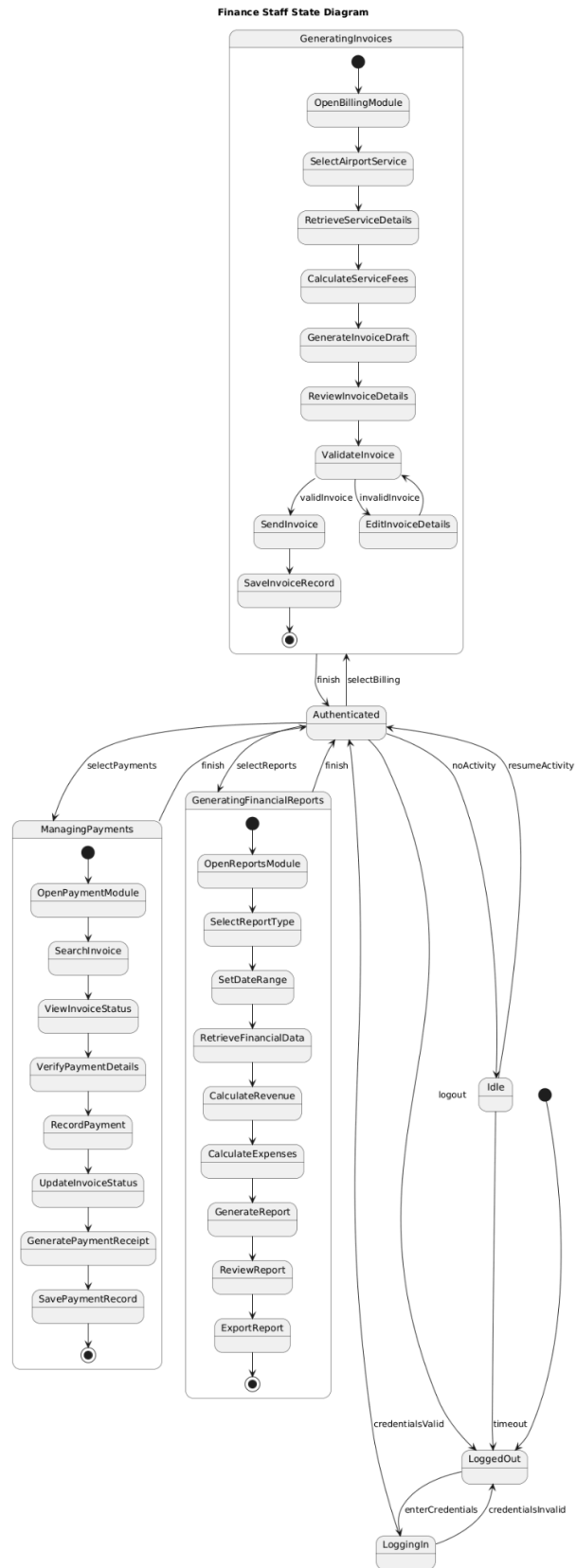
5.5.5 Security Staff - *by Moira Lako*



5.5.6 Maintenance Staff - *by Moira Lako*

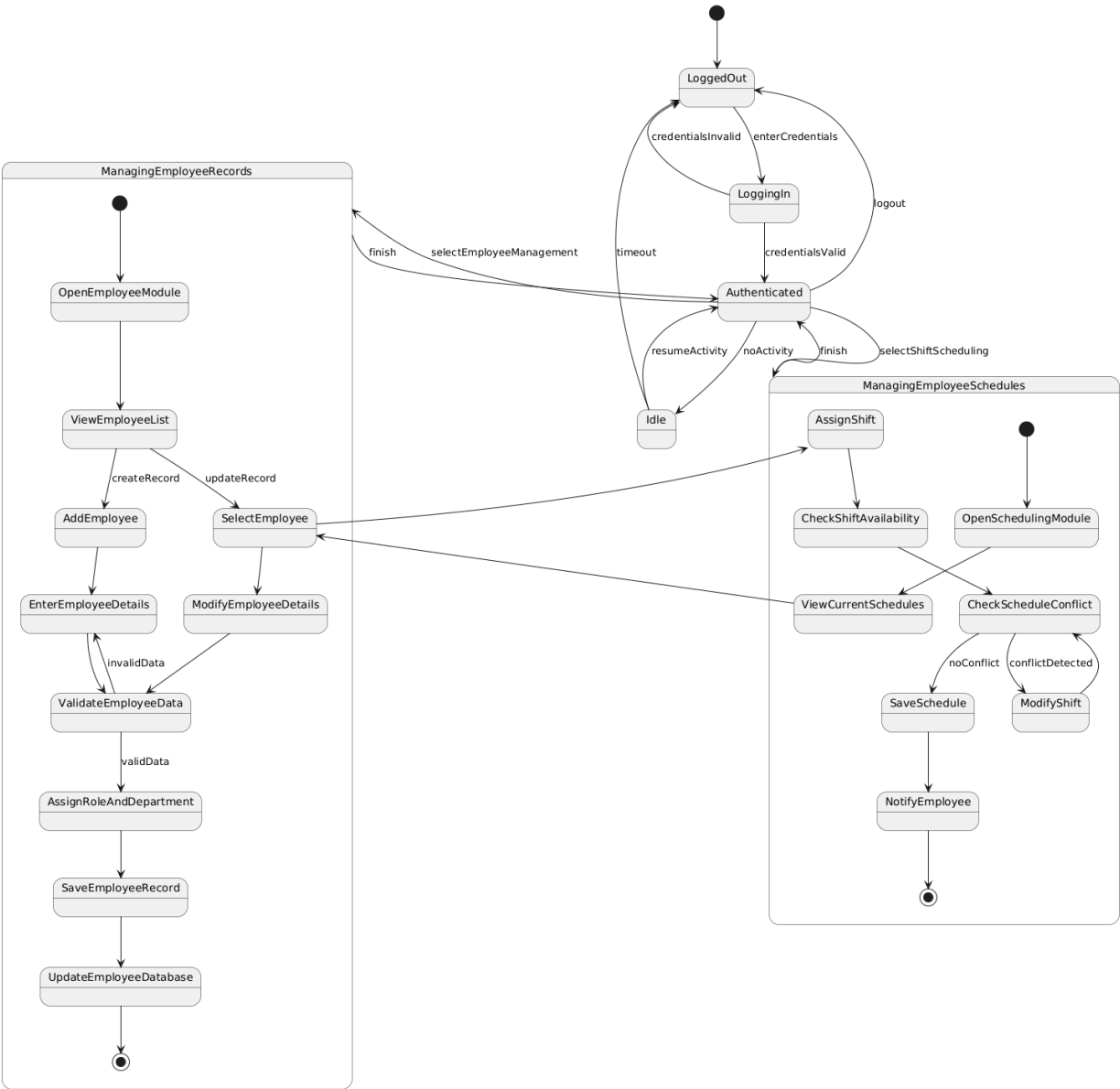


5.5.7 Finance Staff - by Moira Lako



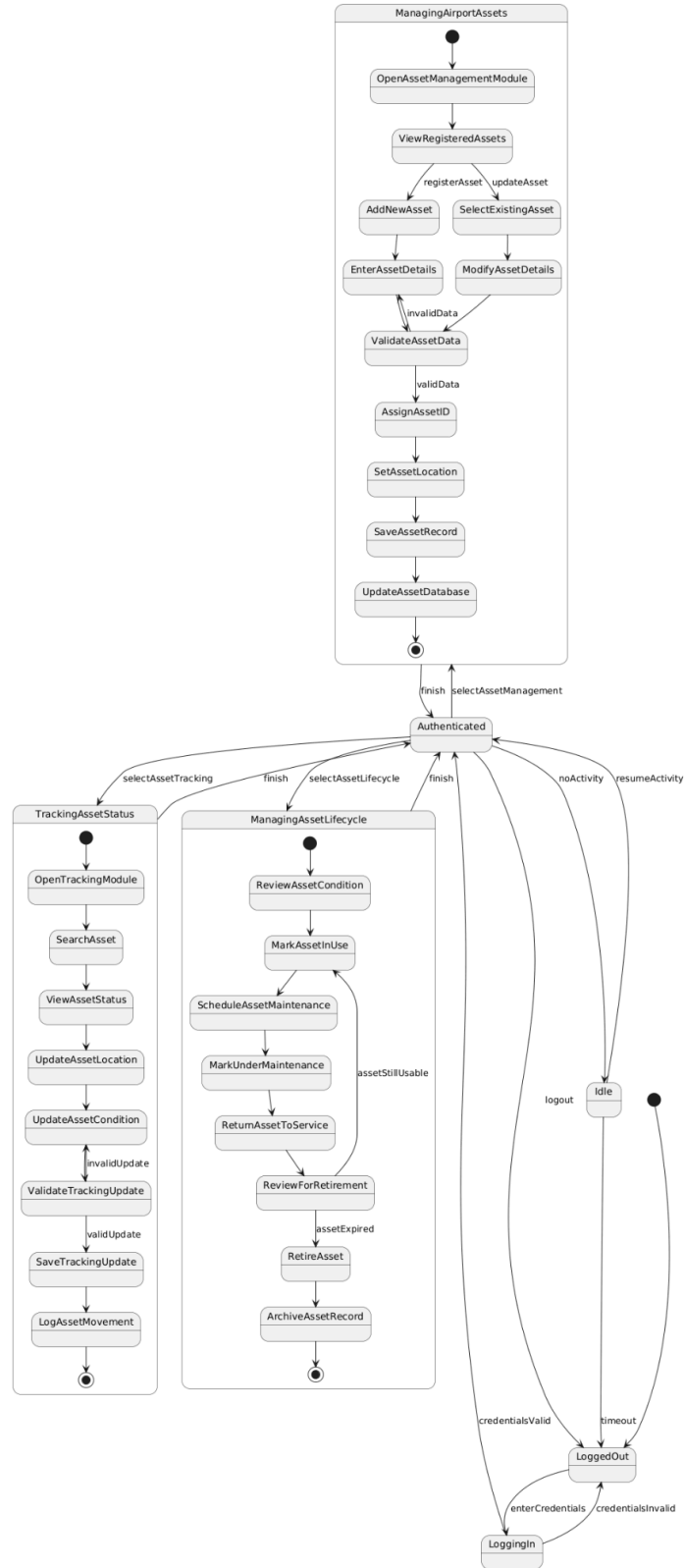
5.5.8 HR Staff - by Moira Lako

HR Staff State Diagram



5.5.9 Asset Management Staff - *by Moira Lako*

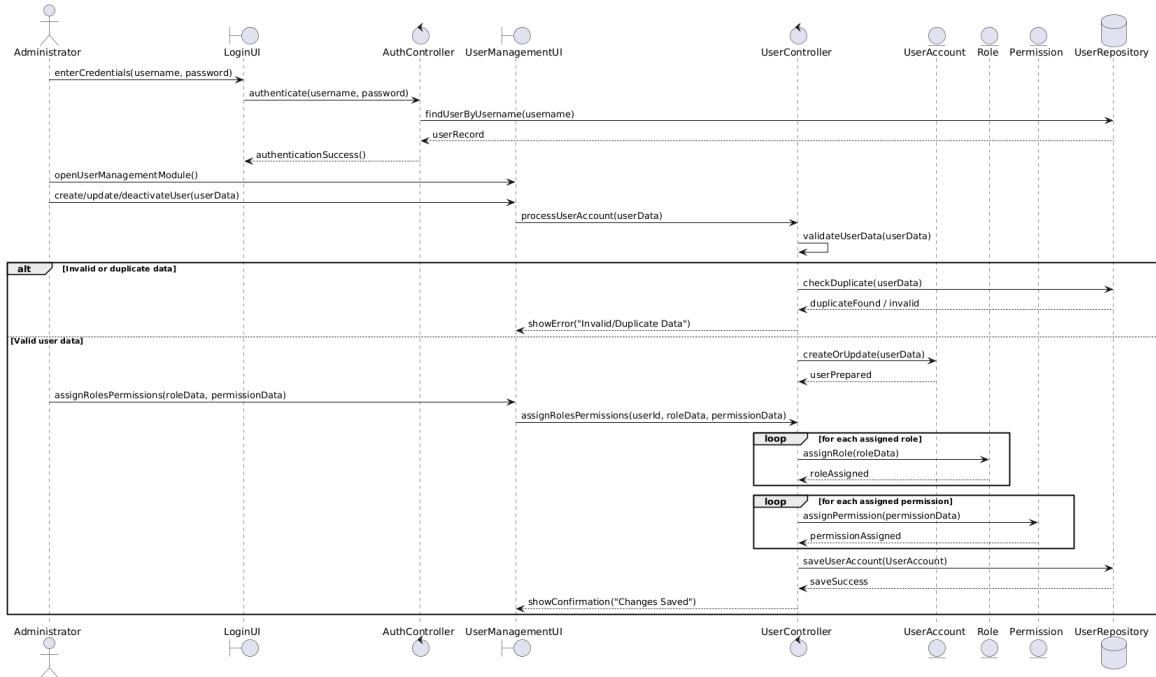
Asset Management Staff State Diagram



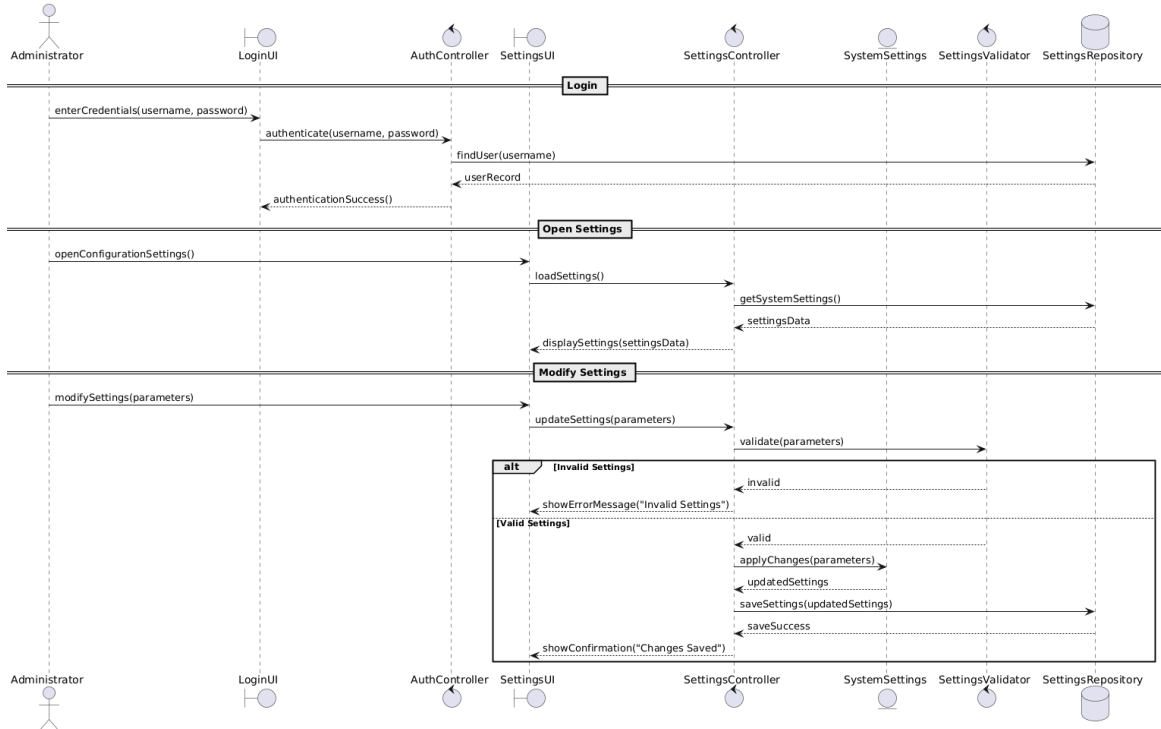
5.6 Sequence Diagrams

5.6.1 Sequence Diagrams UC 01-05 -> Moira Lako

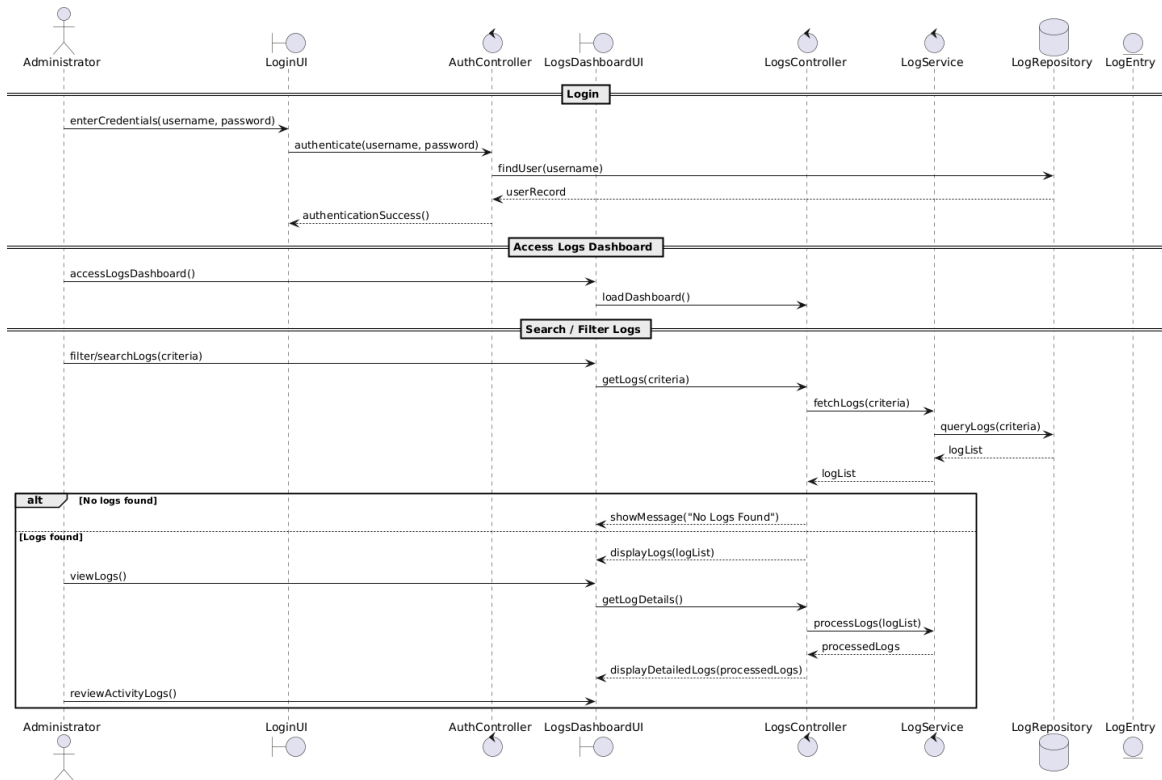
UC-01: Manage User Accounts



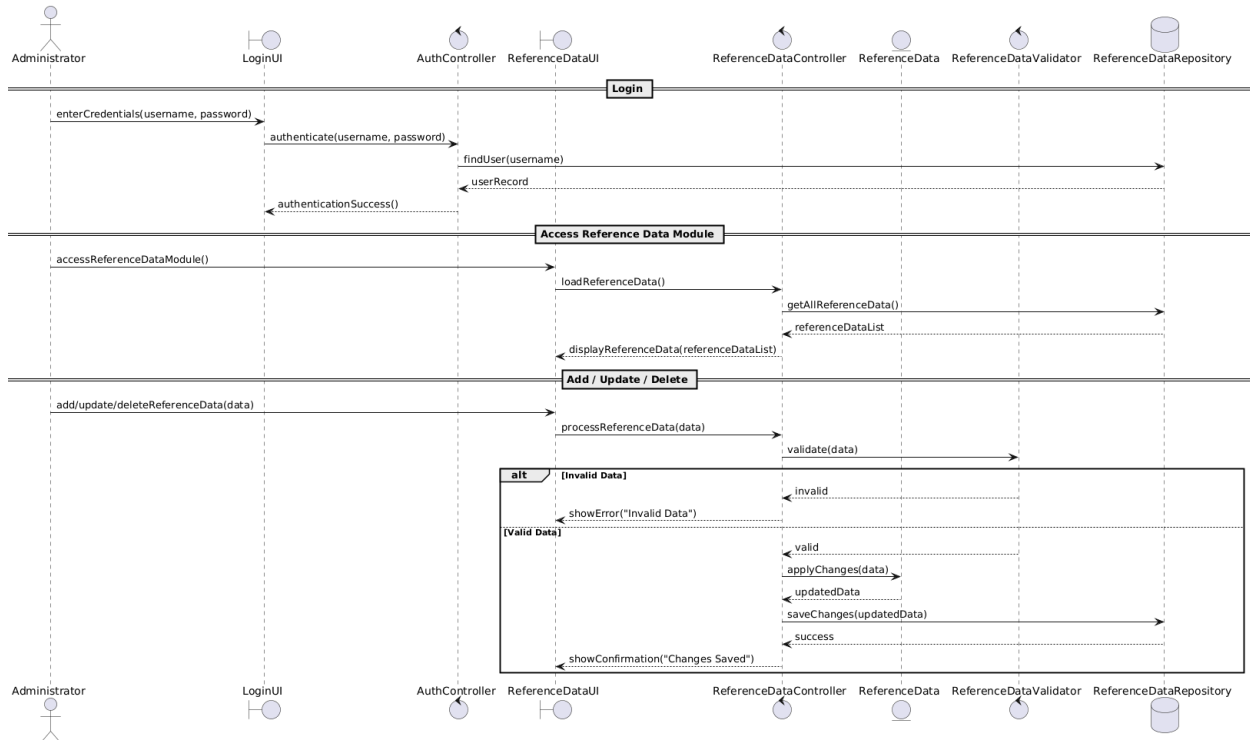
UC-02: Configure System Settings



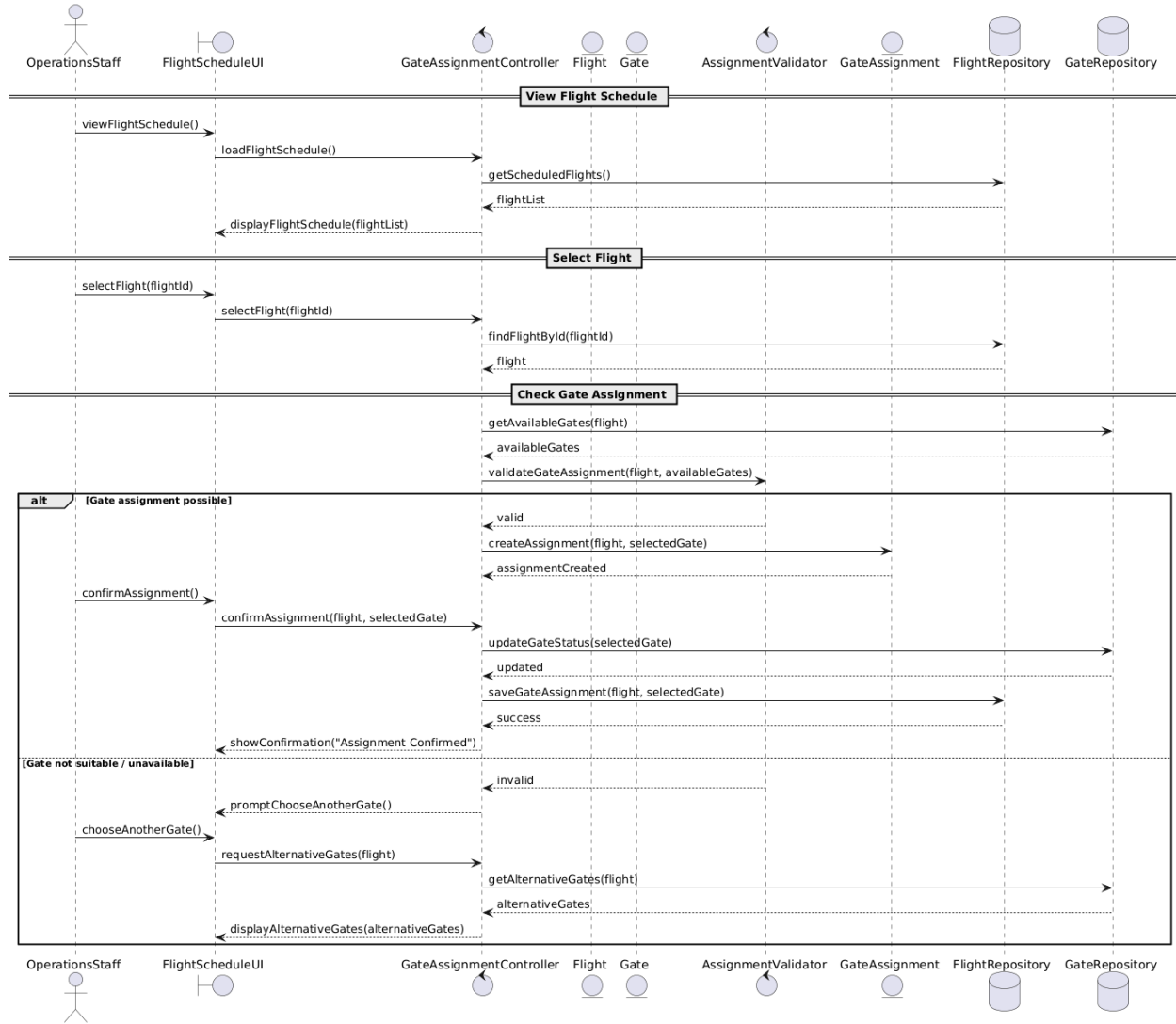
UC-03: Monitor System Logs



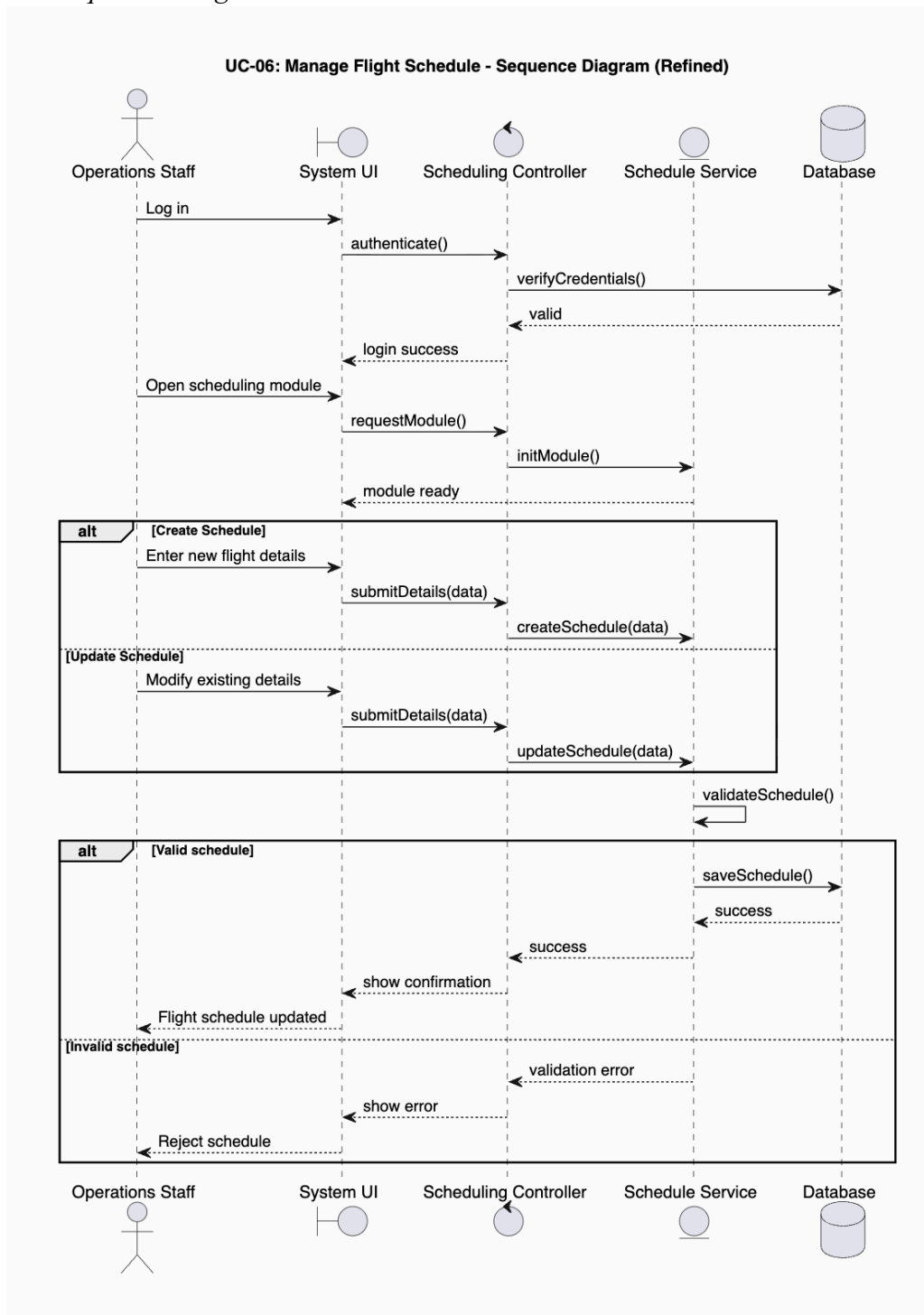
UC-04: Manage Airport Reference Data



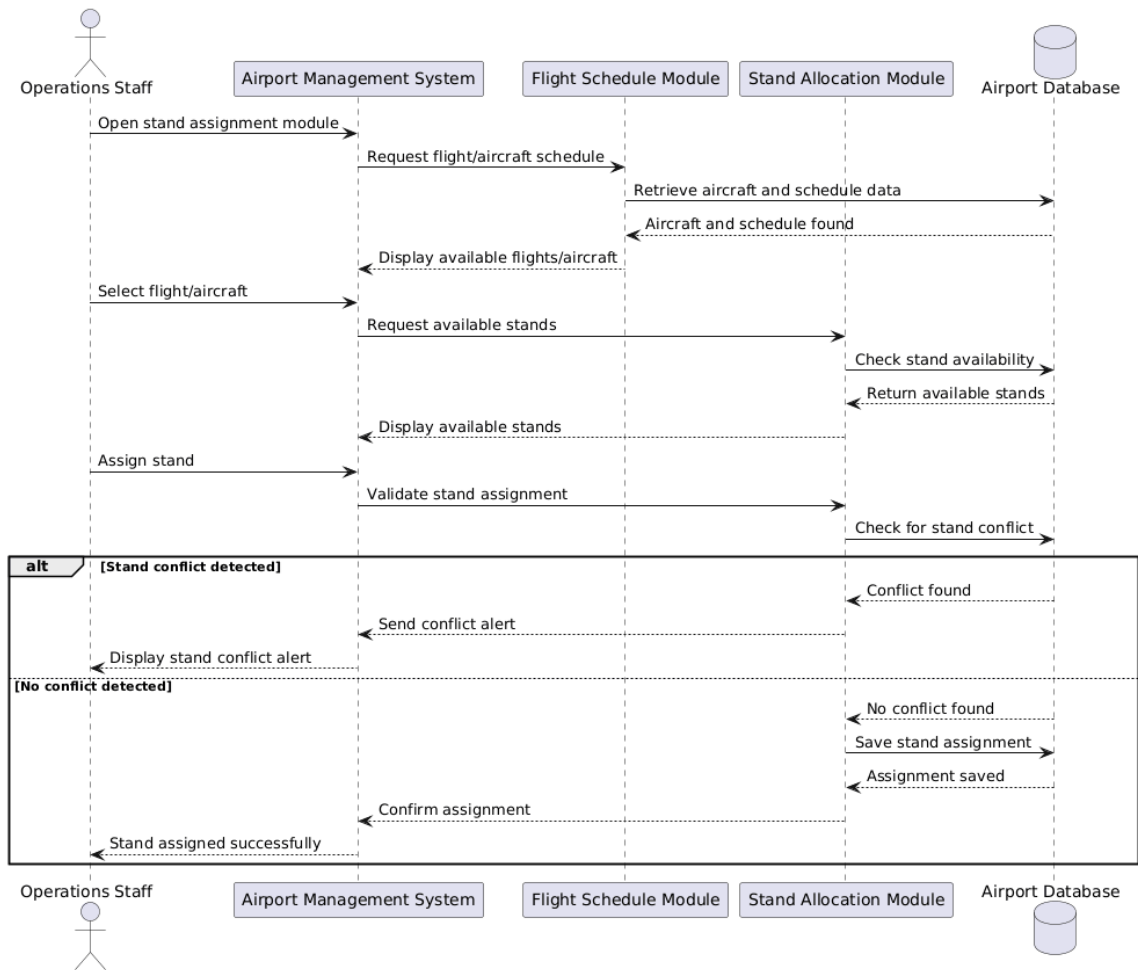
UC-05: Assign Gates to Flights



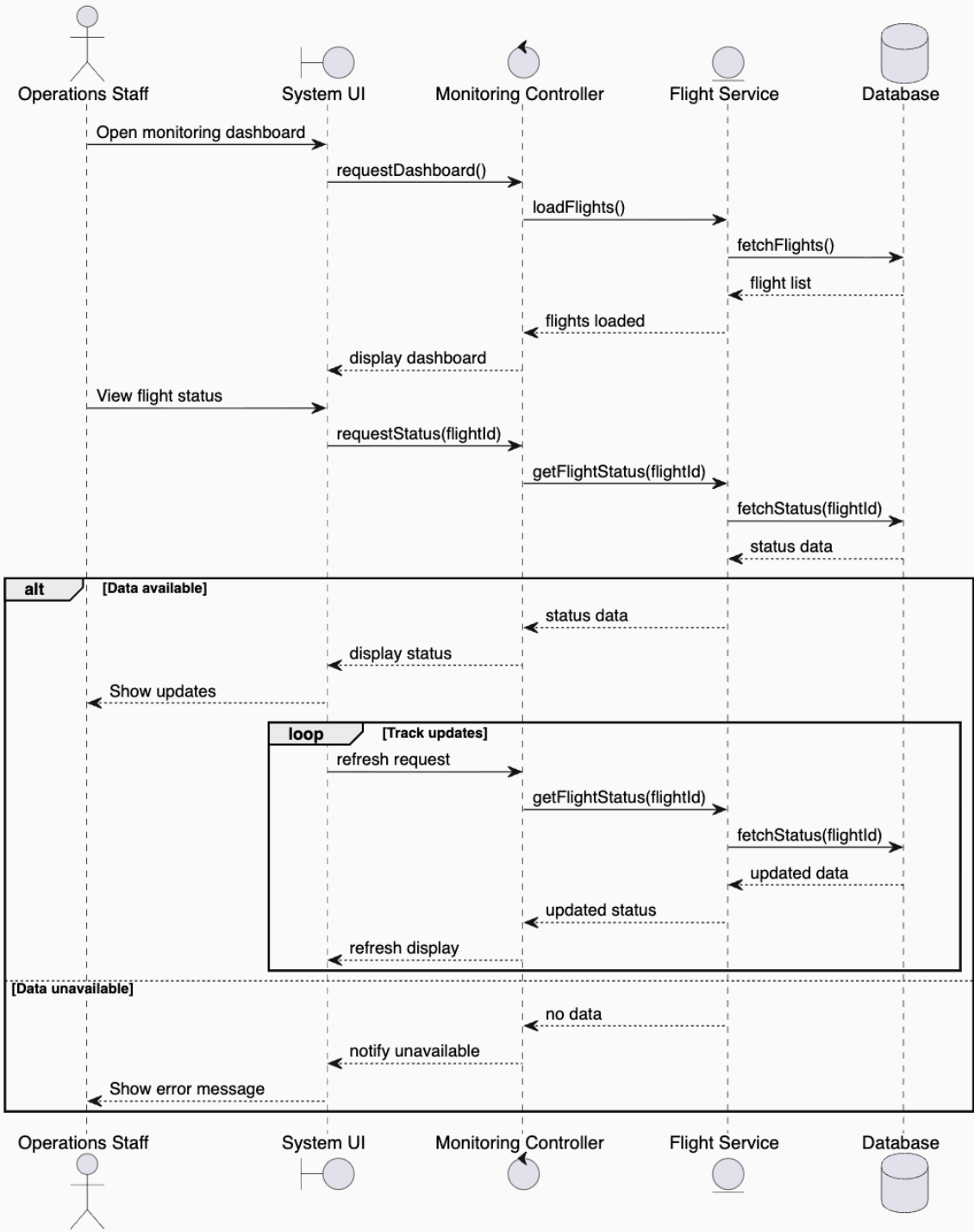
5.6.2 Sequence Diagrams UC 06-10 -> Tea Metani



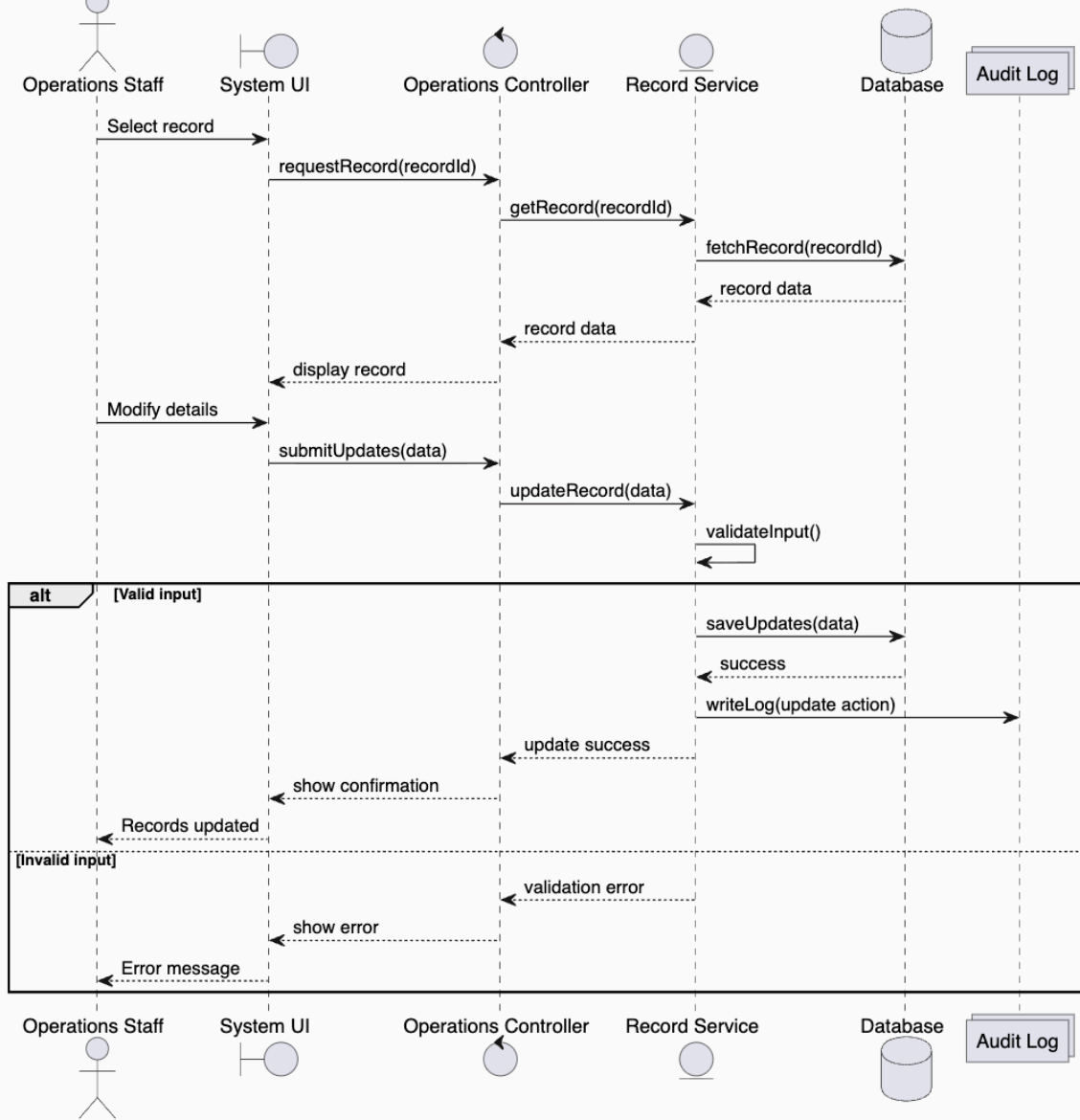
UC-07: Assign Aircraft Stands - Sequence Diagram



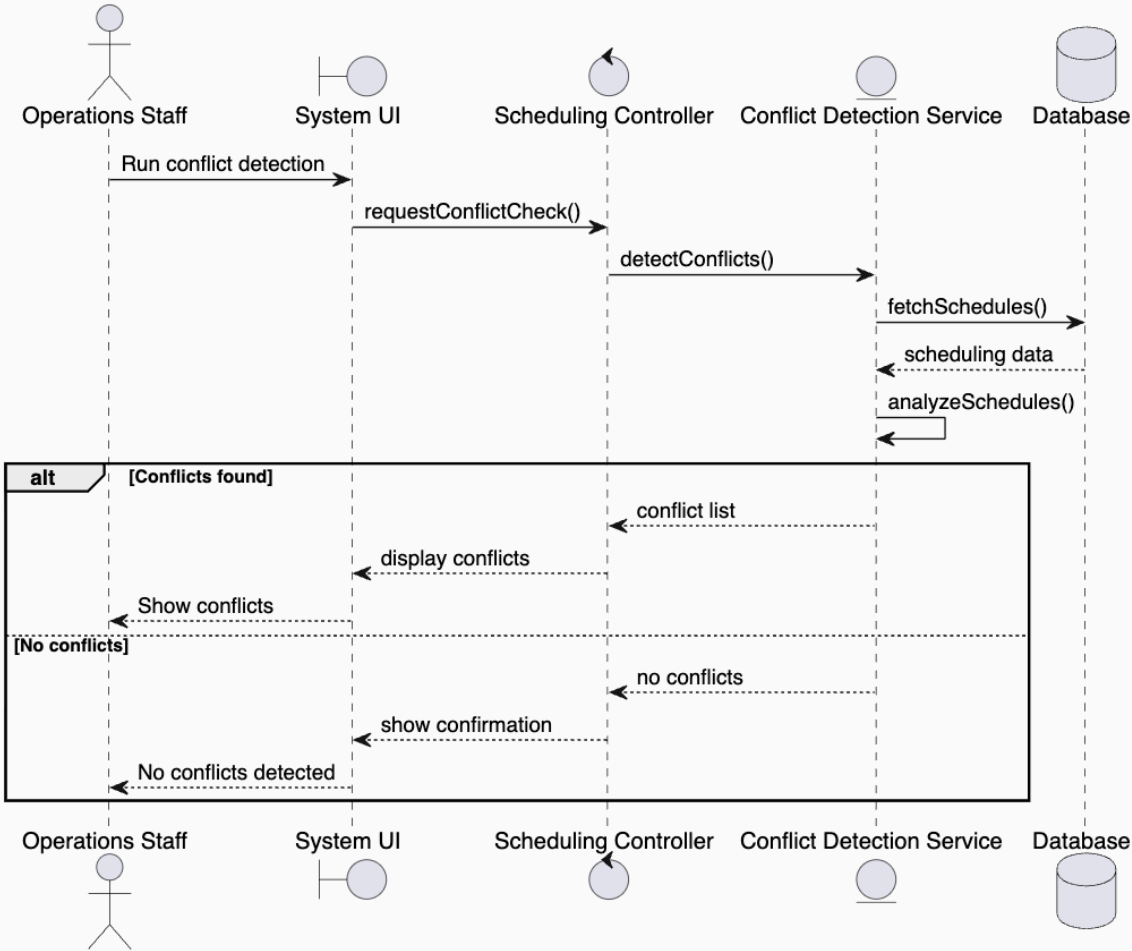
UC-08: Monitor Flight Status - Sequence Diagram



UC-09: Update Operational Records - Sequence Diagram

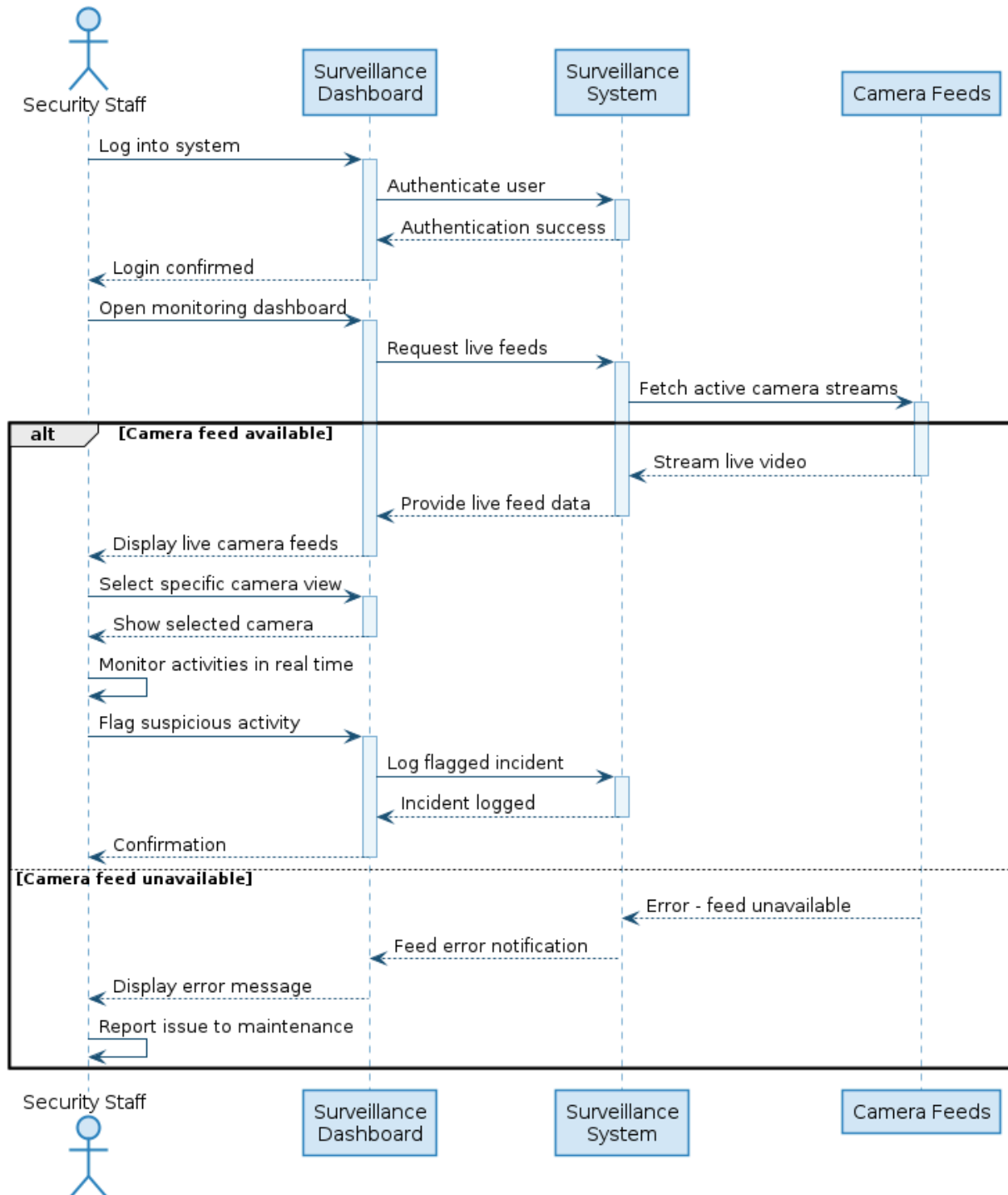


UC-10: Detect Scheduling Conflicts - Sequence Diagram

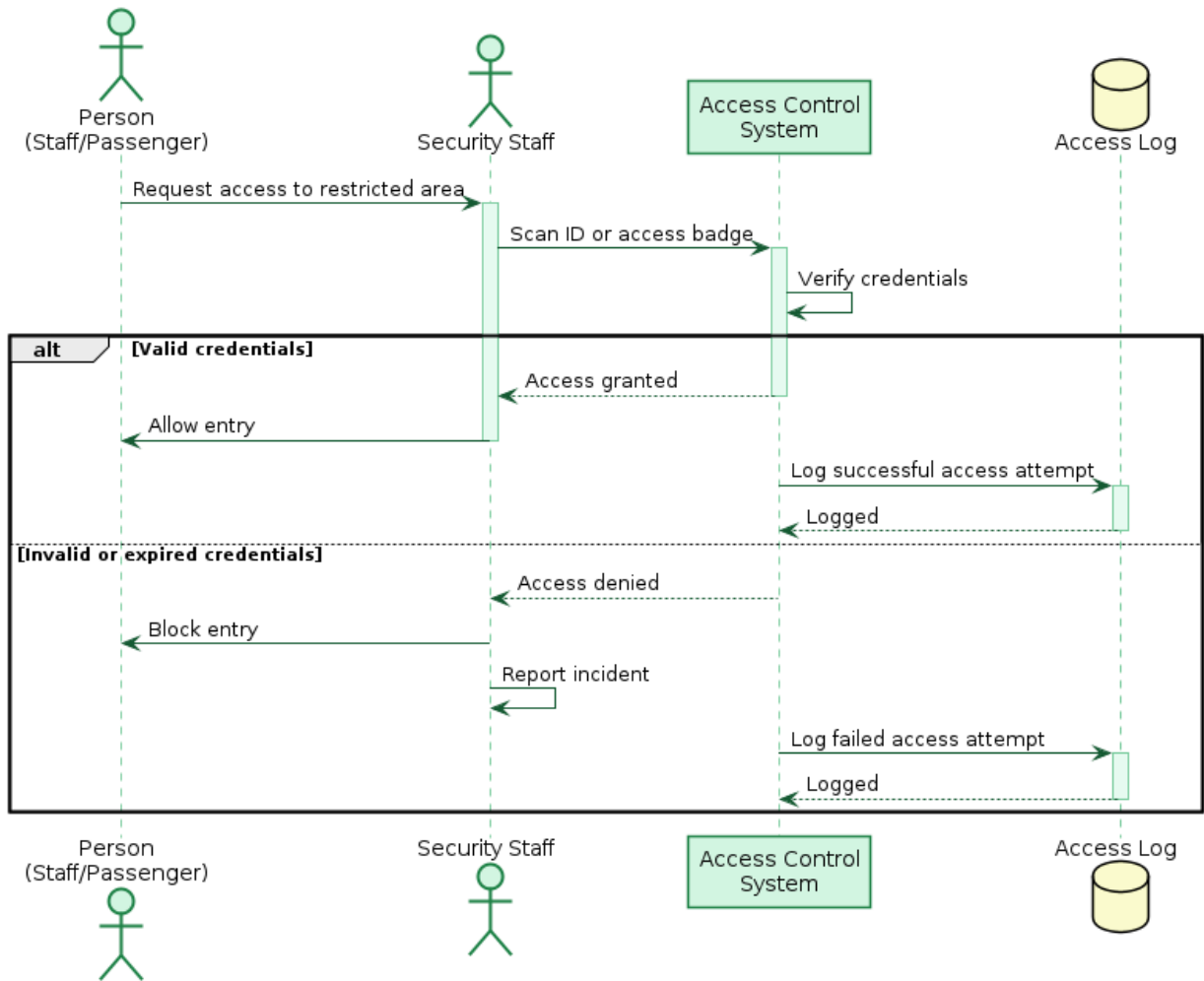


5.6.3 Sequence Diagrams UC 11-15 -> Stelina Spaho

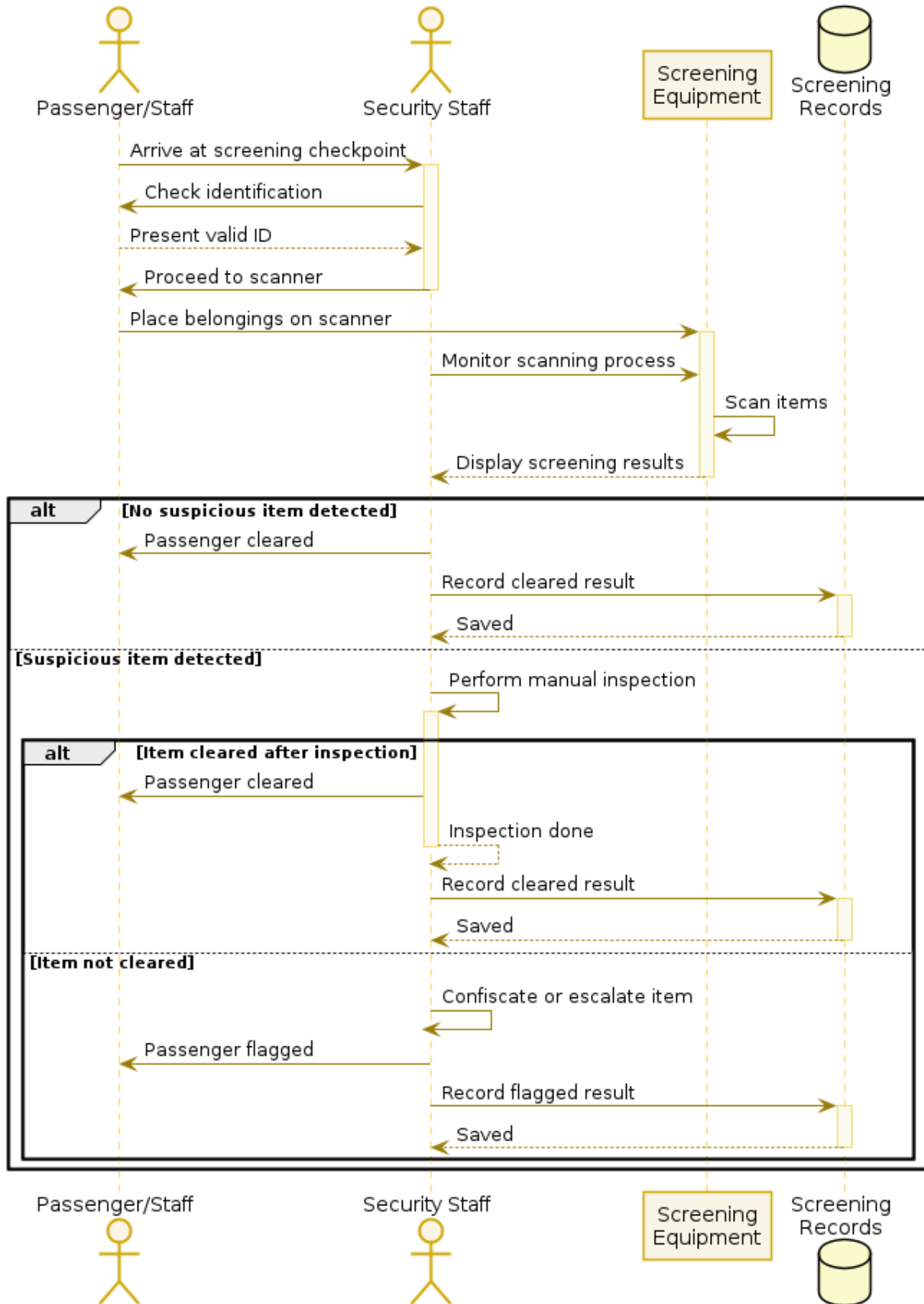
UC-11: Monitor Surveillance Camera — Sequence Diagram



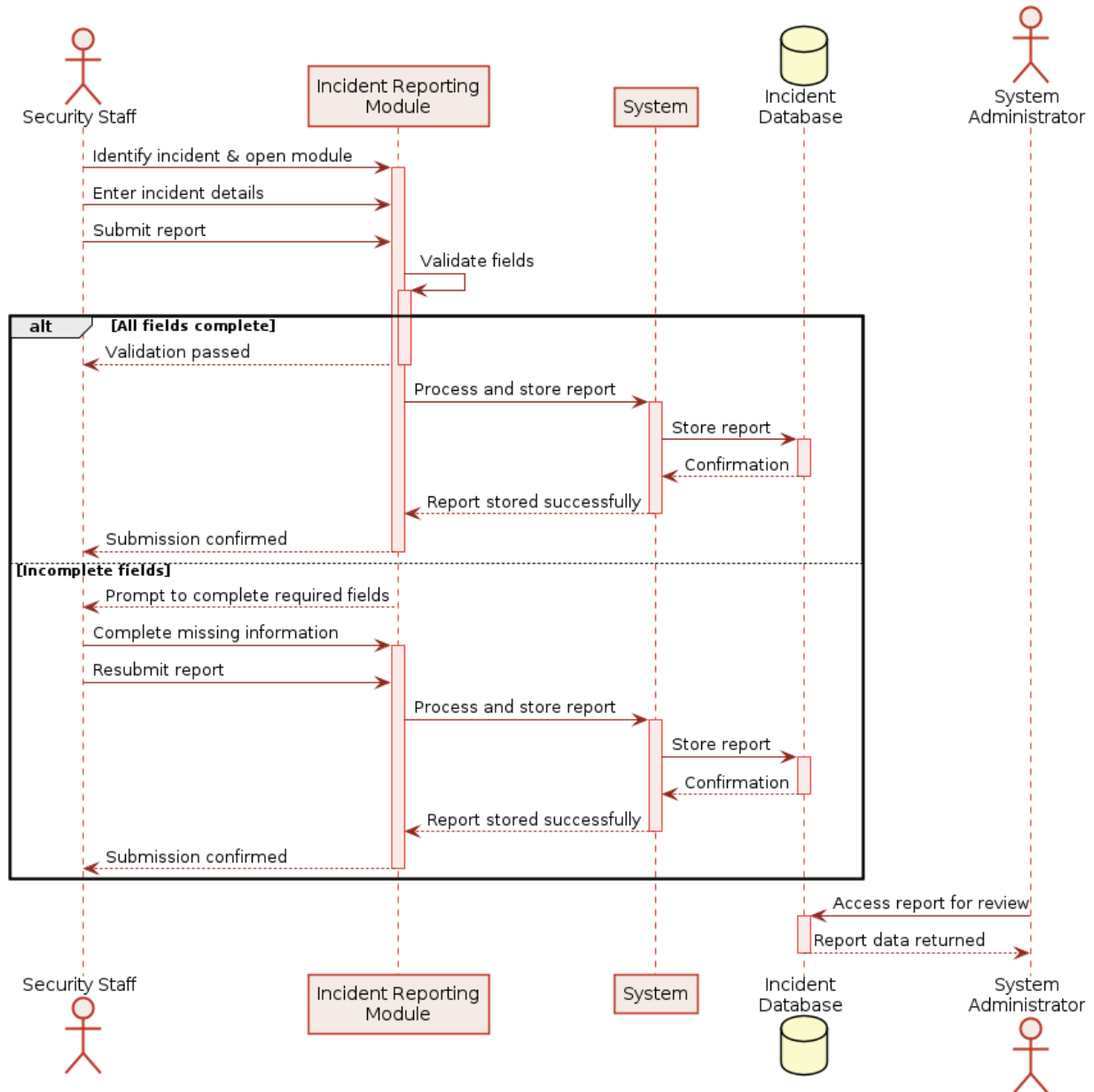
UC-12: Control Access to Restricted Areas — Sequence Diagram



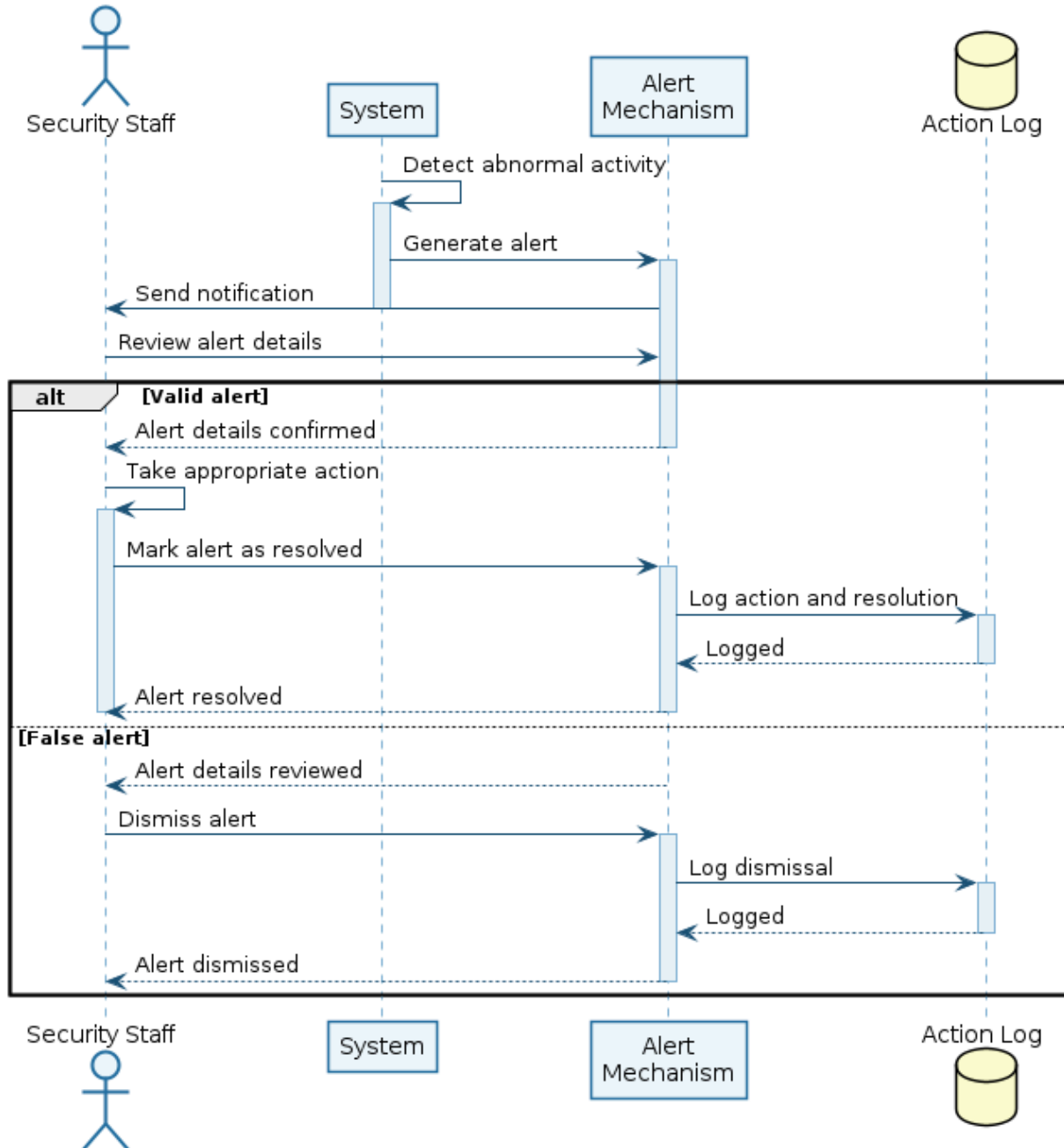
UC-13: Perform Security Screening — Sequence Diagram



UC-14: Record and Report Security Incident — Sequence Diagram

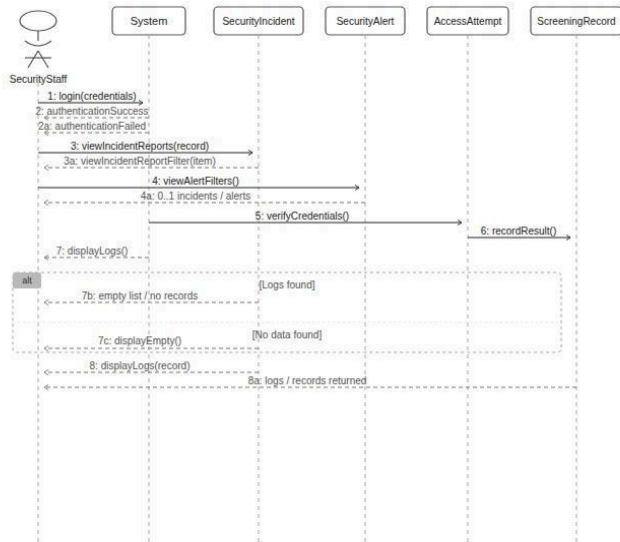


UC-15: Respond to Security Alerts — Sequence Diagram

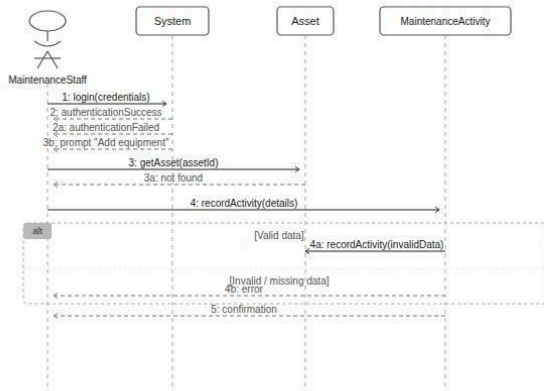


5.6.4 Sequence Diagrams UC 16-20 -> Yeldos Rakhymzhanov

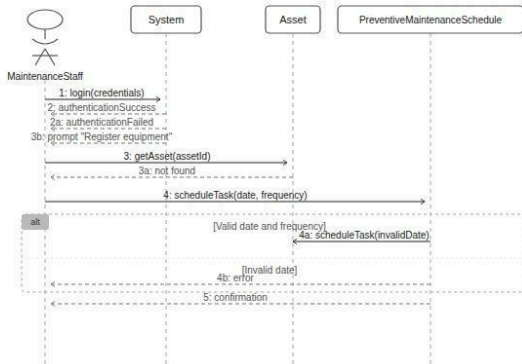
UC-16: Review Security Logs and Reports — Sequence Diagram



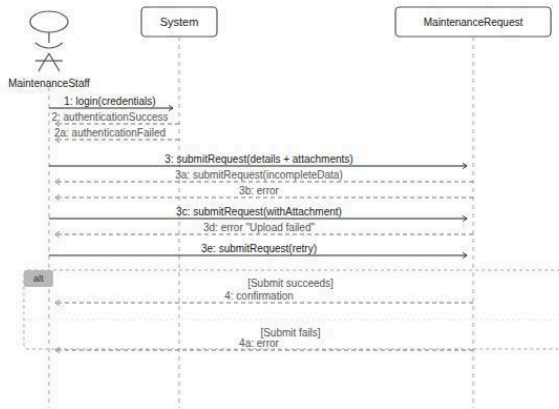
UC-17: Record Maintenance Activities — Sequence Diagram



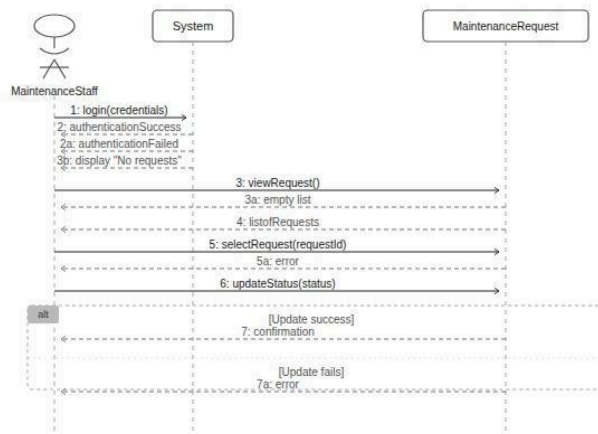
UC-18: Schedule Preventive Maintenance — Sequence Diagram



UC-19: Report Infrastructure Issues — Sequence Diagram

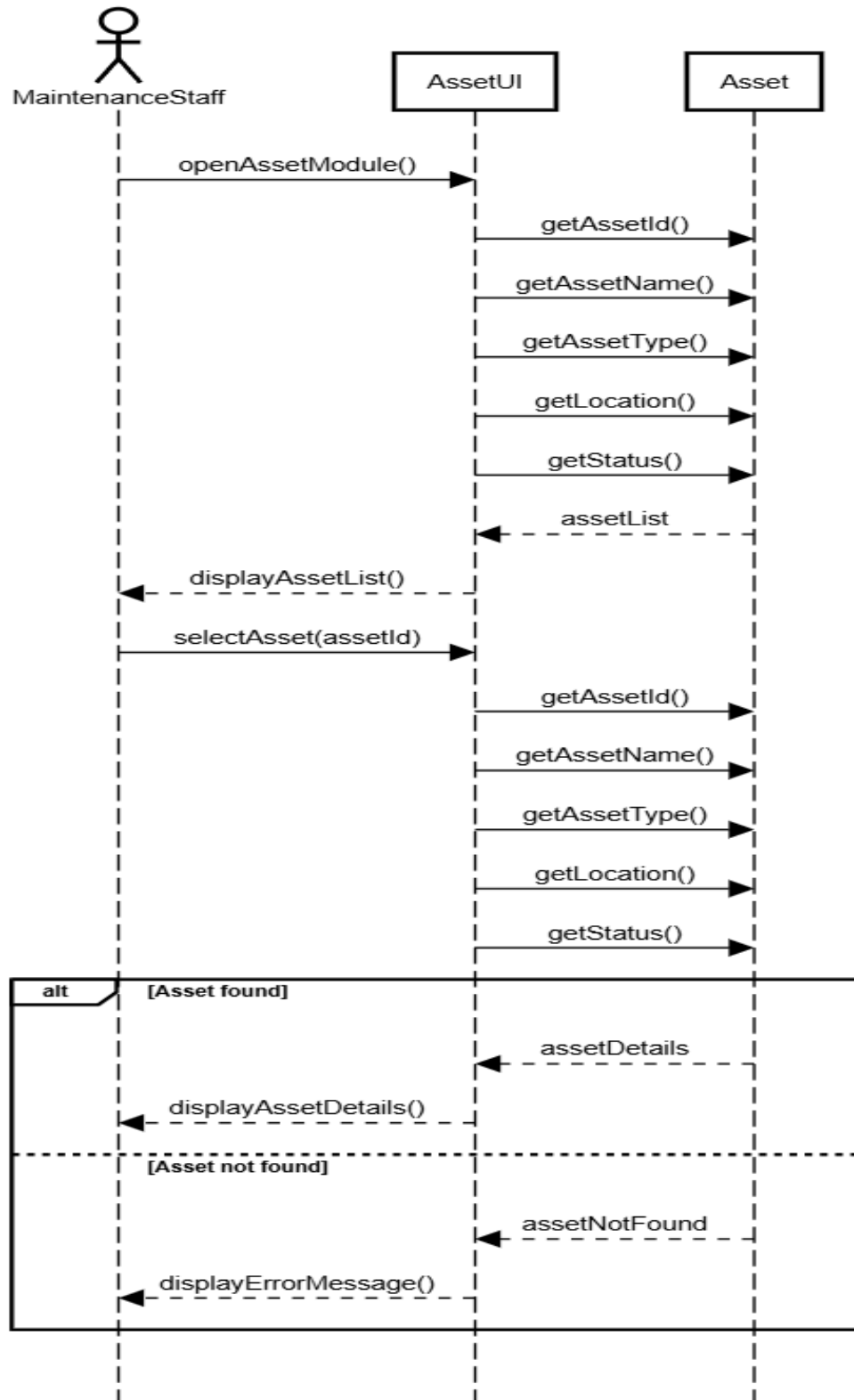


UC-20: Track Maintenance Requests — Sequence Diagram

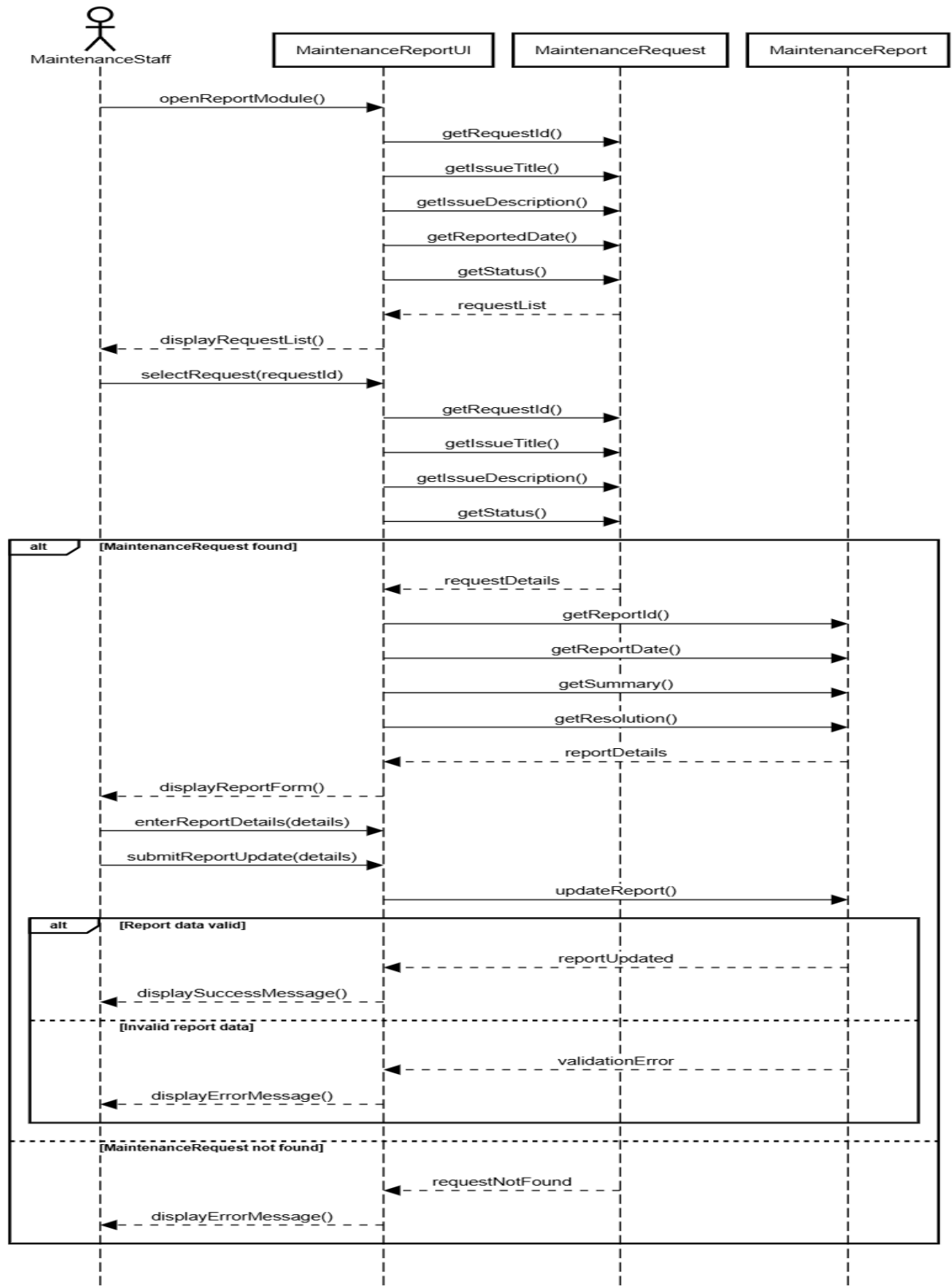


5.6.5 Sequence Diagrams UC 21-25 -> Fatih Gökteke

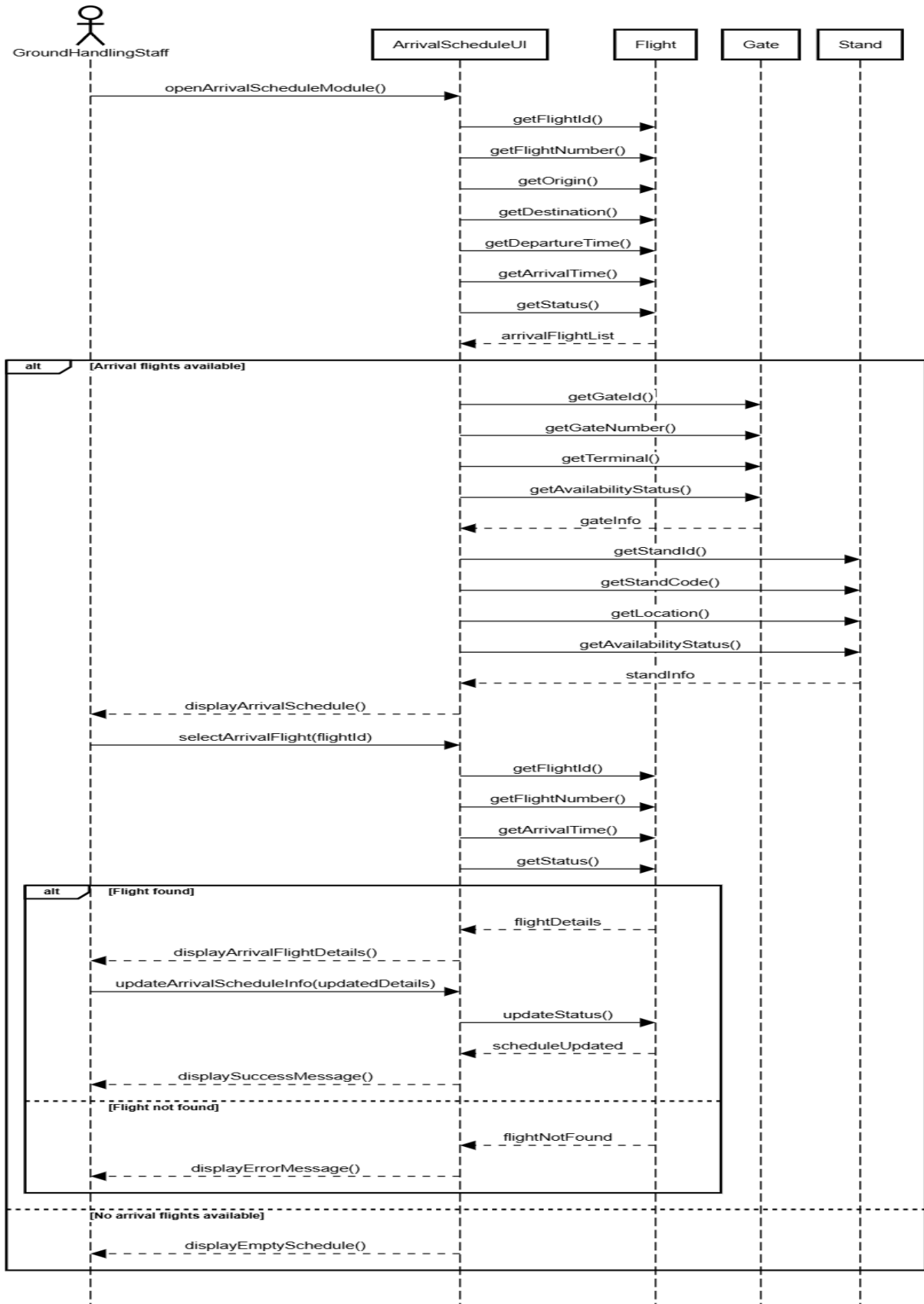
C-21: Access Asset Information - Sequence Diagram



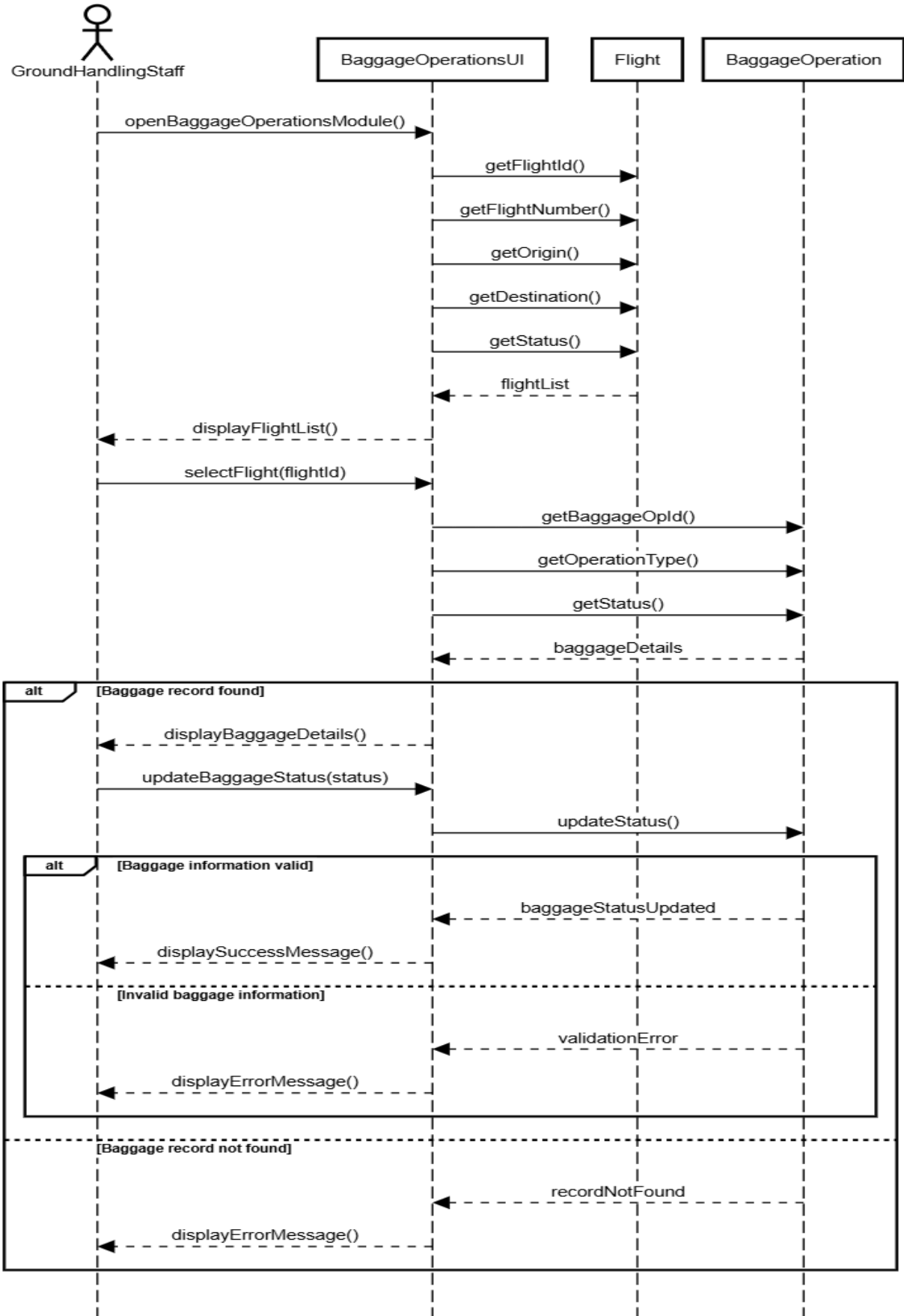
UC-22: Update Maintenance Report - Sequence Diagram



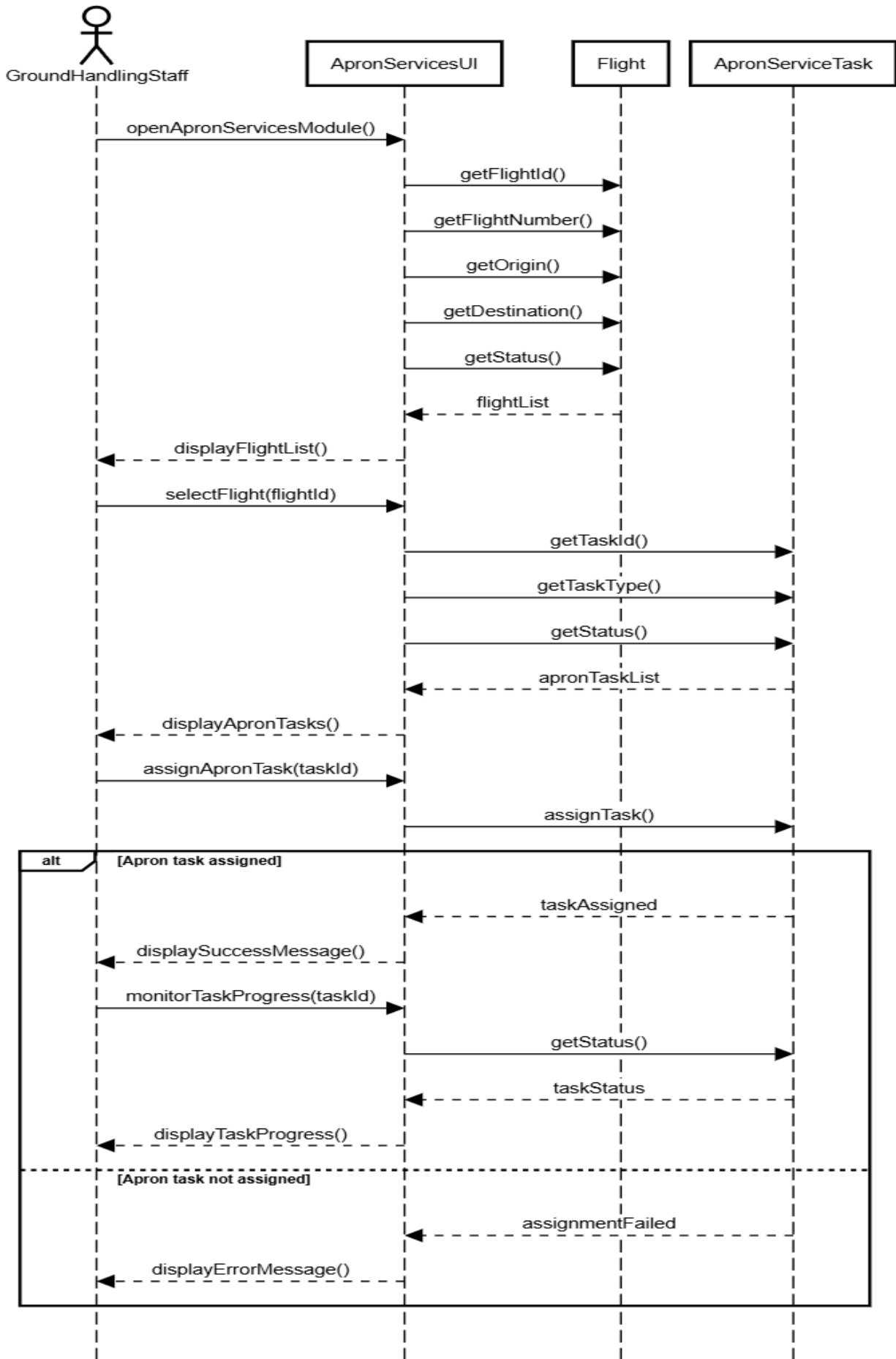
UC-23: Manage Arrival Schedule - Sequence Diagram



UC-24: Manage Baggage Operations - Sequence Diagram

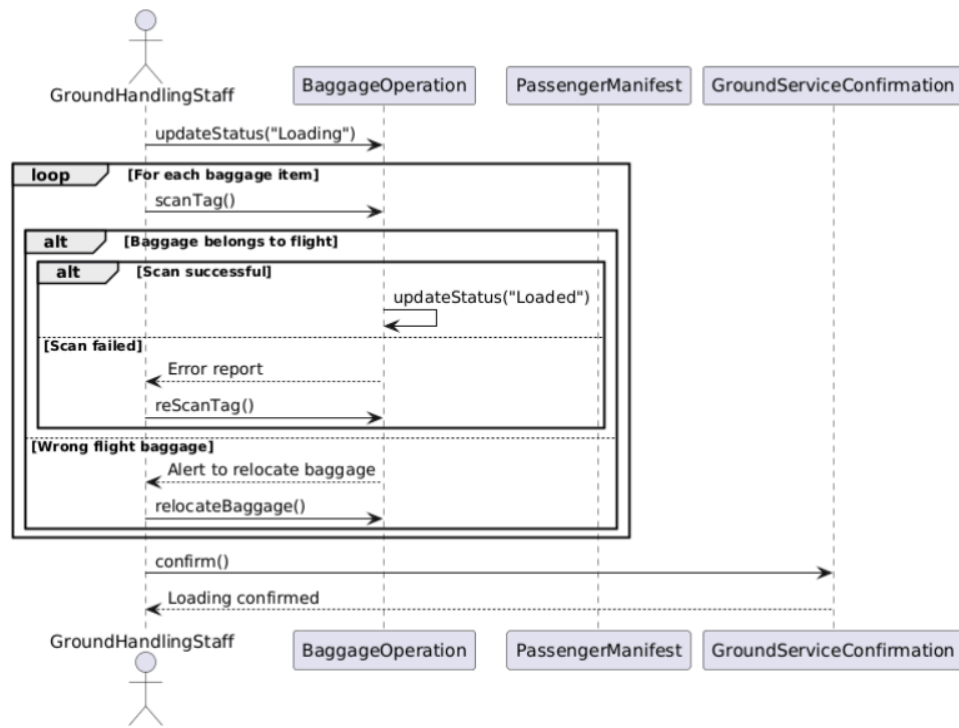


UC-25: Manage Apron Services - Sequence Diagram

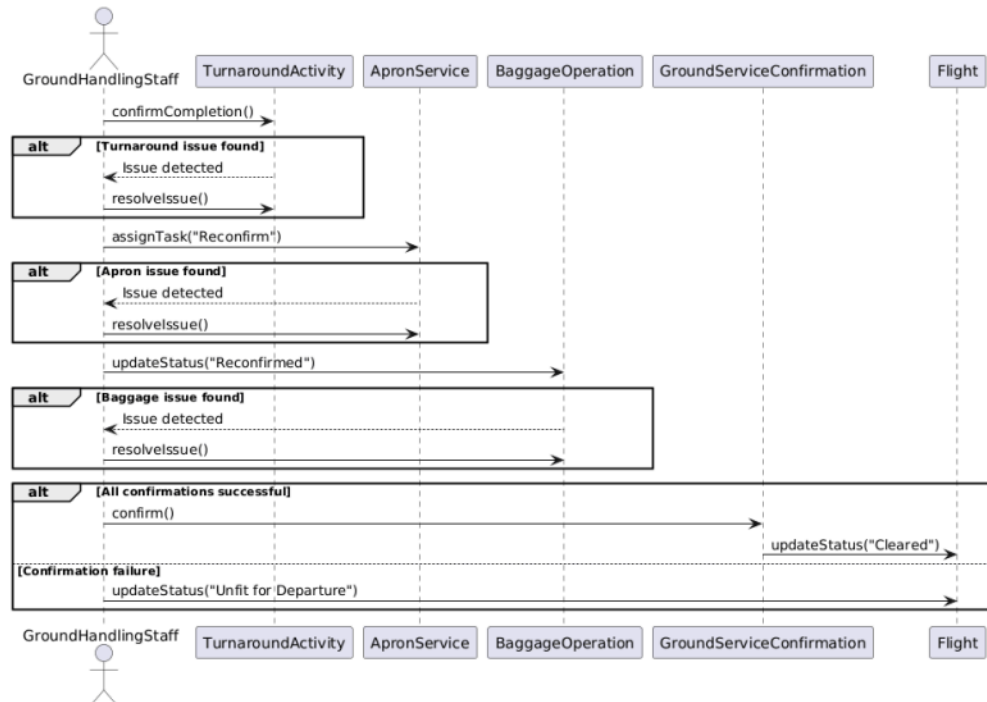


5.6.6 Sequence Diagrams UC 26-30 -> Leart Mukaj

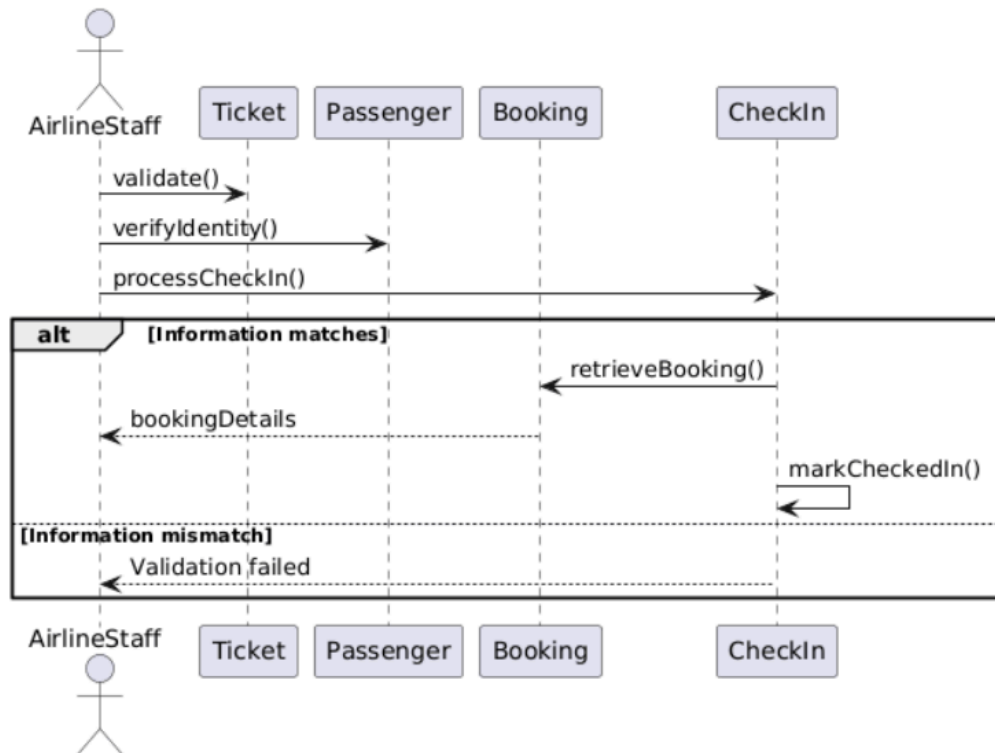
UC-26 Baggage Loading - Sequence Diagram



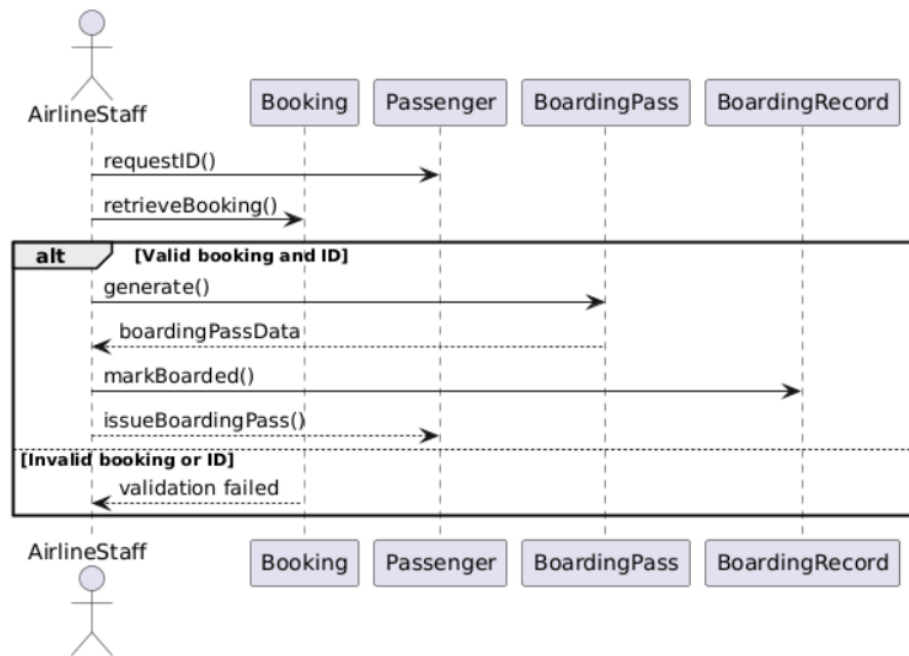
UC-27 Final Confirmation - Sequence Diagram



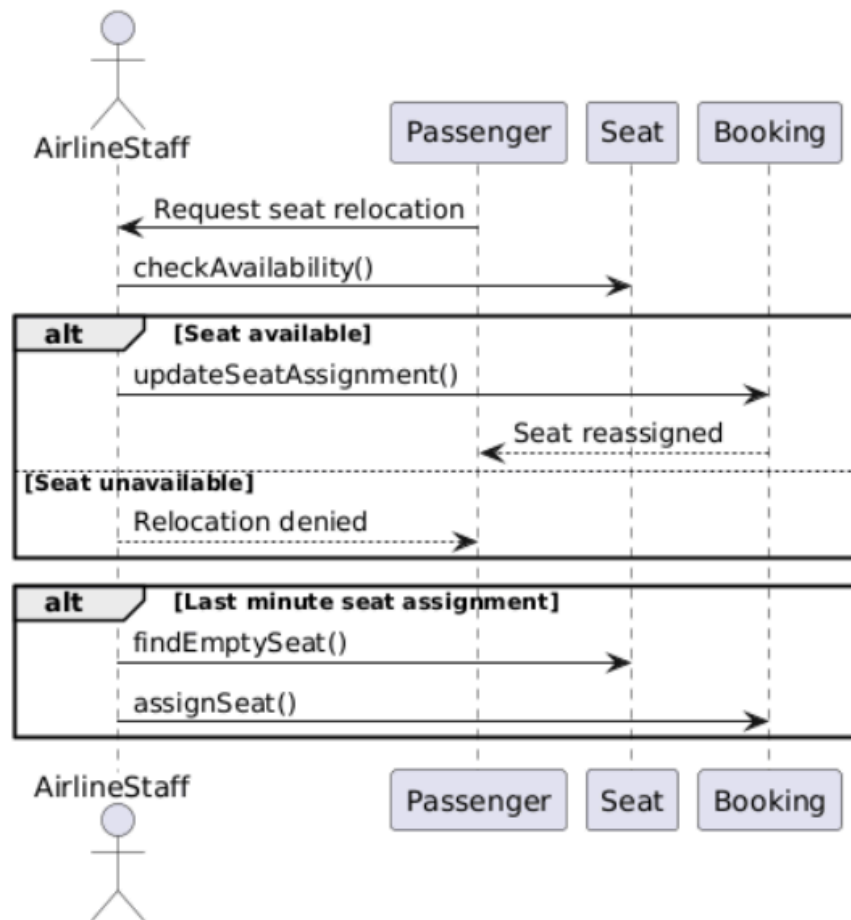
UC-28 Passenger Check-in - Sequence Diagram



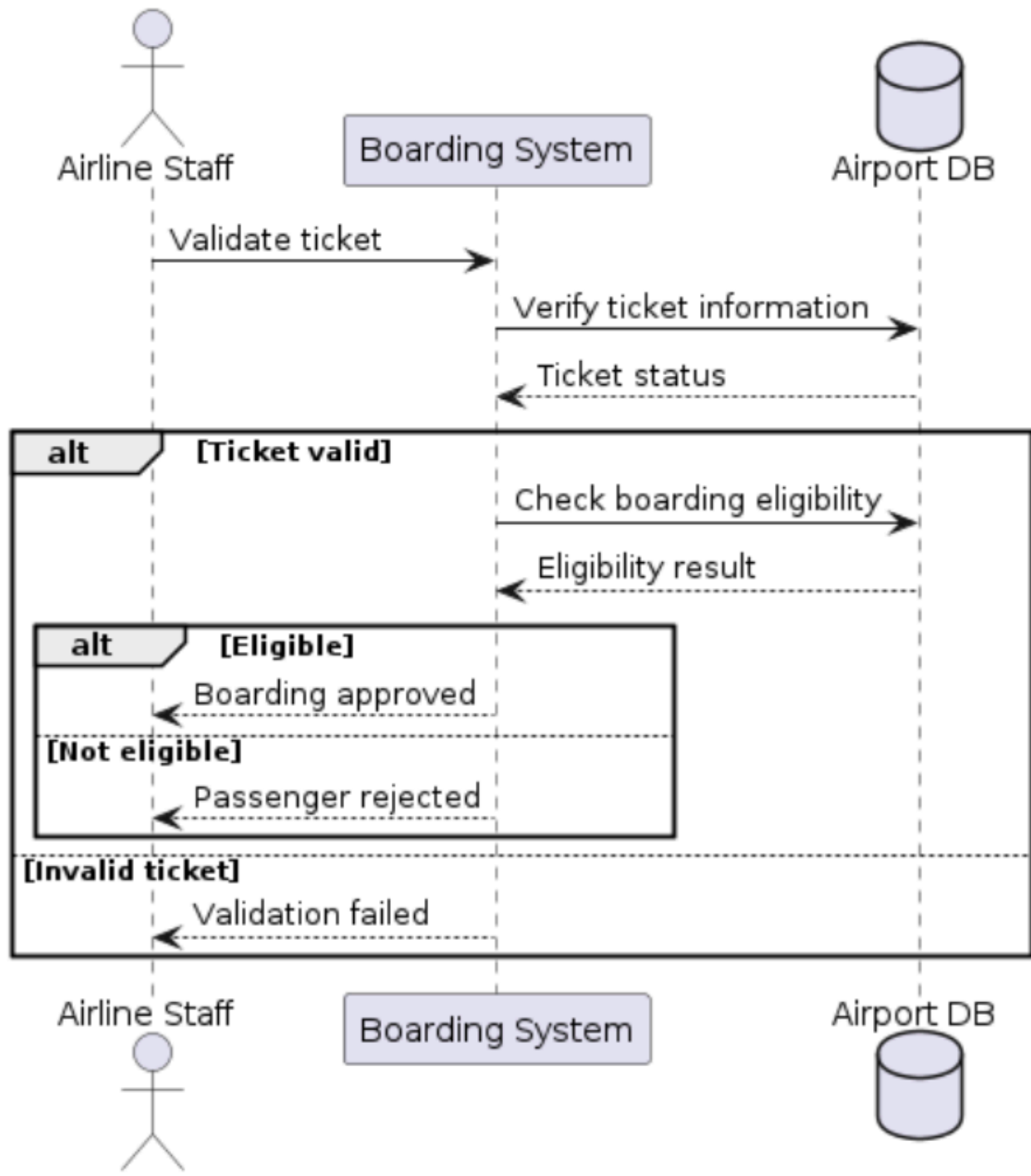
UC-29 Boarding Pass - Sequence Diagram



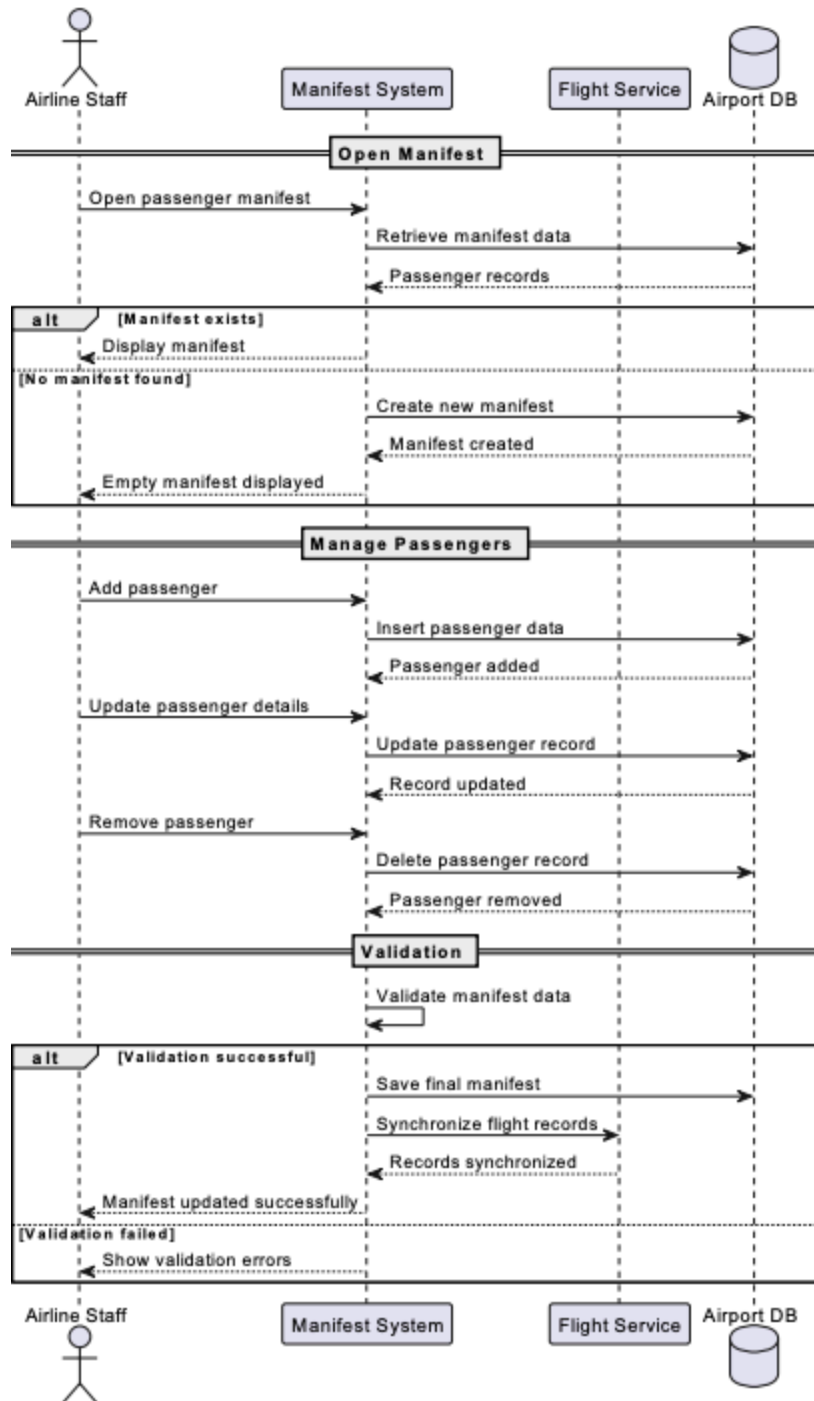
UC-30 Seat Handling - Sequence Diagram



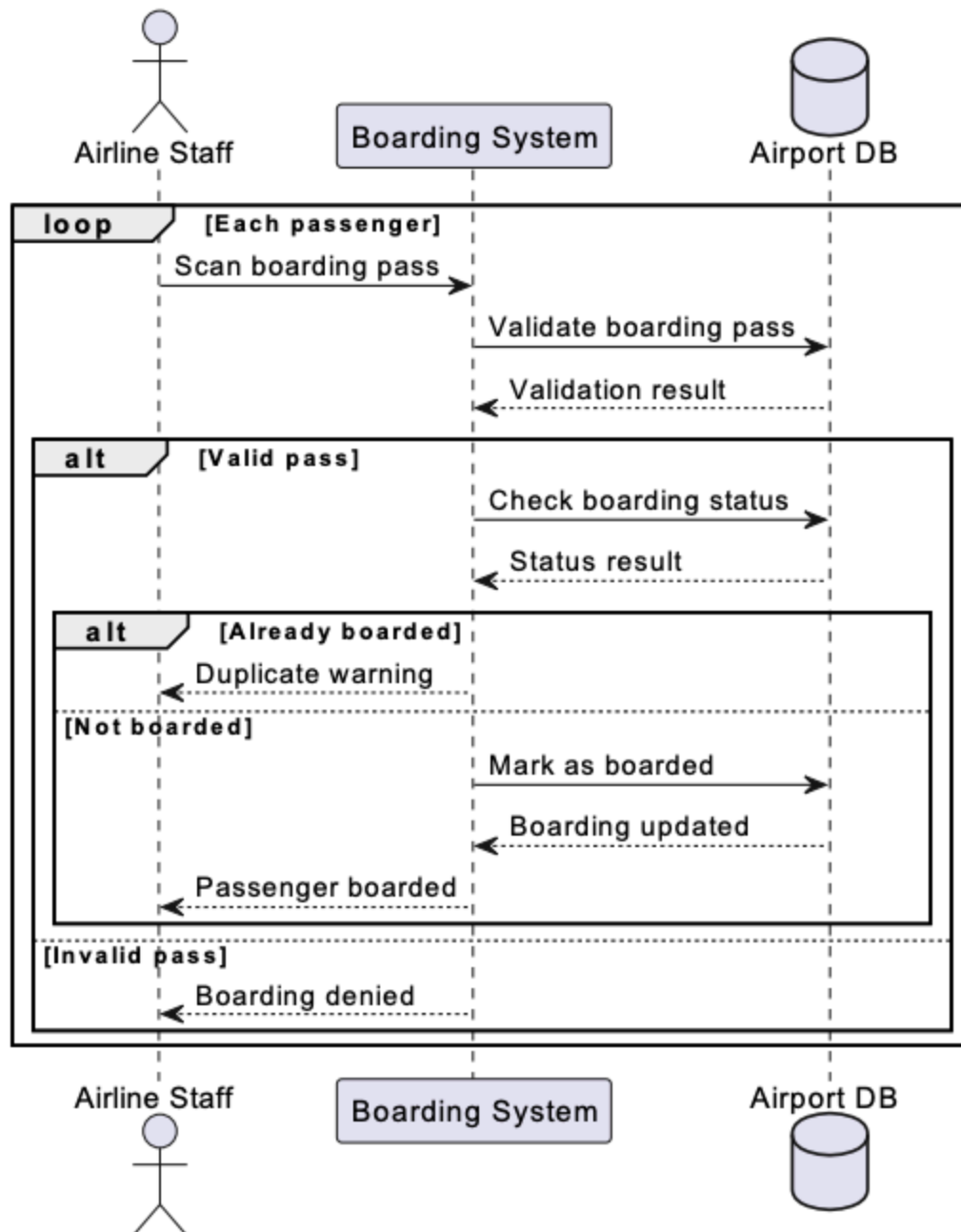
BR-31 Sequence Diagram - Ticket Validation



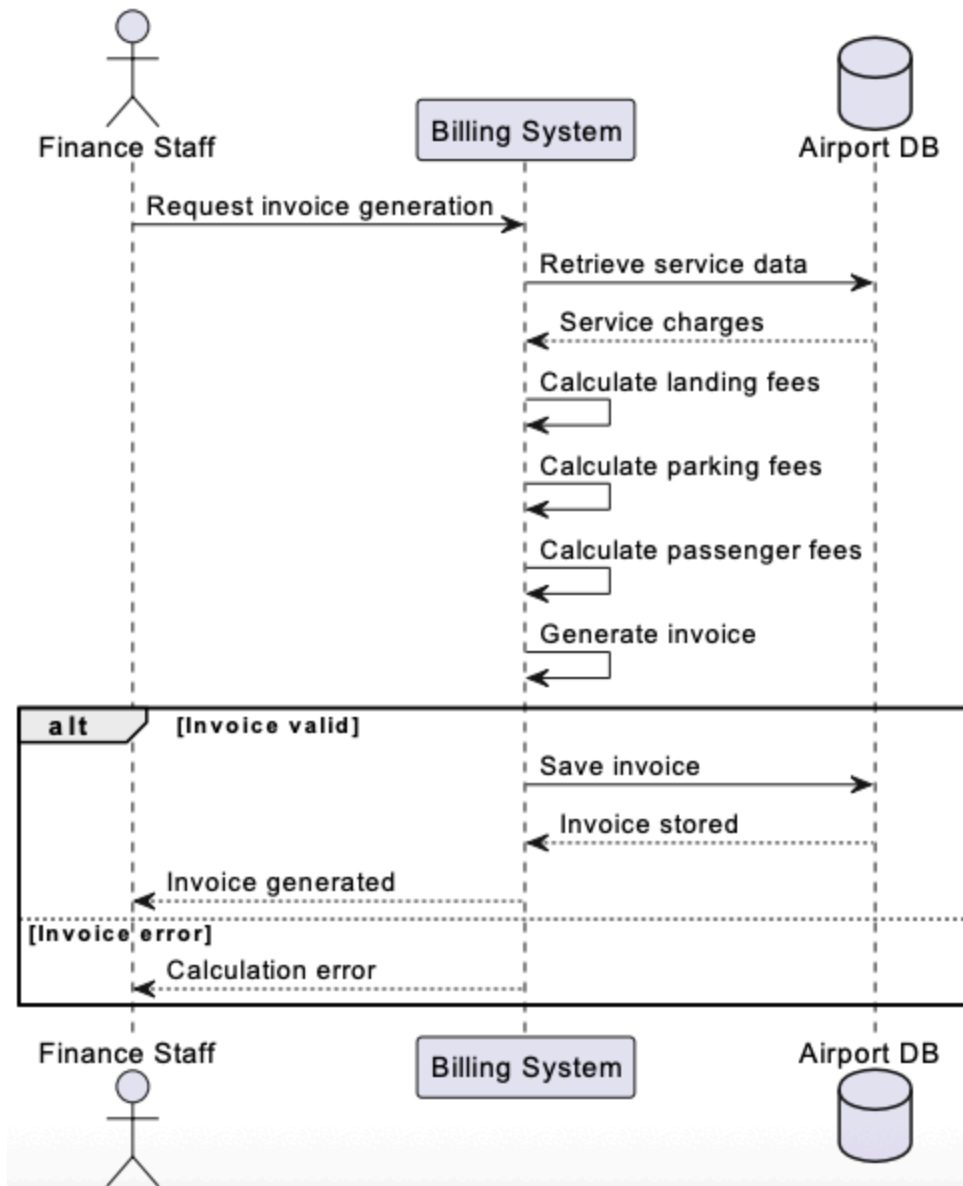
BR-32 Sequence Diagram - Passenger Manifest Management



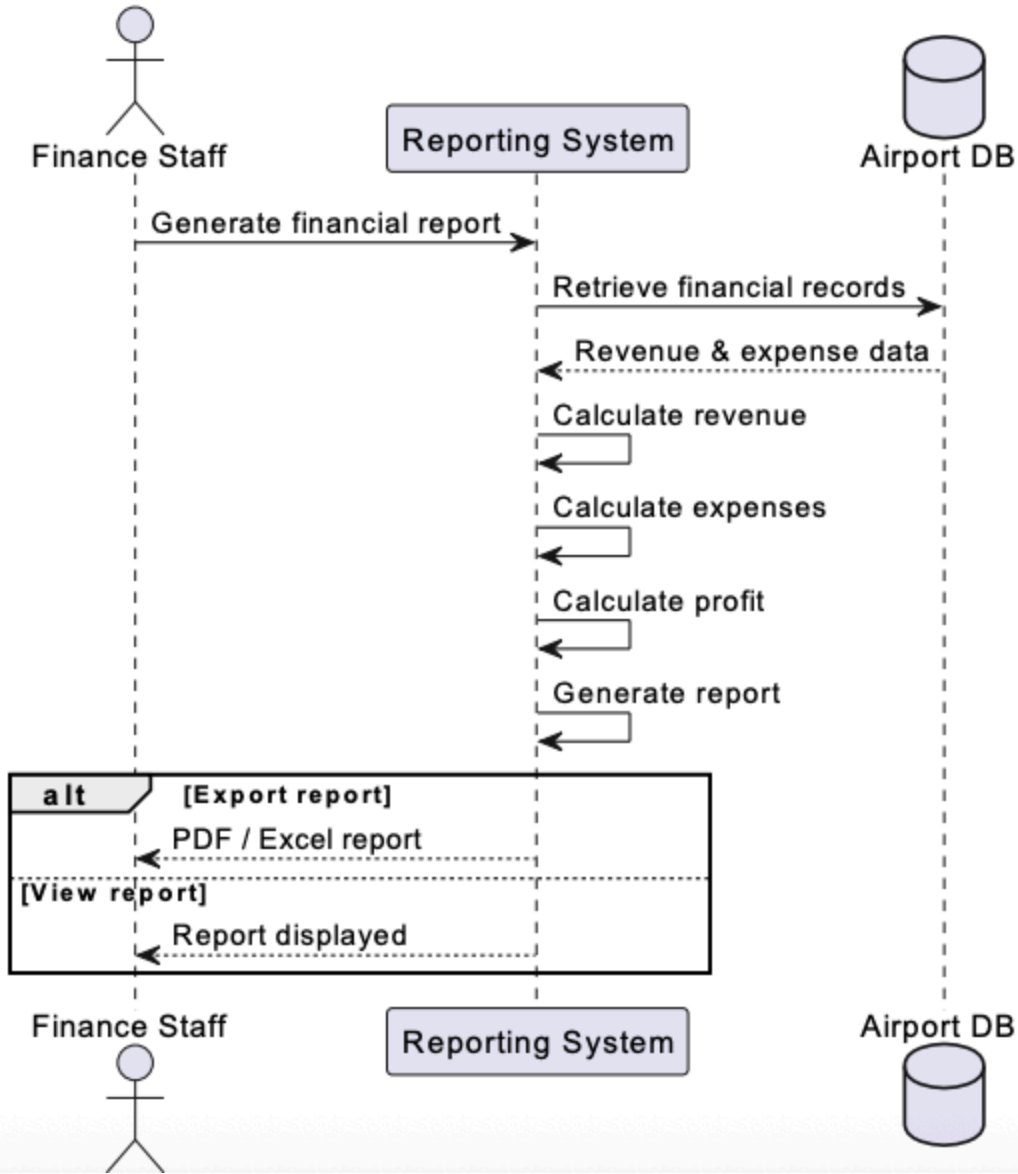
BR-33 Sequence Diagram - Boarding Process



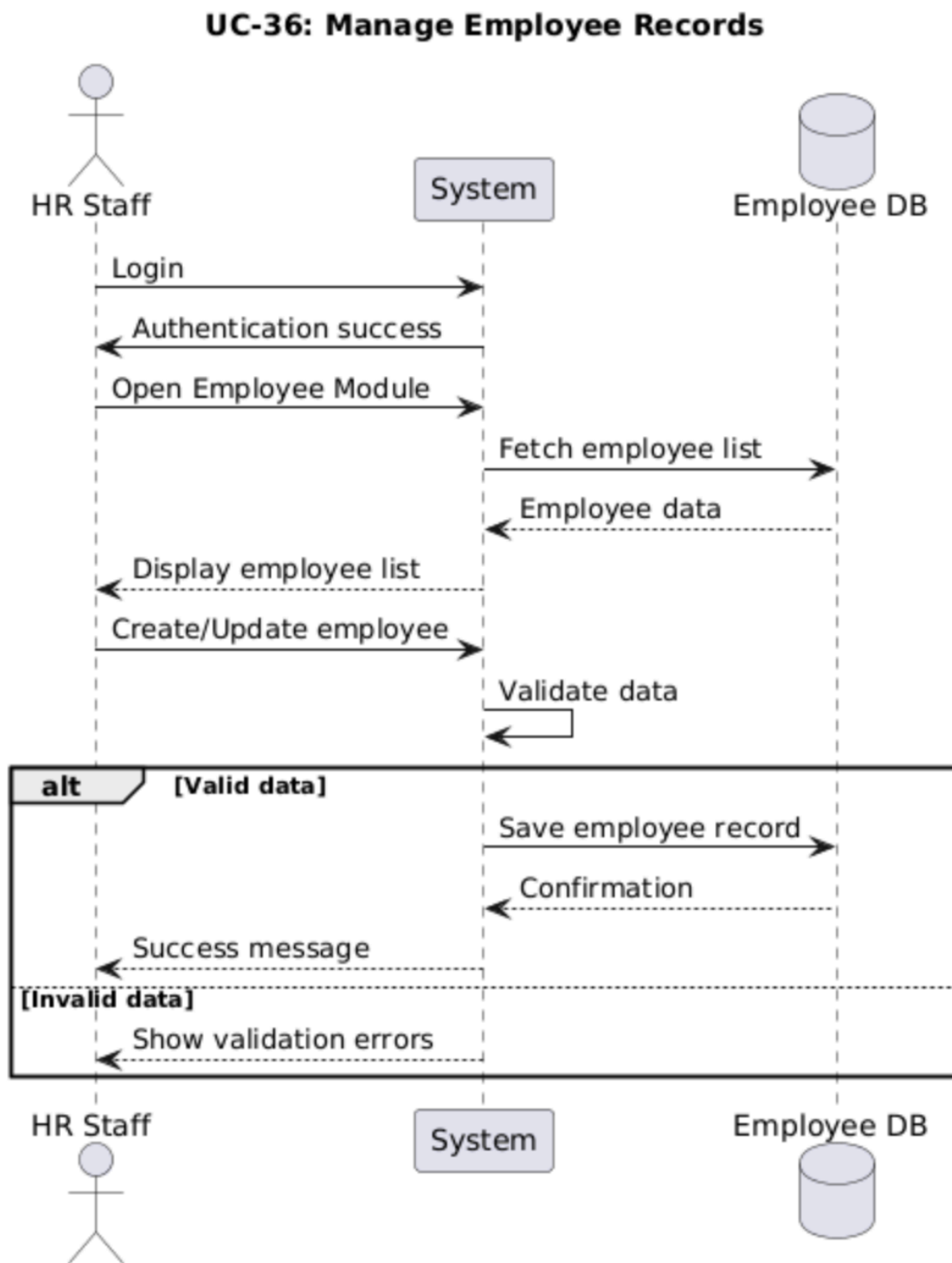
BR-34 Sequence Diagram - Invoice Generation



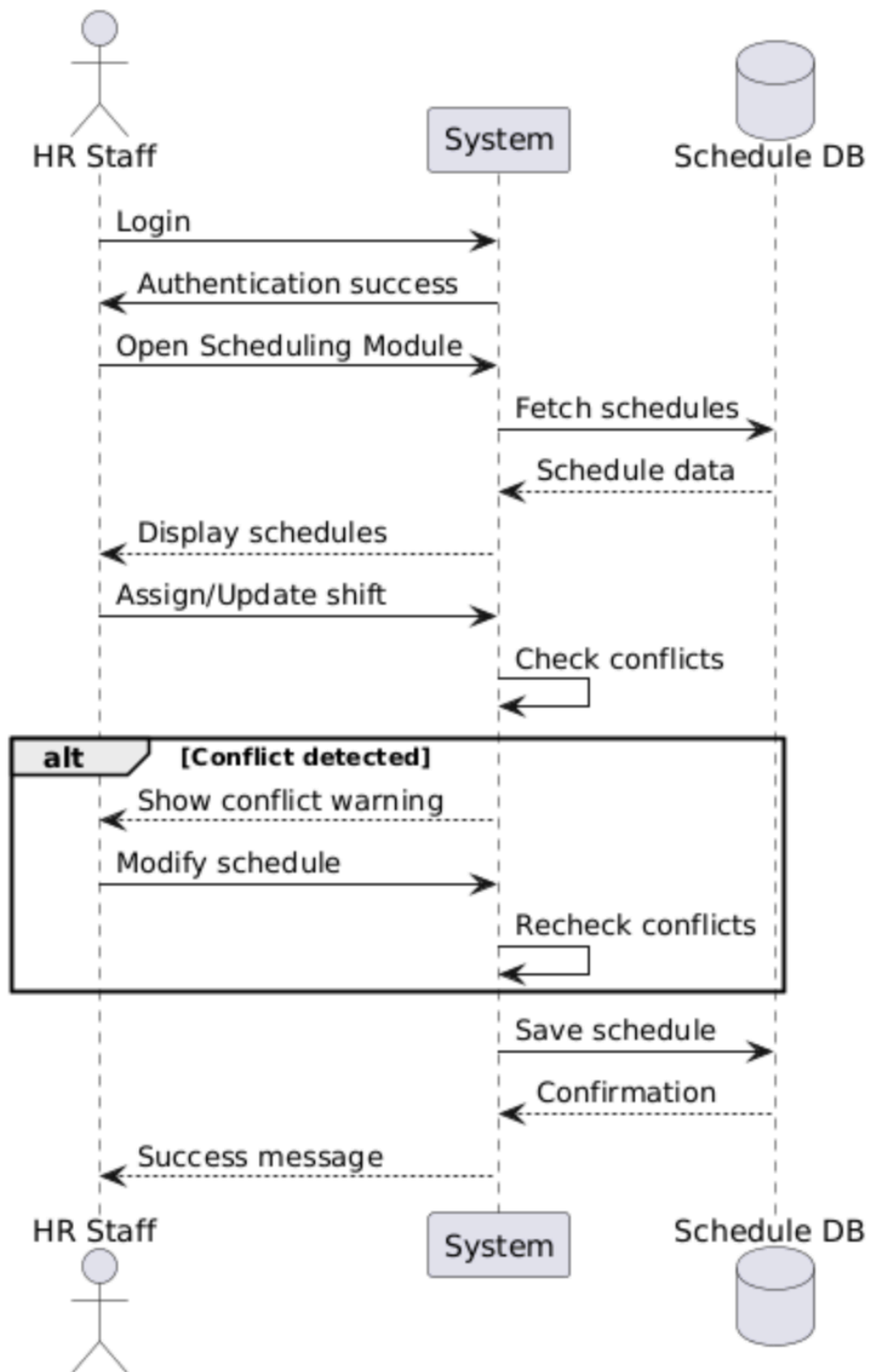
BR-35 Sequence Diagram - Financial Reporting



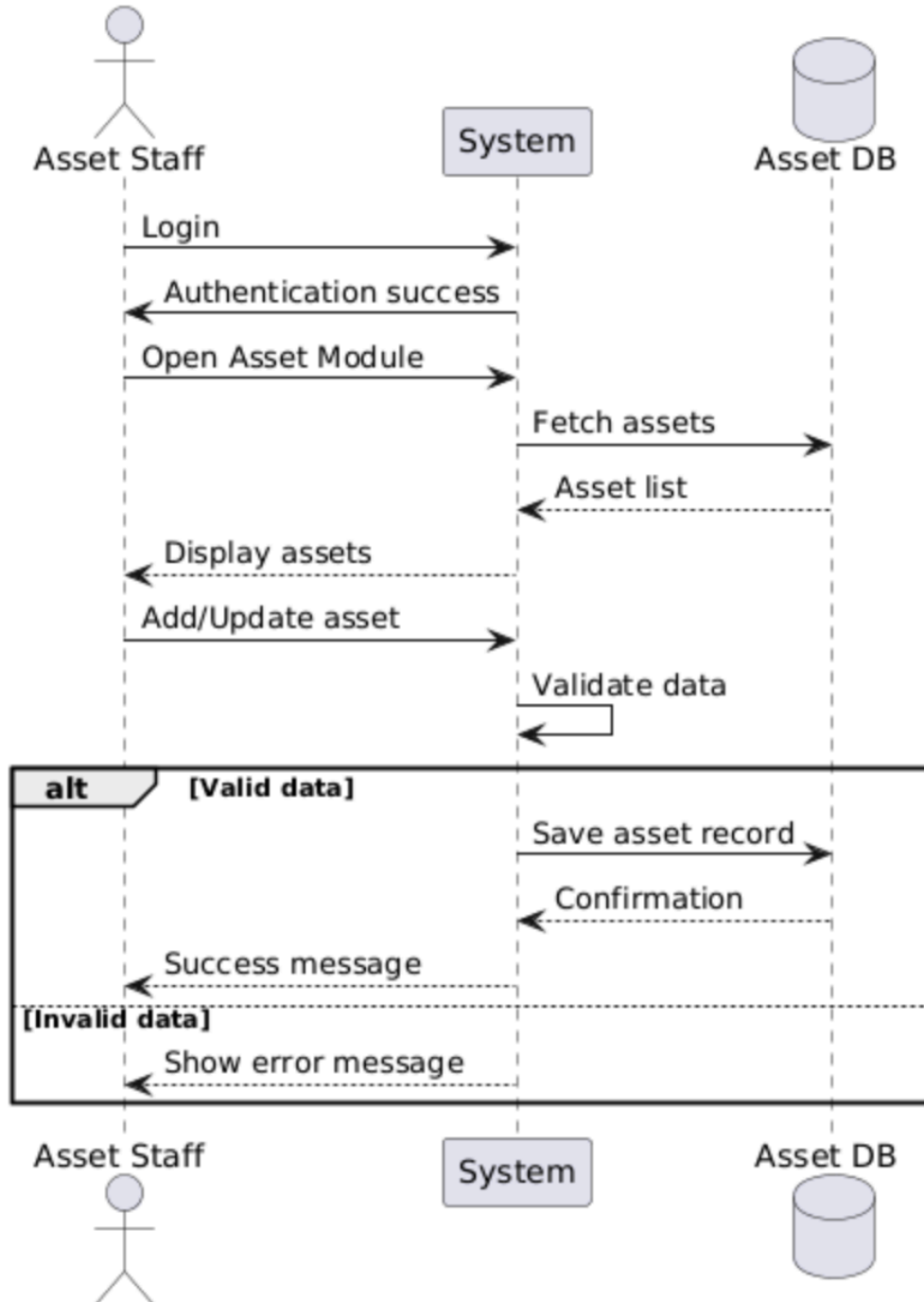
5.6.7 Sequence Diagrams UC 36-38 -> Stelina Spaho



UC-37: Manage Employee Schedules and Shifts



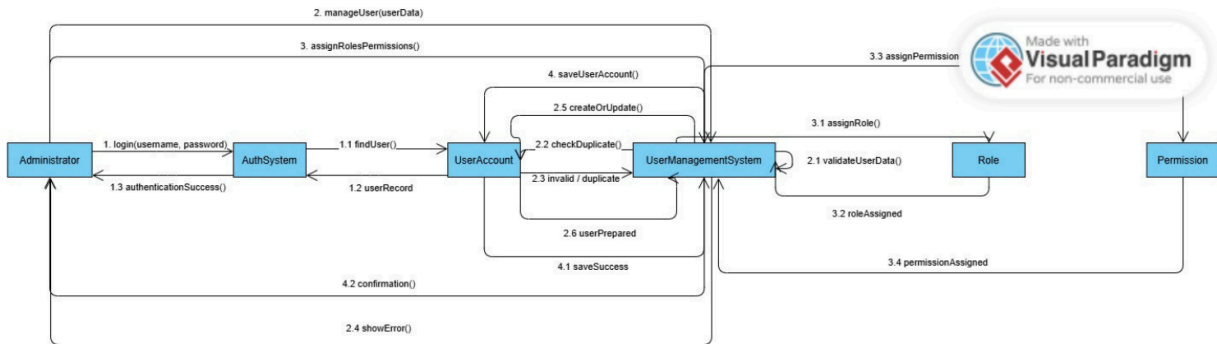
UC-38: Manage Airport Assets



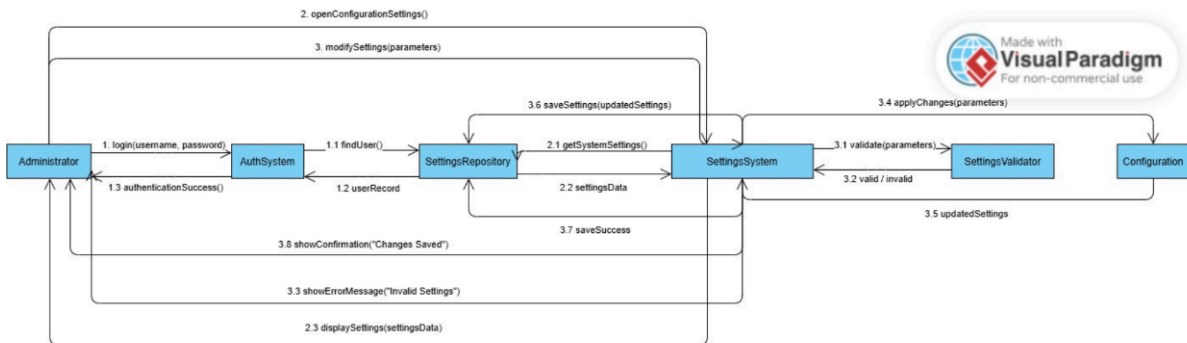
5.7 Collaboration Diagrams

5.7.1 Collaboration Diagrams UC 01-05 -> Moira Lako

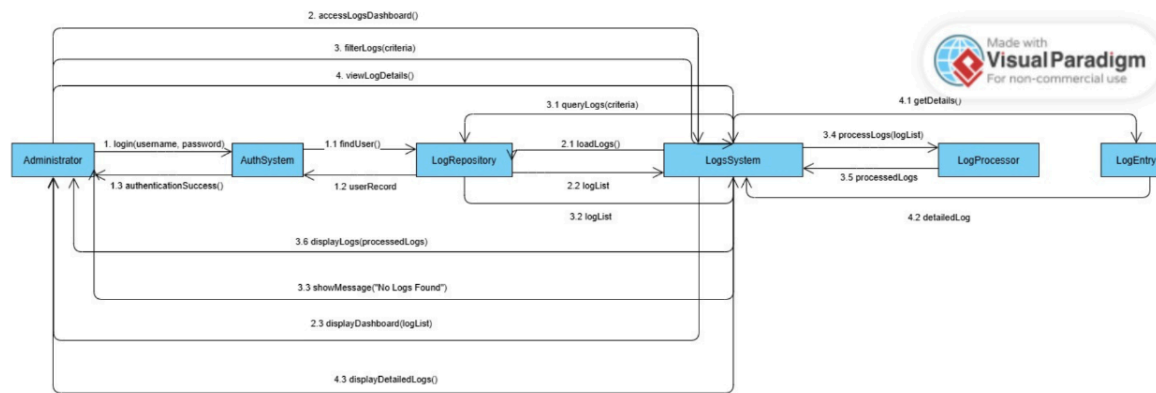
Collaboration Diagram UC-01: Manage User Accounts



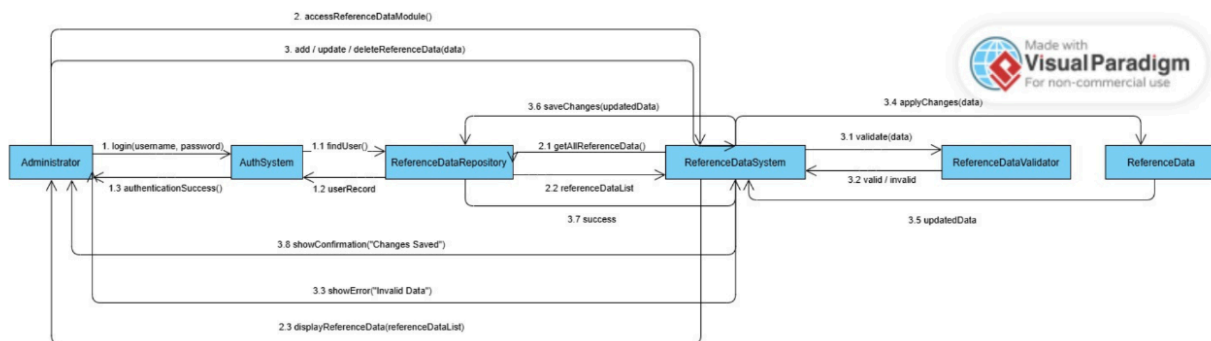
Collaboration Diagram UC-02: Configure System Settings



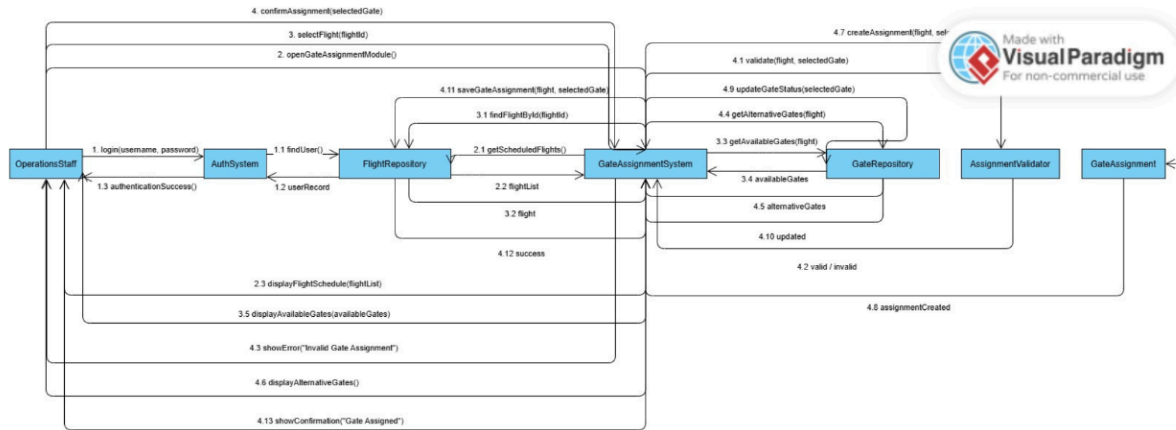
Collaboration Diagram UC-03: Monitor System Logs



Collaboration Diagram UC-04: Manage Airport Reference Data

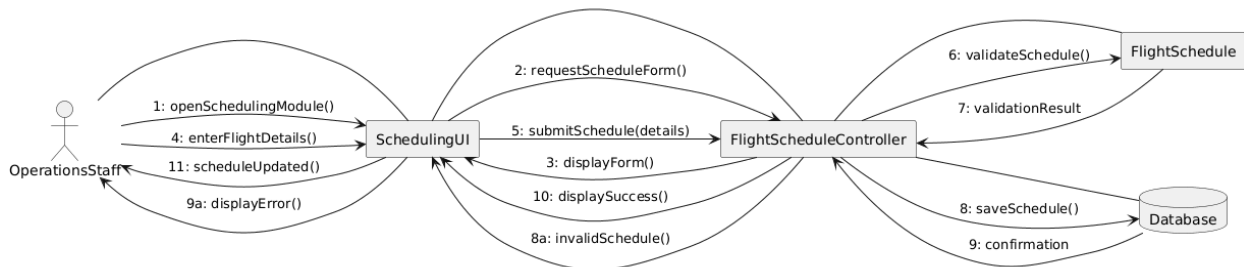


Collaboration Diagram UC-05: Assign Gates to Flights

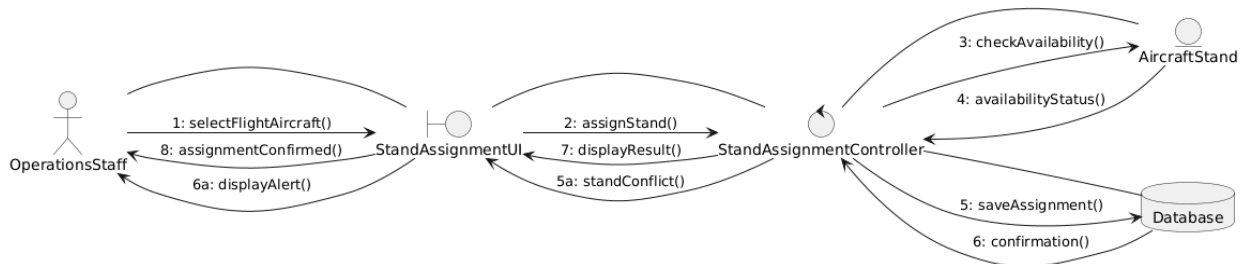


5.7.2 Collaboration Diagrams UC 06-10 -> Tea Metani

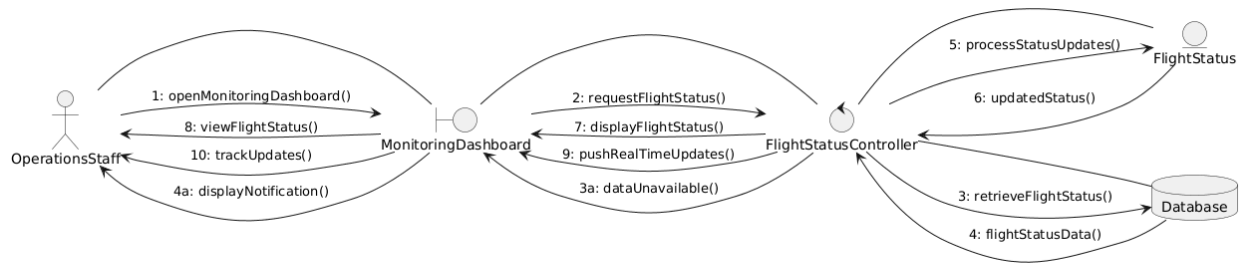
UC-06: Manage Flight Schedule



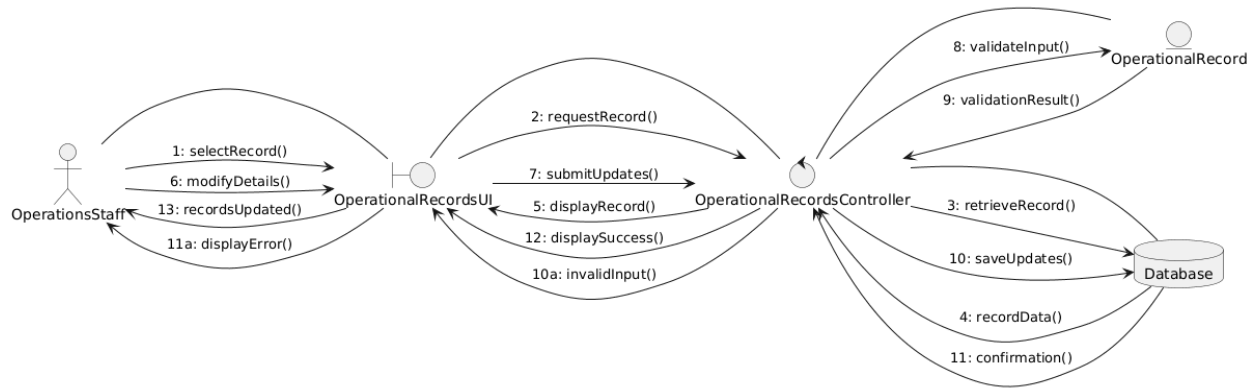
UC-07: Assign Aircraft Stands



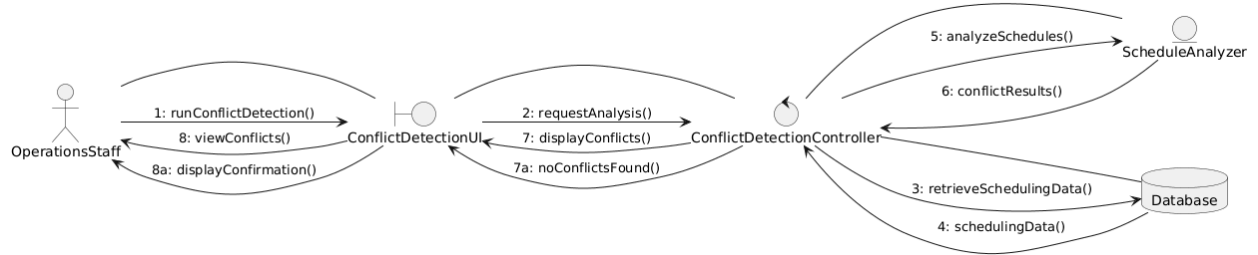
UC-08: Monitor Flight Status



UC-09: Update Operational Records

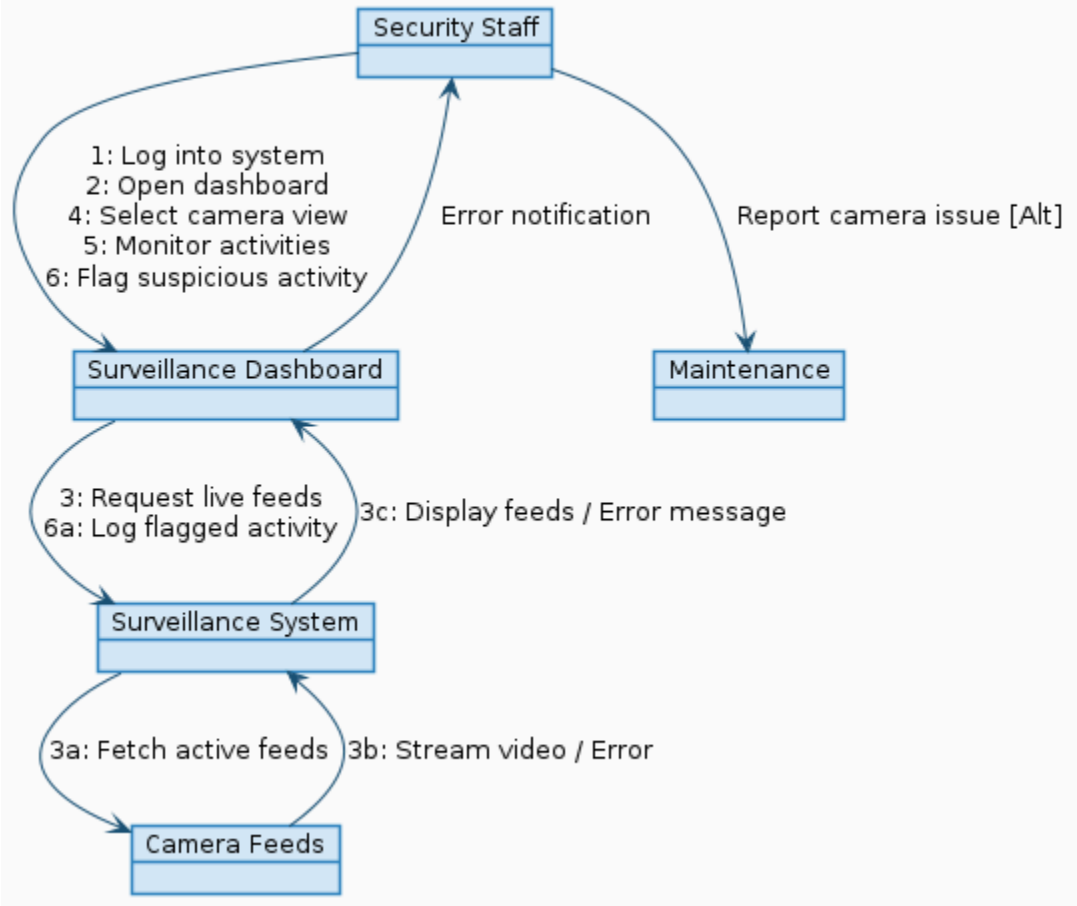


UC-10: Detect Scheduling Conflicts

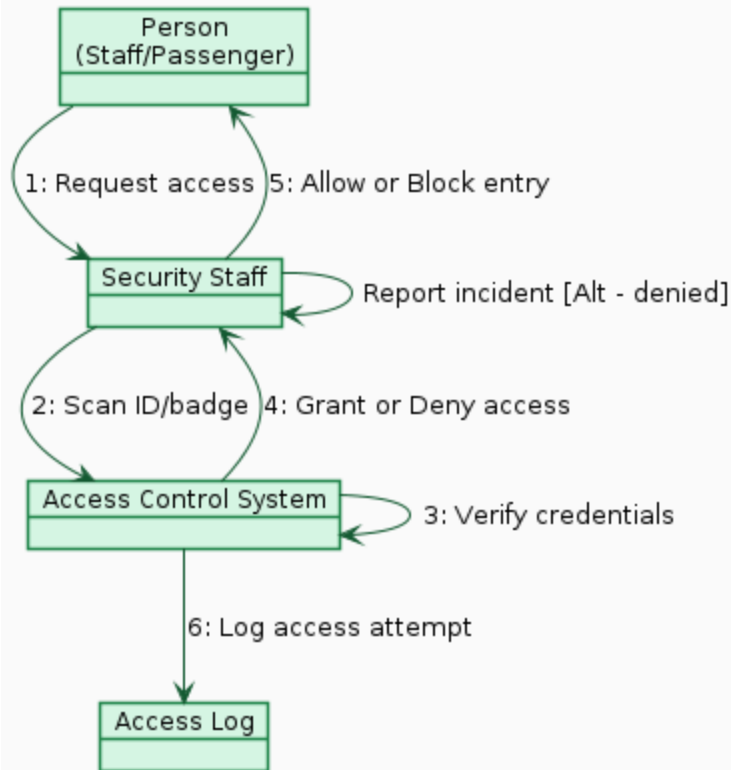


5.7.3 Collaboration Diagrams UC 11-15 -> Stelina Spaho

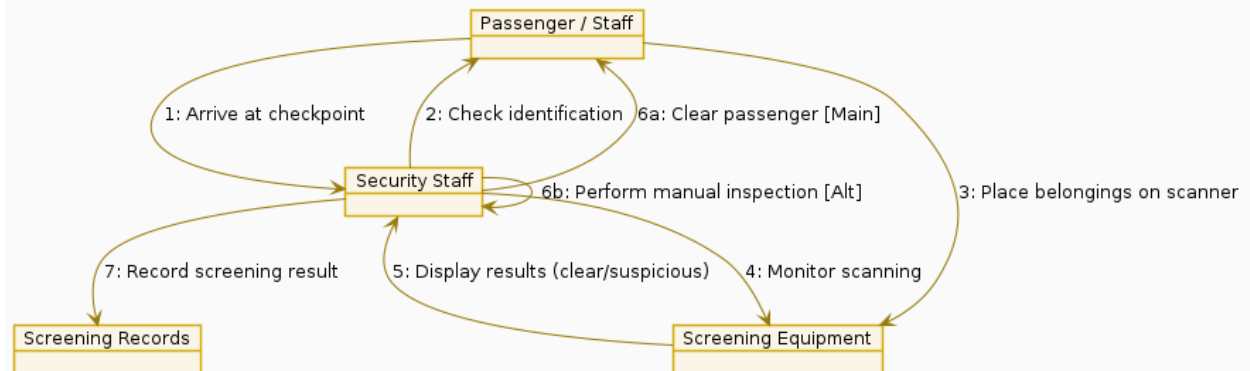
UC-11: Monitor Surveillance Camera Collaboration Diagram



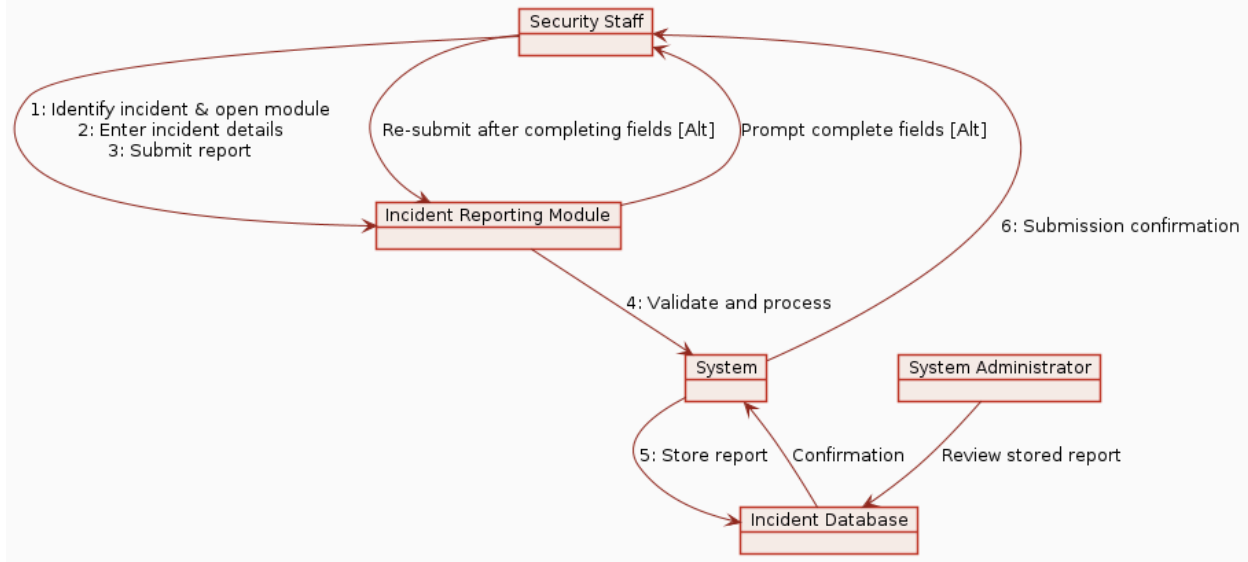
UC-12: Control Access to Restricted Areas Collaboration Diagram



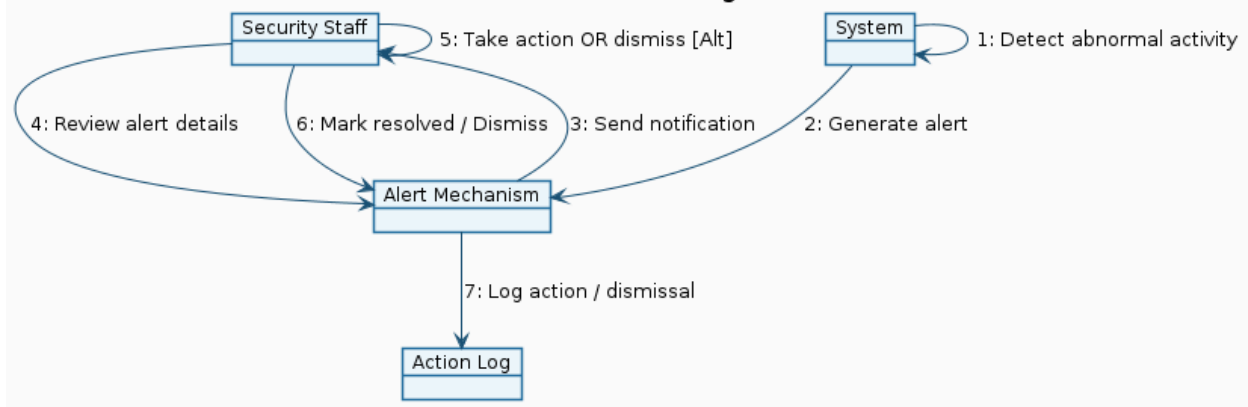
UC-13: Perform Security Screening Collaboration Diagram



UC-14: Record and Report Security Incident Collaboration Diagram

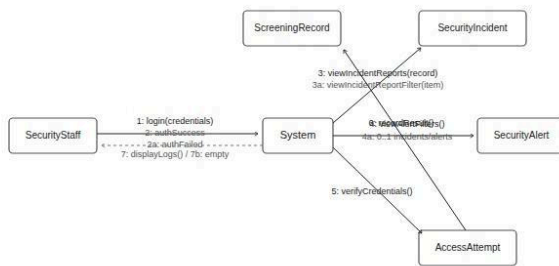


UC-15: Respond to Security Alerts Collaboration Diagram

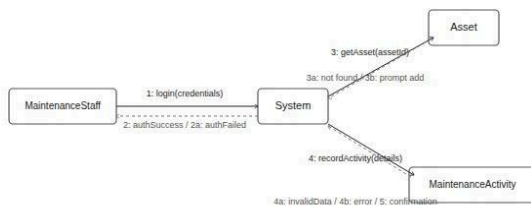


5.7.4 Collaboration Diagrams UC 16-20 -> Yeldos Rakhymzhanov

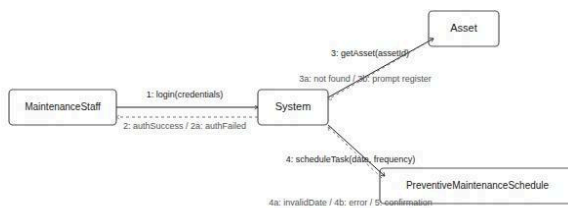
UC-16: Review Security Logs and Reports — Collaboration Diagram



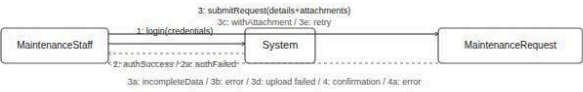
UC-17: Record Maintenance Activities — Collaboration Diagram



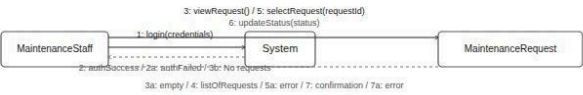
UC-18: Schedule Preventive Maintenance — Collaboration Diagram



UC-19: Report Infrastructure Issues — Collaboration Diagram

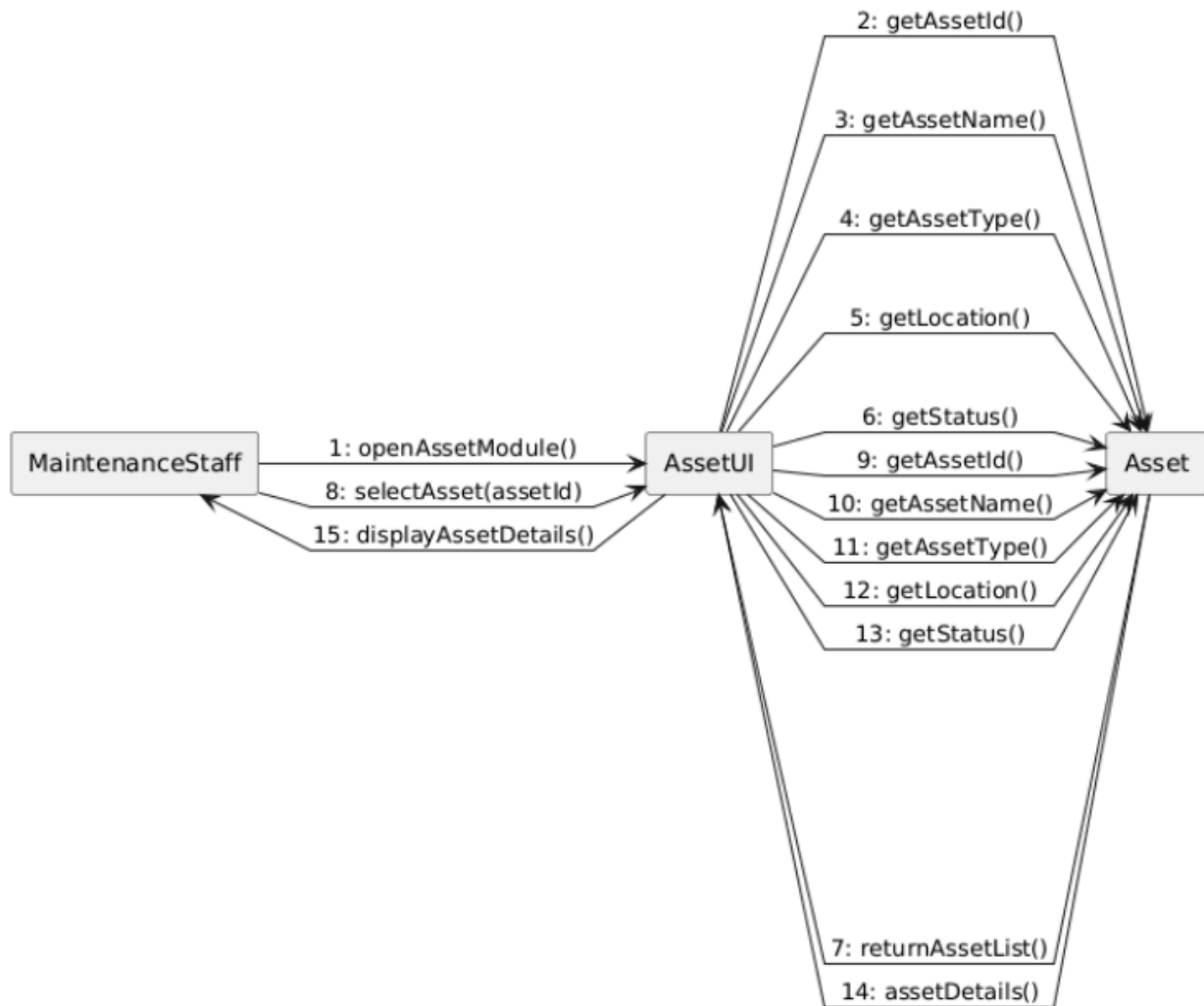


UC-20: Track Maintenance Requests — Collaboration Diagram

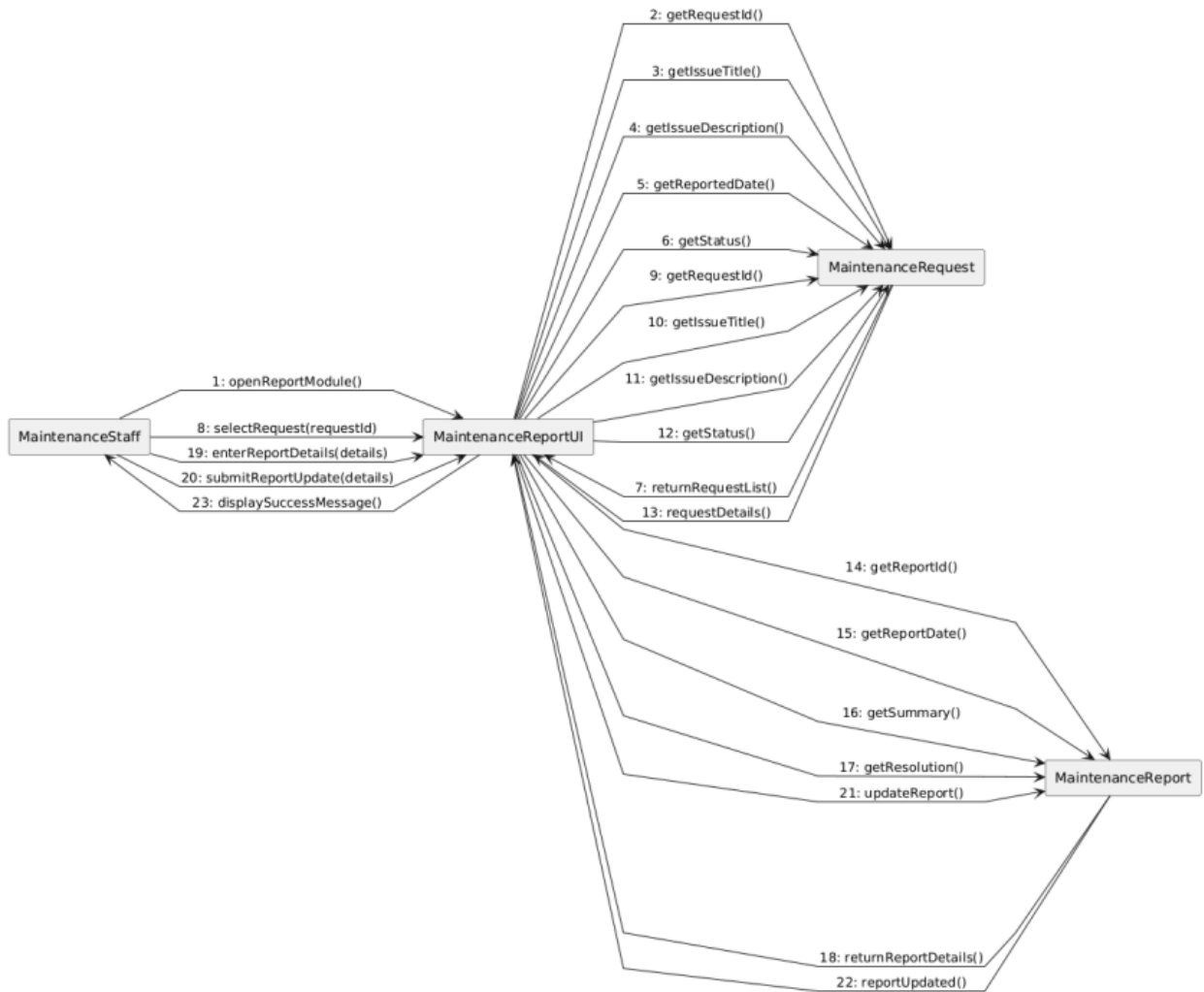


5.7.5 Collaboration Diagrams UC 21-25 -> Fatih Gökteke

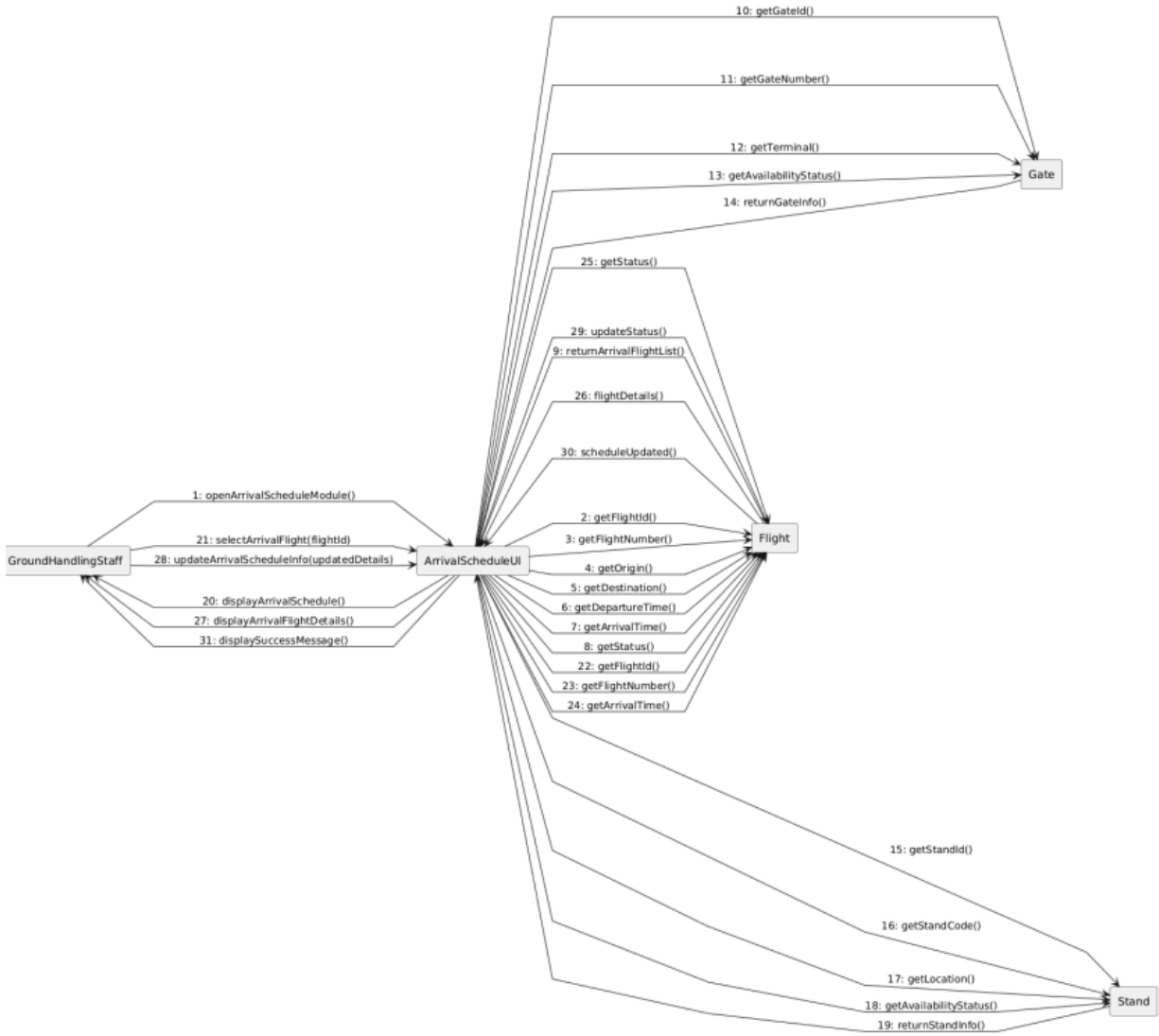
UC-21: Access Asset Information - Communication Diagram



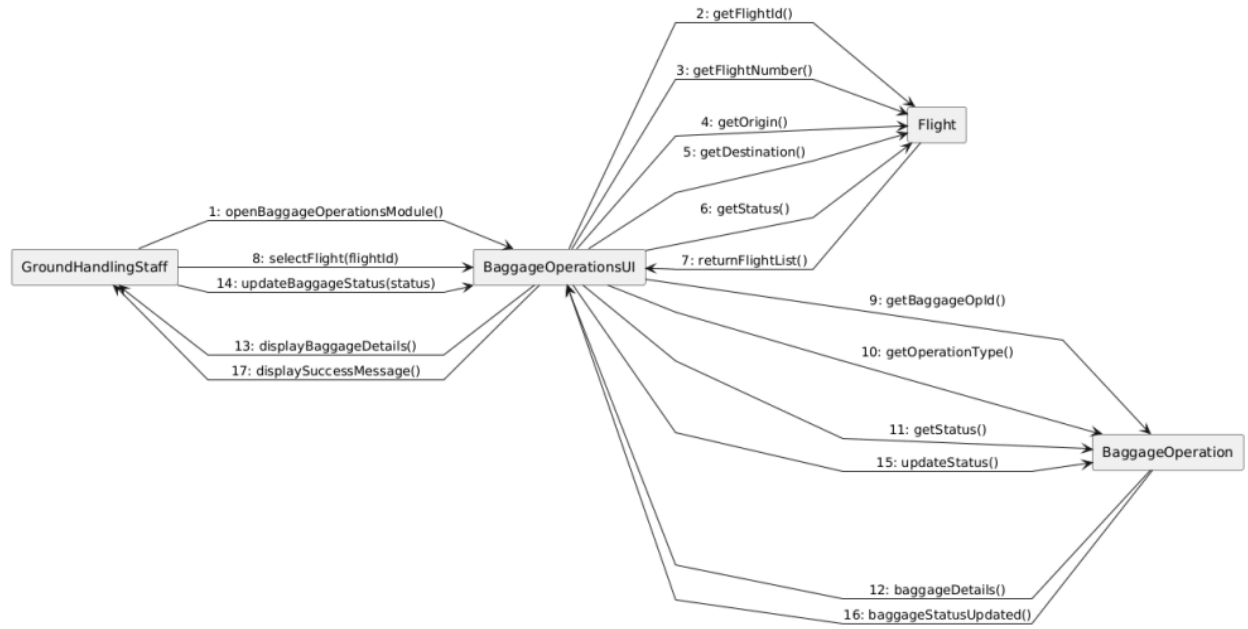
UC-22: Update Maintenance Report - Communication Diagram



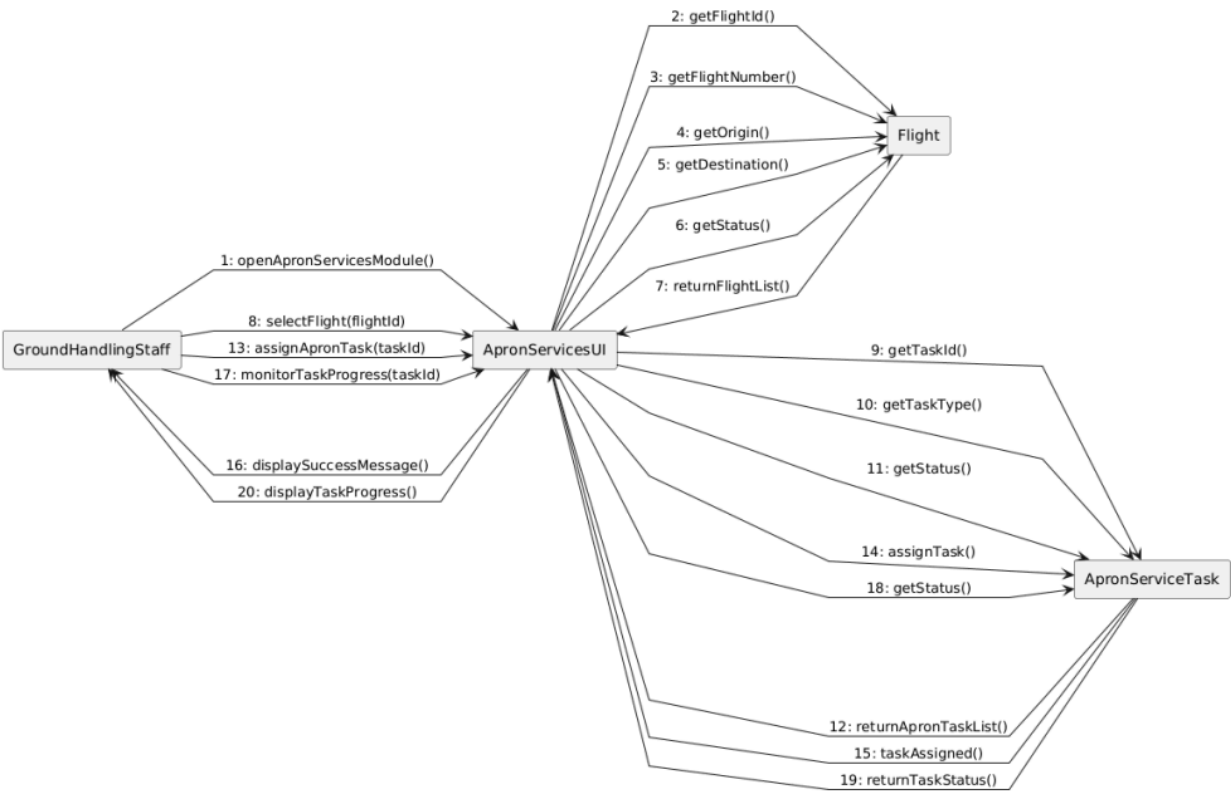
UC-23: Manage Arrival Schedule - Communication Diagram



UC-24: Manage Baggage Operations - Communication Diagram

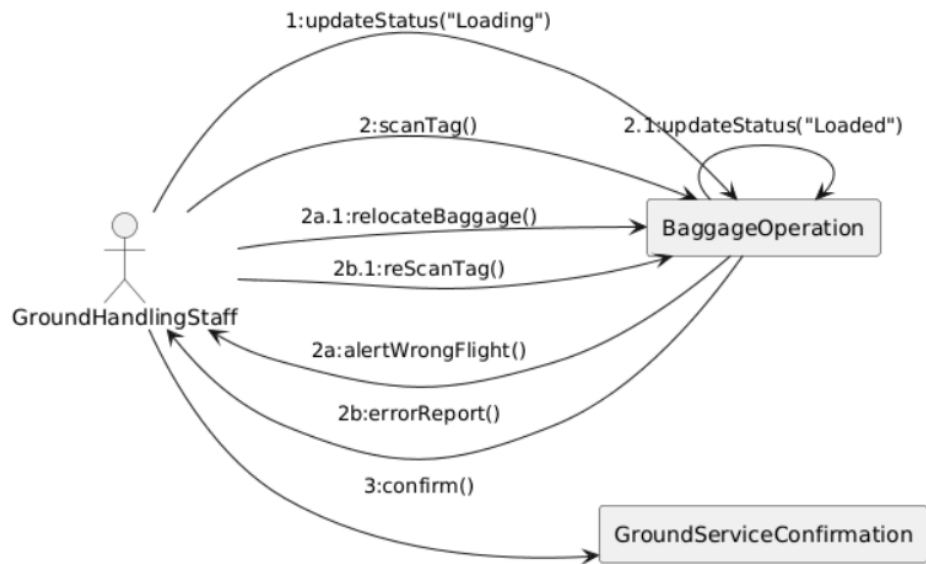


UC-25: Manage Apron Services - Communication Diagram

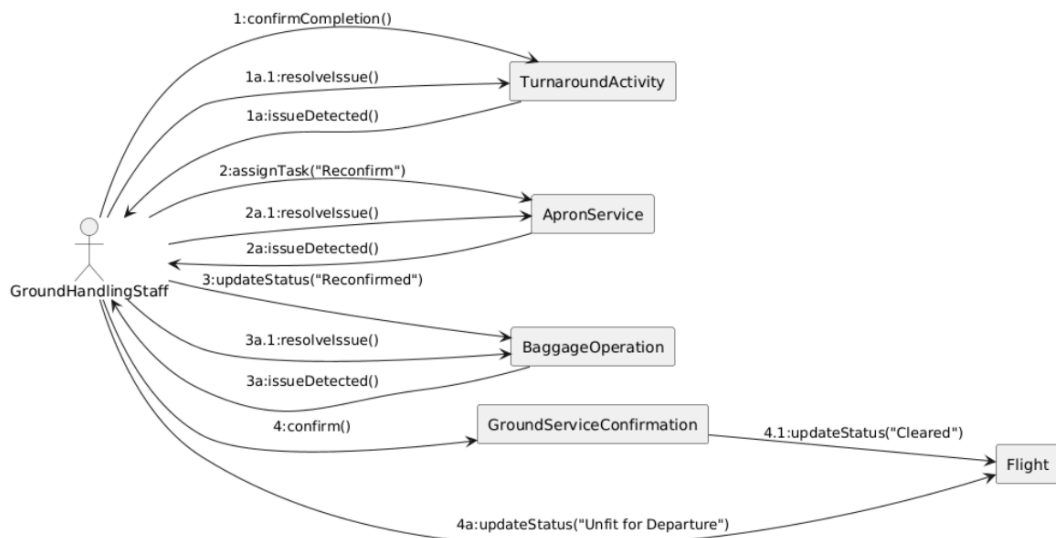


5.7.6 Collaboration Diagrams UC 26-30 -> Leart Mukaj

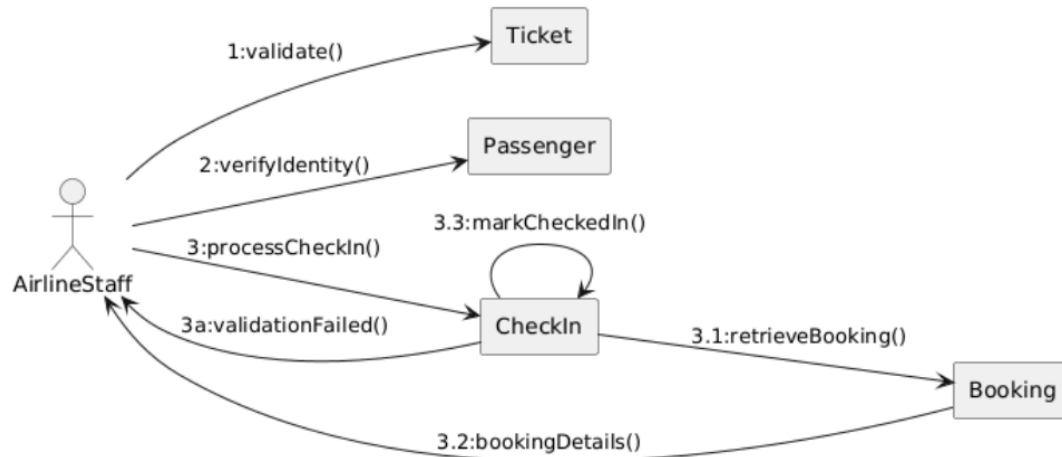
UC-26 Baggage Loading - Collaboration Diagram



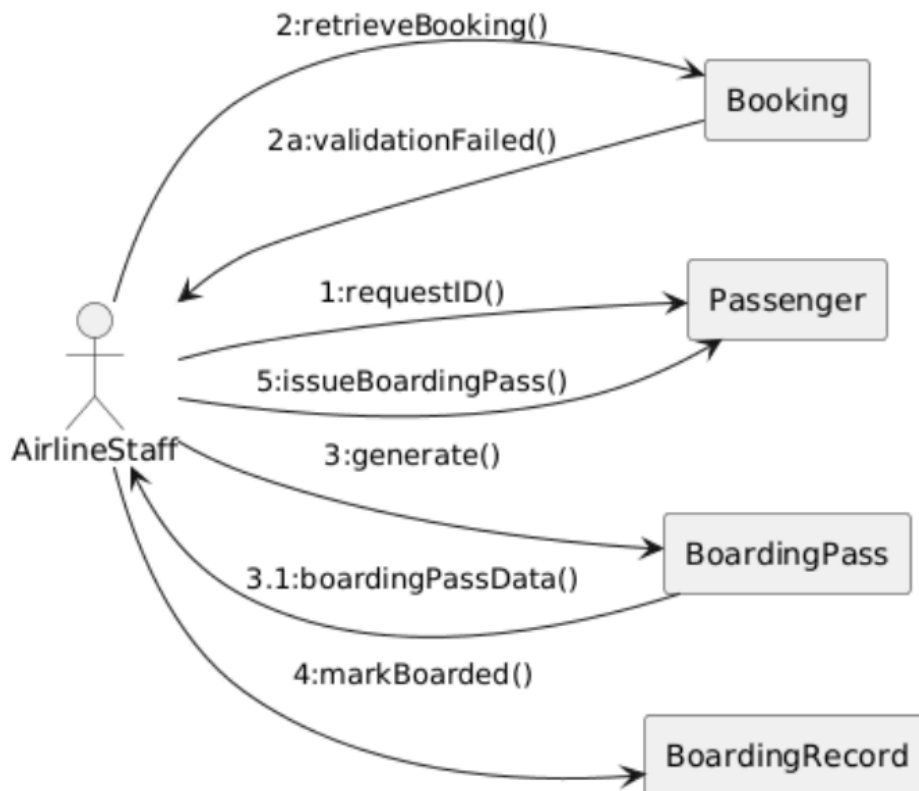
UC-27 Final Confirmation - Collaboration Diagram



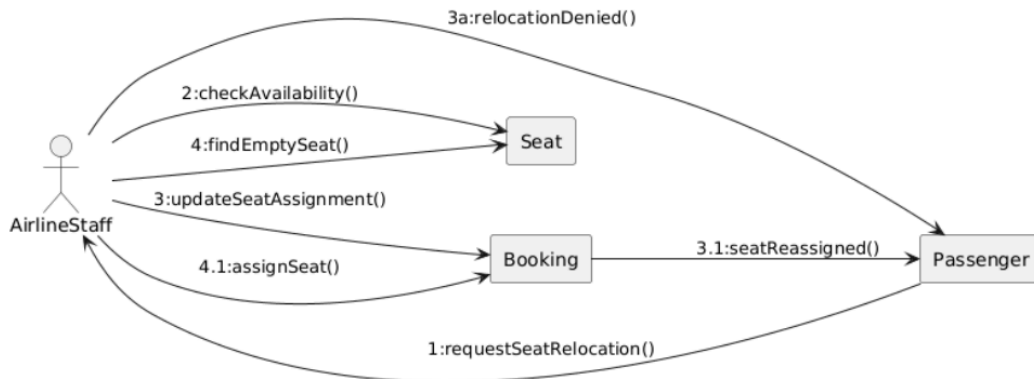
UC-28 Passenger Check-in - Collaboration Diagram



UC-29 Boarding Pass - Collaboration Diagram

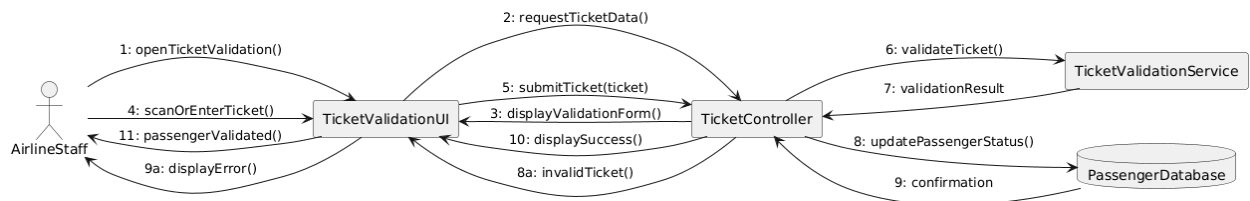


UC-30 Seat Handling - Collaboration Diagram

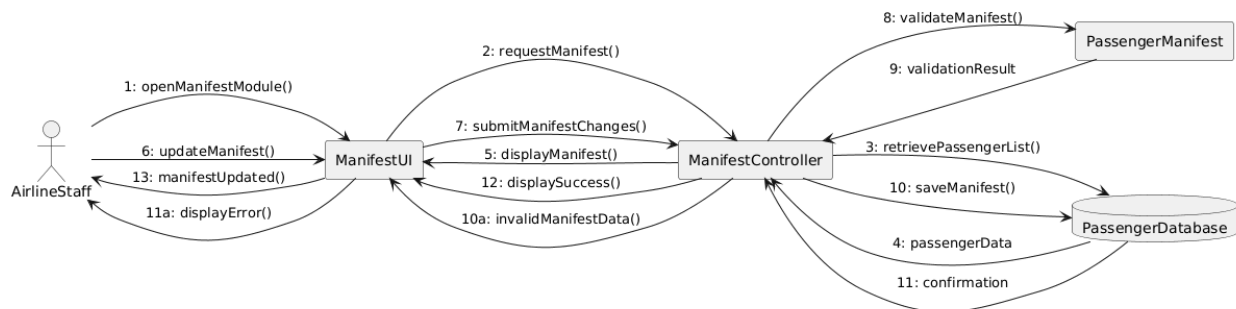


5.7.7 Collaboration Diagrams UC 31-35 -> Erisa Meta

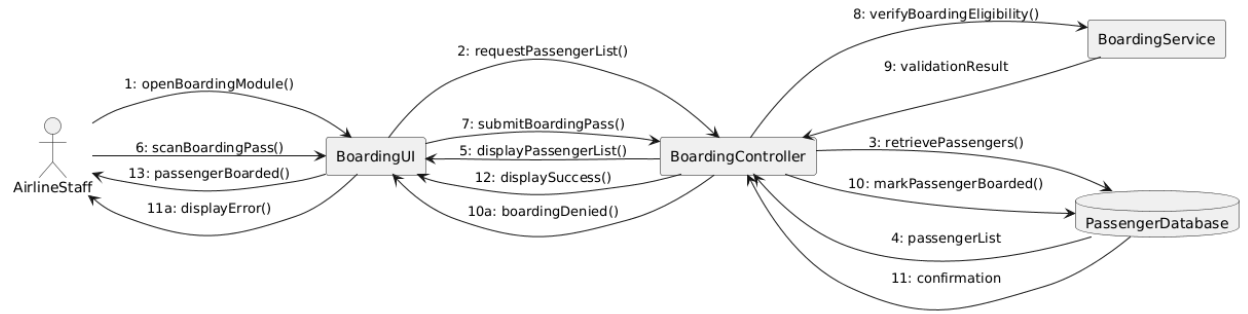
UC-31 Collaboration Diagram - Validate Passenger Tickets



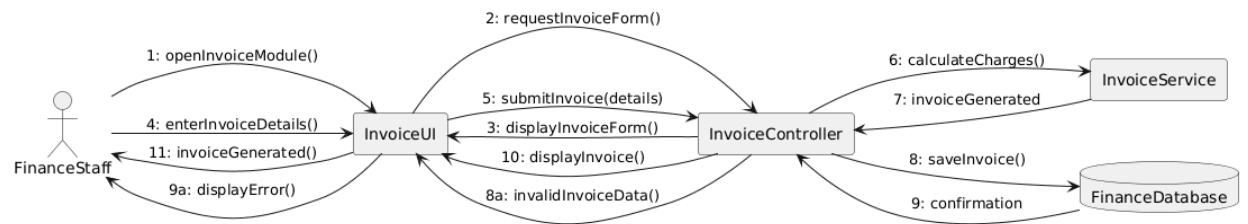
UC-32 Collaboration Diagram - Manage Passenger Manifests



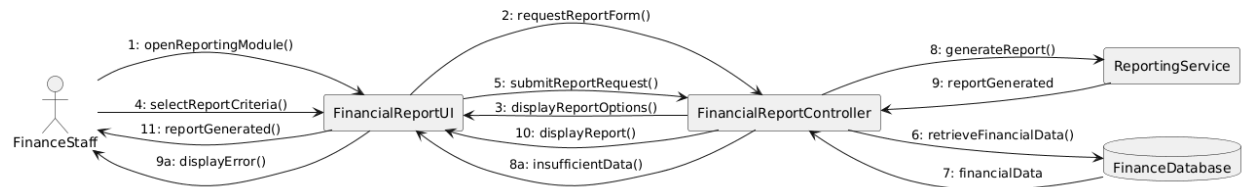
UC-33 Collaboration Diagram - Manage Boarding Process



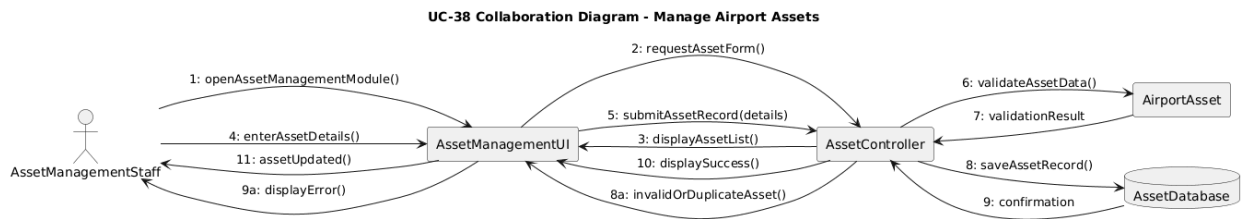
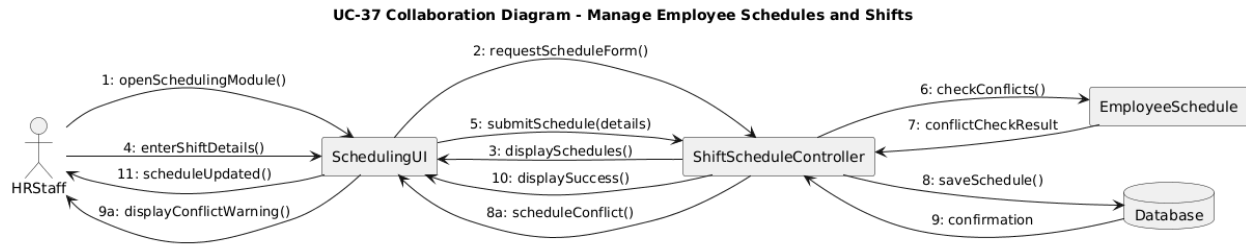
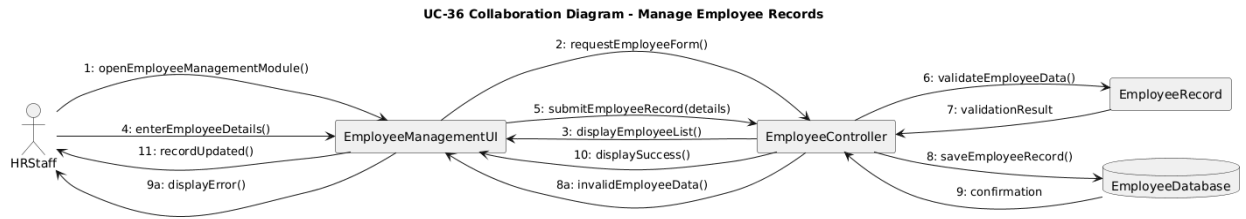
UC-34 Collaboration Diagram - Generate Invoices



UC-35 Collaboration Diagram - Generate Financial Reports

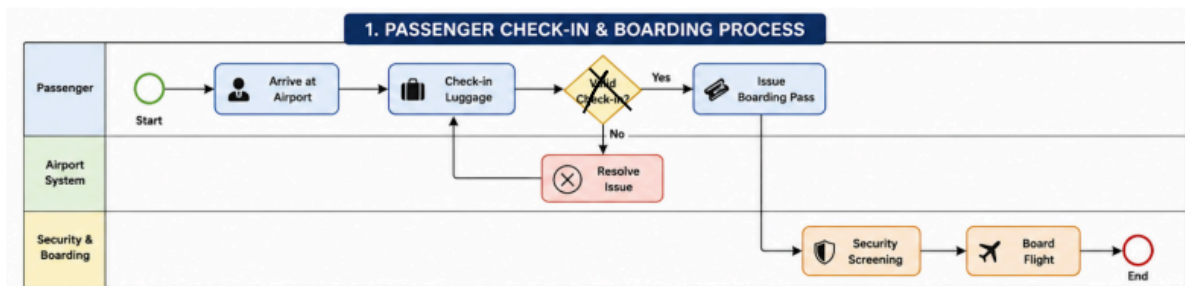


5.7.8 Collaboration Diagrams UC 36-38 -> Moira Lako

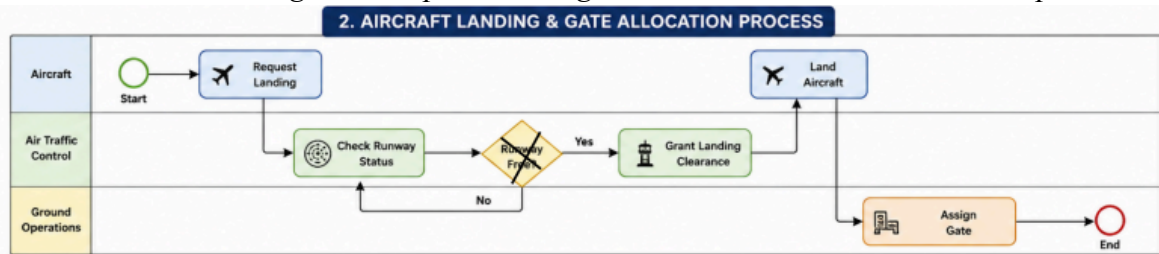


5.8 BPMN Diagrams

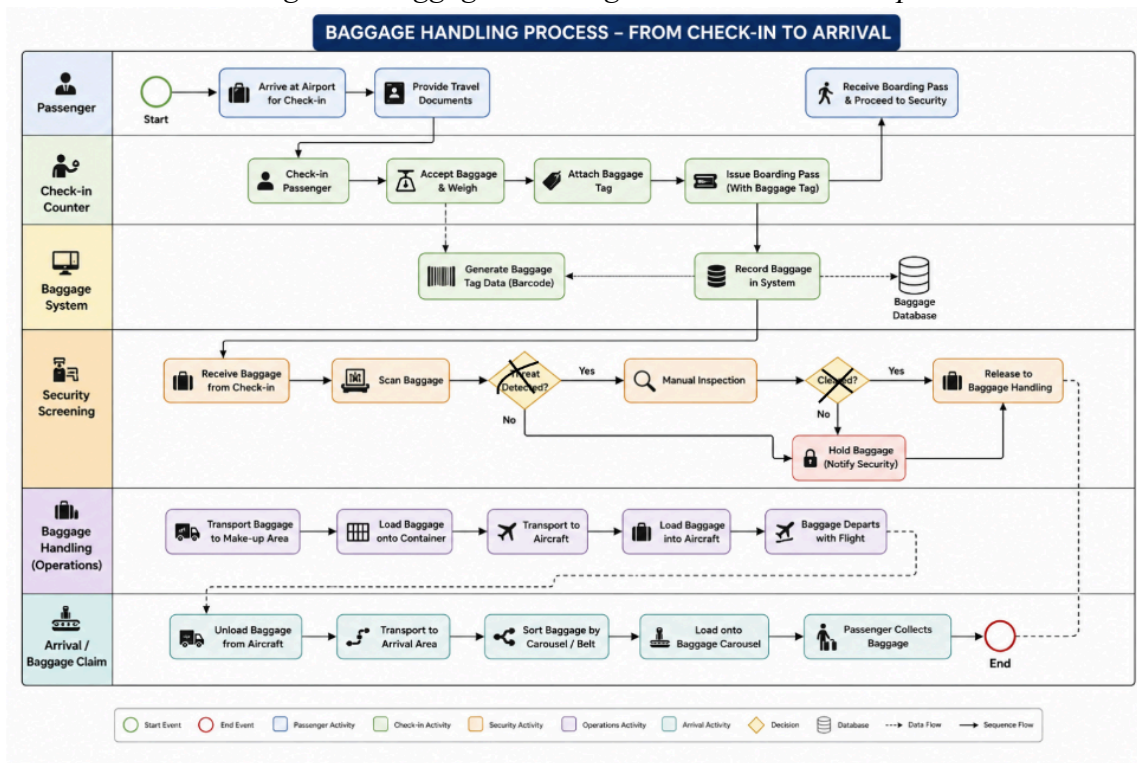
5.8.1 BPMN Diagram - Passenger Check in and Boarding Process -> Stelina Spaho



5.8.2 BPMN Diagram - Airport Landing and Gate Allocation -> Stelina Spaho



5.8.3 BPMN Diagram - Baggage Handling Process -> Stelina Spaho



5.9 Interaction Overview Diagram (IOD) - by Moira Lako



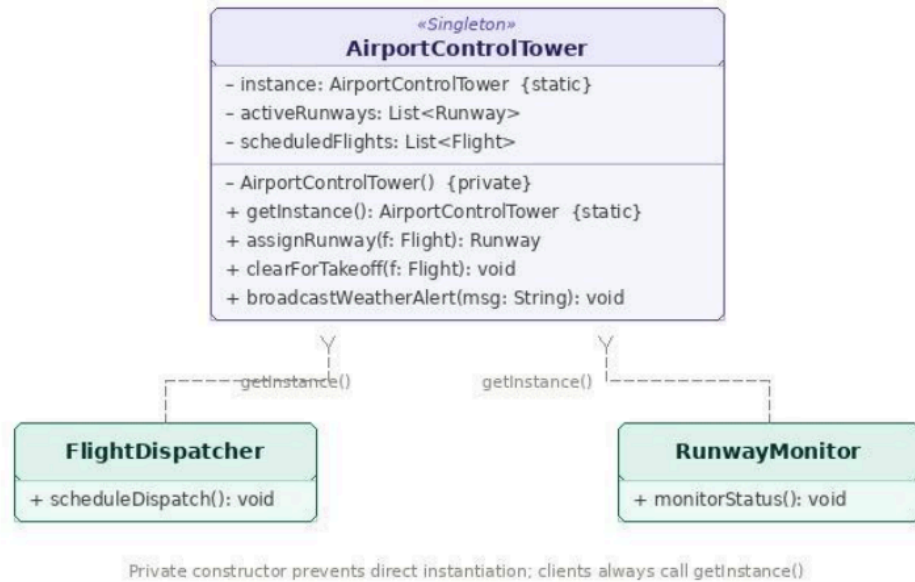
6. Design Patterns

6.1 Singleton — AirportControlTower - by Stelina Spaho

An airport has exactly one control tower coordinating all flight operations. A Singleton ensures a single, globally-accessible instance manages runway assignments, departure clearances, and communication channels, preventing conflicting instructions from multiple instances.

The Singleton diagram shows `AirportControlTower` with a private constructor and a static `getInstance()` method. Both client classes (`FlightDispatcher`, `RunwayMonitor`) access it only through that method — they can never create a second instance.

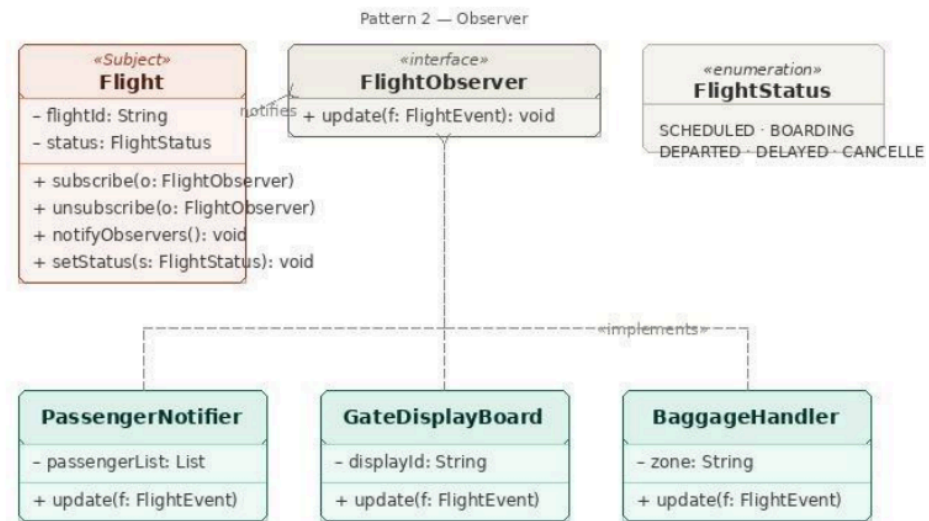
Pattern 1 — Singleton



6.2 Observer — FlightStatusNotifier - *by Stelina Spaho*

Many stakeholders (passengers, gate agents, baggage handlers, airlines) need real-time updates when a flight's status changes (delayed, boarding, cancelled). The Observer pattern decouples the flight entity from its subscribers, so new observers can be added without modifying the flight class.

The Observer diagram shows Flight as the subject holding a list of FlightObserver subscribers. When setStatus() triggers notifyObservers(), all three concrete observers — PassengerNotifier, GateDisplayBoard, and BaggageHandler — receive the FlightEvent through the shared update() interface. New observers (e.g. an airline operations dashboard) can be plugged in without touching Flight.



Any number of observers can be added at runtime without modifying Flight